

III. Otras Resoluciones

Consejería de Agricultura, Ganadería, Pesca y Soberanía Alimentaria

2566 *Instituto Canario de Investigaciones Agrarias.- Resolución de 2 de julio de 2026, del Secretario General, por la que se ordena la publicación del Convenio Específico de Colaboración entre el Instituto Canario de Investigaciones Agrarias y el Cabildo Insular de Tenerife para la realización del Proyecto “Raíces y Tubérculos menores que fomentan la agrobiodiversidad y los servicios ecosistémicos (ROTATES)”.*

De conformidad con lo dispuesto en el artículo 29 apartado 2 y 3 de la Ley 12/2014, de 26 de diciembre, de transparencia y de acceso a la información pública, y en el artículo 20 apartado 1 del Decreto 11/2019, de 11 de febrero, por el que se regula la actividad convencional y se crean y regulan el Registro General Electrónico de Convenios del Sector Público de la Comunidad Autónoma y el Registro Electrónico de Órganos de Cooperación de la Administración Pública de la Comunidad Autónoma de Canarias, los convenios que se celebren por los órganos de la Administración pública de la Comunidad Autónoma de Canarias y de los organismos y entidades vinculadas o dependientes de la misma, con otras administraciones públicas y otros sujetos, públicos o privados, así como las modificaciones, prórrogas y anexos o adendas a los mismos, deberán publicarse en el Boletín Oficial de Canarias dentro de los veinte días siguientes a su firma.

En cumplimiento de los referidos preceptos y con la finalidad de facilitar el público conocimiento de los compromisos asumidos en el ámbito de la actividad convencional del sector público autonómico,

RESUELVO:

Ordenar la publicación en el Boletín Oficial de Canarias del Convenio Específico de Cooperación entre el Instituto Canario de Investigaciones Agrarias y Excmo. Cabildo Insular de Tenerife para la realización del Proyecto “raíces y tubérculos menores que fomentan la agrobiodiversidad y los servicios ecosistémicos (ROTATES)”, cuyo texto se acompaña como anexo.

Santa Cruz de Tenerife, a 2 de julio de 2026.- El Secretario General, Jesús Manuel García Benítez.

ANEXO

CONVENIO ESPECÍFICO DE COOPERACIÓN ENTRE EL INSTITUTO CANARIO DE INVESTIGACIONES AGRARIAS Y EXCMO. CABILDO INSULAR DE TENERIFE PARA LA REALIZACIÓN DEL PROYECTO “RAÍCES Y TUBÉRCULOS MENORES QUE FOMENTAN LA AGRODIVERSIDAD Y LOS SERVICIOS ECOSISTÉMICOS (ROTATES)”.

REUNIDOS

De una parte, Dña. Janira Gutiérrez Peraza, en calidad de Presidenta del Instituto Canario de Investigaciones Agrarias, con CIF Q8850003H (en adelante ICIA), con domicilio en Carretera del Boquerón, s/n, Valle de Guerra, 38270 - La Laguna, actuando en nombre y representación de la misma, en virtud del Decreto 324/2023, de 1 de agosto, por el que se dispone su nombramiento, y conforme a las atribuciones que le confiere el artículo 7, letra c), de la Ley 4/1995, de 27 de marzo, de creación del Instituto Canario de Investigaciones Agrarias y del Consejo Asesor de Investigaciones Agrarias, y los artículos 31 y 32 del Decreto 77/2002, de 17 de junio, por el que se aprueba el Reglamento de dicho Instituto.

De otra parte, la Excm. Sra. Dña. Rosa Elena Dávila Mamely, con DNI ***7951**, en calidad de Presidenta del Excmo. Cabildo Insular de Tenerife (en adelante ECIT), domiciliado a estos efectos en el Ed. del Palacio Insular, Plaza de España, n.º 1, de Santa Cruz de Tenerife; en calidad de Presidenta (se adjunta certificado de D. Domingo Jesús Hernández Hernández, Secretario General del Pleno del Excmo. Cabildo Insular de Tenerife).

En la representación que ostentan, las partes se reconocen capacidad jurídica y de obrar necesaria para el otorgamiento y formalización del presente Convenio de Cooperación, a cuyos efectos,

EXPONEN

Primero.- El ICIA es un Organismo Autónomo de carácter administrativo de la Comunidad Autónoma de Canarias, que ostenta personalidad jurídica y patrimonio propios y plena capacidad de obrar para el cumplimiento de sus fines, adscrito a la Consejería competente en materia de Agricultura.

De conformidad con el artículo 3 de la Ley 4/1995, de 27 de marzo, de Creación del Instituto Canario de Investigaciones Agrarias y del Consejo Asesor de Investigaciones Agrarias, tiene entre sus fines los de programación, ejecución y seguimiento de las actividades de investigación y el desarrollo y transferencia de tecnologías agrarias en el ámbito de la Comunidad Autónoma de Canarias, así como aquellos otros que en materia de investigación y desarrollo tecnológico agrarios le atribuyan los órganos competentes de la Administración Pública de la Comunidad Autónoma de Canarias.

En atención a lo dispuesto en el artículo 31 del Reglamento del Instituto Canario de Investigaciones Agrarias, aprobado por Decreto 77/2002, de 17 de junio, “El ICIA podrá establecer relaciones de colaboración con las universidades, organismos públicos de

investigación, cabildos insulares y empresas públicas o privadas para el mejor desarrollo de sus funciones, sin detrimento de sus competencias, obligaciones y actuaciones en el marco del Plan Sectorial Agrario y Alimentario. Dichas relaciones se instrumentalizarán a través de los correspondientes contratos o convenios de colaboración y cooperación, que serán suscritos por el Presidente del ICIA.

Segundo.- El Cabildo de Tenerife, como Corporación Local Territorial, tiene atribuido el gobierno, la administración y representación de cada isla y goza de plena autonomía para el ejercicio de sus competencias propias que corresponden a las islas y las que le sean delegadas por otras administraciones públicas de acuerdo con lo establecido en la legislación de régimen local y en su legislación específica. El Cabildo Insular de Tenerife es una Entidad Local, conforme a la Constitución; la Ley 7/1985, de 2 de abril, Reguladora de las Bases del Régimen Local, establece en su artículo 41, apartado 1, que los Cabildos Insulares, como órganos de gobierno, administración y representación de cada isla, se rigen por las normas contenidas en la disposición adicional decimocuarta de dicha ley y, supletoriamente, por las normas que regulan la organización y funcionamiento de las Diputaciones, asumiendo las competencias de estas, sin perjuicio de lo dispuesto en el Estatuto de Autonomía de Canarias. La legislación específica viene constituida, fundamentalmente por la Ley 8/2015, de 1 de abril, de cabildos insulares.

Tercero.- En base a la Ley 14/2011, de 1 de junio, de la Ciencia, la Tecnología y la Innovación, en su artículo 34.1 establece “ Los agentes públicos de financiación o ejecución del Sistema Español de Ciencia, Tecnología e Innovación, incluidas las Administraciones Públicas, las universidades públicas, los organismos públicos de investigación de la Administración General del Estado, los consorcios y fundaciones participadas por las administraciones públicas, los organismos de investigación de otras administraciones públicas, y los centros e instituciones del Sistema Nacional de Salud, podrán suscribir convenios sujetos al derecho administrativo. Podrán celebrar estos convenios los propios agentes públicos entre sí, o con agentes privados que realicen actividades de investigación científica y técnica, nacionales, supranacionales o extranjeros, para la realización conjunta de las siguientes actividades:

- a) Proyectos y actuaciones de investigación científica, desarrollo e innovación.
- b) Creación o financiación de centros, institutos, consorcios o unidades de investigación, e infraestructuras científicas.
- c) Financiación de proyectos científico-técnicos singulares.
- d) Formación de personal científico y técnico.
- e) Divulgación científica y tecnológica.
- f) Uso compartido de inmuebles, de instalaciones y de medios materiales para el desarrollo de actividades de investigación científica, desarrollo e innovación”.

Cuarto.- La Resolución de 23 de abril de 2025, de la Presidenta del Instituto Canario de Investigaciones Agrarias por la que se publica en el Boletín Oficial de Canarias el Protocolo general de actuación entre el Excmo. Cabildo Insular de Tenerife y el Instituto Canario

de Investigaciones Agrarias para el desarrollo de actividades científico-técnicas en el área de investigación, extensión y capacitación agraria así como en el ámbito de los recursos genéticos agrarios, por el que se regulan las actuaciones conjuntas a desarrollar por el ICIA y el ECIT, las cuales se ejecutarán mediante convenios específicos de colaboración.

Quinto.- La colaboración entre el ICIA y el Cabildo de Tenerife surge en el ámbito del proyecto de investigación ROTATES, MINOR ROOTS AND TUBERS FOSTERING AGROBIODIVERSITY AND ECOSYSTEM SERVICES (ROTATES-Raíces y tubérculos menores que fomentan la agrobiodiversidad y los servicios ecosistémicos) concedido en el marco de los proyectos europeos HORIZON en la convocatoria HORIZON-CL6-2024-BIODIV-02-3-two-stage [Promoting minor crops in farming systems] con duración desde el 1 de julio de 2025 al 30 de junio de 2029.

ROTATES tiene como objetivo mejorar de manera sostenible la agrobiodiversidad y los servicios ecosistémicos mediante la introducción y promoción de cultivos de raíces y tubérculos con alto contenido de almidón, que actualmente se consideran cultivos menores y están infrautilizados en Europa, a pesar del creciente interés por parte de los consumidores. Las raíces y tubérculos menores ricos en almidón (MRT), como la batata (*Ipomoea batatas*), el ñame (*Dioscorea spp.*), la yuca (*Manihot esculenta*) y el taro (*Colocasia esculenta*), se cultivan tradicionalmente en regiones tropicales.

Este proyecto tiene alto interés para las partes, pues el ICIA está muy interesado en todos aquellos proyectos que pongan de manifiesto la potencialidad de las variedades locales de cultivos tradicionales canarios, como son en este caso la batata y el ñame, permitiéndole este proyecto valorizar esta riqueza a nivel internacional. Para el Cabildo de Tenerife implica un beneficio directo, ya que le permitirá evaluar parte de su material en las diferentes áreas de trabajo del proyecto, arrojando una valiosa información para la sostenibilidad del cultivo en la isla. Con este fin el Cabildo suministrará el material vegetal necesario procedente del Centro de Conservación de la Biodiversidad Agrícola de Tenerife, además de aportar su personal para asegurar hacerlo en las mejores condiciones.

Sexto.- Como información complementaria a este Convenio se adjunta memoria del proyecto de investigación ROTATES. El Acuerdo de Transferencia de Material (ATM) se hará de acuerdo a la normativa vigente entre el organismo donante y el receptor.

Séptimo.- Se consideran criterios adecuados para suscribir el presente Convenio Específico el interés científico previsible derivado de la posible colaboración y cualquier otro que contribuya a aportar valor a la actividad científica dentro del sector agrario en Canarias.

Octavo.- No se consideran adecuadas otras posibles alternativas por cuanto que el material vegetal necesario para desarrollar el proyecto se encuentra custodiado por el Cabildo de Tenerife (a través del Centro de Conservación de la Biodiversidad Agrícola de Tenerife, CCBAT), firmante del Convenio. El convenio es un instrumento que permite llevar a cabo estas actividades de una forma ágil, en condiciones de igualdad entre las partes y respetando los principios de transparencia y accesibilidad.

El Convenio que se pretende firmar y la actividad amparada por el mismo no tiene naturaleza contractual por cuanto que se persigue la consecución de un fin común de interés público y no existe el elemento de contraprestación económica. El mismo se ajusta

a las previsiones establecidas en la Ley 40/2015, de 1 de octubre, de Régimen Jurídico del Sector Público, así como a las especialidades contenidas en la legislación autonómica de Canarias, contenidas en el Decreto 11/2019, de 11 de febrero, por el que se regula la actividad convencional y se crean y regulan el Registro General Electrónico de Convenios del Sector Público de la Comunidad Autónoma y el Registro Electrónico de Órganos de Cooperación de la Administración Pública de la Comunidad Autónoma de Canarias.

Noveno.- Por parte de la Comisión Científica del ICIA se ha informado de forma favorable la tramitación del Convenio de Cooperación entre el ICIA y el Cabildo de Tenerife, a la vista del interés científico valorado.

Por su parte, el Cabildo de Tenerife con fecha 26 de marzo de 2025 aprobó en Consejo de Gobierno Insular un protocolo general de actuación entre el Cabildo Insular de Tenerife y el Instituto Canario de Investigaciones Agrarias para el desarrollo de actividades científico-técnicas en el área de investigación, extensión y capacitación agrarias, así como en el ámbito de los recursos fitogenéticos agrarios. En este protocolo se establece un marco que regula las actuaciones conjuntas a desarrollar por el ICIA y el ECIT, las cuales se ejecutarán mediante convenios específicos de colaboración.

Por todo ello, las partes firmantes manifiestan su voluntad de formalizar el presente Convenio Específico de Cooperación con arreglo a las siguientes,

CLÁUSULAS

Primera.- Objeto y objetivos del Convenio.

En términos generales, el presente Convenio tiene por objeto regular la cooperación y colaboración entre el ICIA y el Cabildo de Tenerife, con el fin de establecer un marco de actuación conjunta en el proyecto de investigación “ROTATES” (101181532), financiado por el programa HORIZON de la Comisión Europea, en el que el ICIA participa como entidad socia y donde mediará para la distribución y correcta utilización del material vegetal suministrado de las colecciones de la entidad para ser evaluado durante la realización del proyecto.

Más concretamente, la participación del ICIA consiste en la intermediación entre los Bancos canarios y el resto de socios de forma que puedan ser vinculados formalmente al proyecto ROTATES. Sus funciones serán principalmente la de recibir el material vegetal por parte de los Bancos y tramitar su llegada a los socios que participan en el paquete de trabajo WP3, WP4, WP5 del proyecto.

Las actividades que se realizarán en el proyecto se engloban en diferentes paquetes de trabajo (Ver Anexo 1), pero en resumen las acciones más importantes se llevarán a cabo en:

a) Paquete de trabajo 3 (WP3): estudiar la diversidad del germoplasma y las estrategias de saneamiento para facilitar el acceso a MRT, y fomentar iniciativas de mejora genética innovadoras.

- Identificar nuevos materiales MRT y caracteres valiosos para los programas de mejora genética, así como aumentar la adaptación del cultivo a las condiciones agroecológicas, las limitaciones climáticas y la calidad de la producción.

- Optimizar la selección de híbridos MRT mediante la identificación y combinación de alelos favorable, mejorando así la eficiencia y precisión del proceso de mejora genética

- Diseñar enfoques integrados de sistema para eliminar y controlar patógenos transmitidos por semilla.

b) Paquete de trabajo 4 (WP4): Co-construcción de nuevas prácticas de manejo agroecológico basadas en cultivos MRT.

- Estudio de los nuevos manejos basados en cultivos MRT frente a diferentes factores externos (sequía, biocontrol, deforestación).

- Monitoreo y evaluación de los servicios ecosistémicos respaldados por la integración de cultivos MRT en los sistemas agrícolas.

c) Paquete de trabajo 5 (WP5): Aprovechar el potencial de los cultivos MRT como alimento humano y animal.

- Explotar el potencial de los cultivos MRT para alimentación humana.

- Explotar el potencial de los cultivos MRT como forraje.

Segunda.- Actuaciones necesarias.

El efectivo cumplimiento del Convenio queda reflejado en el siguiente documento anexo al mismo:

- Memoria del proyecto de investigación - Anexo I. En este anexo se recoge todo el protocolo a desarrollar por el consorcio en el proyecto de investigación, y en especial lo referido a las Tareas:

3.1: Recolección y caracterización de germoplasma para la conservación de cultivos MRT raros y de élite

3.2: Implementación de estudios avanzados de fenotipado y genética para la mejora vegetal.

5.3: Procesamiento para la alimentación animal y evaluación de la calidad de los productos ganaderos derivados

Tercera.- Obligaciones de las partes.

a) Para la ejecución del presente Convenio, el ICIA queda obligado a:

1.- Determinar a partir de las consultas con el Banco las variedades de batata y taro (denominado como ñame en Canarias) que se utilizarán en el proyecto, para su elección de mutuo acuerdo.

2.- Establecer el tipo de material vegetal (estacas) y la cantidad de cada una que se enviarán para cada especie.

3.- Gestionar administrativamente el envío del material vegetal en coordinación con el Servicio de Sanidad Vegetal de la Comunidad Autónoma de Canarias.

4.- El punto 3 implica cumplir con los requisitos fitosanitarios establecidos por el país de destino (Portugal) en cuanto a la certificación de ausencia de plagas y enfermedades y, en su caso, llevar a cabo los análisis oportunos para descartarlas.

5.- Ejecutar el envío del material vegetal en tiempo y forma para garantizar en la medida de lo posible su viabilidad al llegar a destino.

6.- Cultivar las variedades de batata previamente seleccionadas, evaluar su procesamiento como alimento animal y estudiar su inclusión en dietas ganaderas y su efecto en la calidad de las producciones del ganado aviar y caprino.

7.- Mantener puntualmente informado al Banco de todo el proceso y de los resultados obtenidos en las distintas actividades a través de los interlocutores designados.

b) Por su parte, el Cabildo Insular de Tenerife, queda obligado a:

1.- Entregar al ICIA el material vegetal en el momento, la cantidad y forma acordado previamente. El Cabildo Insular de Tenerife entregará el material vegetal en la medida de lo posible, asegurándose en primer lugar poder garantizar la conservación en el Banco, para lo que mantendrá en todo momento una cantidad suficiente en sus colecciones.

2.- Facilitar la información disponible de cada variedad de batata y taro (ñame en Canarias) que pueda ser relevante para el proyecto.

3.- En la medida de lo posible, facilitar nuevo material en el caso de que una parte de él no llegue en condiciones a destino.

4.- Cumplimentar el acuerdo de transferencia de material (ATM) entre los Bancos de Germoplasma.

Cuarta.- Plazo de vigencia.

El Convenio surtirá efectos desde la fecha de la firma del mismo y tendrá una vigencia de cuatro años, pudiendo ser prorrogada, si se solicita por una o ambas partes con un mes de antelación a la finalización de la vigencia de la misma.

Quinta.- Aspectos económicos.

No existen gastos ni aportaciones económicas de las partes para la ejecución del proyecto. Los fondos necesarios para la realización de las tareas de coordinación, gestión y desarrollo del proyecto provienen del programa HORIZON de la Comisión Europea, en el que el ICIA participa como entidad socia.

Por lo tanto, la financiación de los gastos para la ejecución del presente Convenio no tendrá repercusión económica para el Cabildo de Tenerife.

Sexta.- Modificación del Convenio.

Para la modificación del Convenio en cualquiera de sus condiciones, se requerirá propuesta razonada de la Comisión de Seguimiento, de oficio o bien a petición de cualquiera de las partes firmantes, siendo necesario el acuerdo unánime de ambas. Las modificaciones particulares al presente Convenio se reflejarán en la correspondiente adenda al mismo.

Séptima.- Mecanismos de seguimiento, vigilancia y control de la ejecución del Convenio. la Comisión de Seguimiento: composición.

Para el seguimiento, planificación y control de las acciones derivadas del presente Convenio, así como resolver las dudas que puedan plantearse en la interpretación del mismo, se crea una Comisión de Seguimiento paritaria compuesta por al menos dos miembros designados por cada una de las partes.

A dicha Comisión podrán incorporarse otros miembros que sean designados por acuerdo de las partes, así como otras personas con funciones de carácter consultivo, estas últimas con voz, pero sin voto.

El nombramiento de esta Comisión se realizará en el mes siguiente a la firma del convenio y se reunirá al menos dos veces al año y cuantas veces sean necesarias en función del buen desarrollo de este convenio. Las reuniones se convocarán con antelación mínima de una semana, por escrito, por cualquiera de las partes.

Esta Comisión estará encargada de resolver por consenso las incidencias de interpretación y ejecución que pudieran plantearse con ocasión del cumplimiento del convenio y desarrollará las actuaciones que se contienen en el mismo. En concreto, serán funciones de la Comisión de Seguimiento las Siguientes:

- Vigilancia y control del cumplimiento de las actuaciones previstas
- Dictar las instrucciones que sean precisas para asegurar su realización
- Informar a las partes de las incidencias acaecidas durante la ejecución del Convenio.
- Emisión de informe anual para evaluar la conveniencia de continuar con la relación pactada, en atención a los siguientes aspectos:
 - Valoración de los resultados obtenidos de acuerdo con el cronograma del trabajo establecido y documentación aportada por los participantes. Será preciso haber alcanzado, al menos un 80% de los resultados previstos, salvo que se establezca otro porcentaje por las partes firmante al comienzo de la vigencia del Convenio o se den causas de fuerza mayor aceptadas por la propia Comisión.

Para la emisión del informe anual y el informe final, la Comisión de Seguimiento se apoyará en los documentos presentados por las entidades y el personal investigador directamente adscrito al proyecto, que consistirán básicamente en:

- Planificación del trabajo inicial.
- Informes de seguimiento con periodicidad semestral.
- Informe final, con el resultado y conclusiones definitivas del proyecto.

Con el objetivo de que la Comisión pueda ejercer sus labores de seguimiento, deberá facilitarse a la misma por las partes, cuando así lo requiera, la documentación necesaria para tal fin.

Octava.- Confidencialidad de la información.

A los efectos del presente Convenio, se entenderá por Información Confidencial, toda aquella información y toda aquella documentación de naturaleza técnica o de cualquier otra naturaleza que a continuación se relaciona, y que incluye, -sin que esta lista signifique limitación alguna- la siguiente: “know how”, patentes, marcas, diseños, fórmulas, procesos, resultados de investigaciones, planos, fotografías, informes, listados de clientes, de proveedores, de fabricantes, datos sobre precios, ideas, invenciones o estudios, hardware, componentes, aplicaciones informáticas, que se encuentre claramente clasificada como confidencial y que sea transmitida por cualquier soporte tanto escrito, informático o electrónico como oral.

Con la finalidad de identificar la información como confidencial, las partes acuerdan marcar el conocimiento secreto con la palabra “confidencial”.

En los supuestos en los que la información haya sido divulgada de forma oral o visual, será considerada confidencial siempre que sea identificada como tal por el propietario cuando se revela y posteriormente se proceda a su transcripción en soporte legible siendo marcada con el sello de “confidencial” en el plazo de treinta (30) días de su transmisión oral o visual.

A efectos de este Convenio no se considerará información confidencial aquella que:

a) Sea o se convierta en dominio público por alguna causa que no sea producida por vulneración de las estipulaciones contenidas en el presente Convenio.

b) Sea conocida por la contraparte como consecuencia de la transmisión de esa información por un tercero ajeno al presente Convenio, con legitimación suficiente para su transmisión libre y sin estar sujeto a limitaciones derivadas de previos acuerdos de confidencialidad.

c) La Receptora poseyera o conociera antes de recibirla de la parte Transmisora sin obligación alguna de mantener su confidencialidad, siempre que notifique este hecho a la otra parte con carácter previo.

Las partes garantizan ser titulares de la información confidencial a intercambiar ostentando la plena disposición sobre la misma.

Respecto aquella información confidencial a la que tengan acceso, las partes se obligan mutuamente a:

a) No utilizar la información confidencial para otro propósito distinto que el establecido en este Convenio, no pudiendo ser utilizada, directa o indirectamente para beneficio propio ni de terceros.

b) No revelar la información confidencial a terceros o a sus empleados, colaboradores o subcontratistas, con excepción de aquellos empleados o colaboradores que necesiten conocer la información para desarrollar el objeto de este Convenio.

c) Garantizar y responder en todo caso del cumplimiento efectivo por parte del personal y colaboradores de la Receptora de las obligaciones que se derivan del presente Convenio y de los daños causados de forma directa o indirecta por la vulneración del secreto por parte de aquel a quien la Receptora o las personas por las que este responde hayan revelado la información en contra de lo previsto en este Convenio.

d) Adoptar las medidas de seguridad de tipo organizativo, personal o técnico necesarias para salvaguardar la integridad y confidencialidad de la información facilitada por la Transmisora, así como para evitar accesos no autorizados a la misma, garantizando el mismo grado de diligencia que utilizan en el manejo de su propia documentación confidencial.

e) Cualquier uso de la información diferente de las expresamente fijadas en las estipulaciones contenidas en el presente Convenio, deberá previamente notificarse expresamente y por escrito a la parte Transmisora para su aceptación, sin que en ningún caso la falta de comunicación pueda ser considerada como aceptación.

f) No disponer ni hacer uso de la información confidencial después de la extinción o resolución anticipada de este Convenio, procediendo a su devolución inmediata ante el requerimiento de la parte Transmisora y en su caso a la destrucción de todo aquello que no pueda ser devuelto, sin dejar ningún soporte sea cual sea el mismo con toda o parte de la citada información confidencial.

Novena.- Derechos de los resultados, propiedad intelectual e industrial.

Los derechos de propiedad intelectual o industrial (DPI) preexistentes a la firma del Convenio pertenecerán a la parte que los aporte para la realización de actividades conjuntas.

Se considerarán resultados del proyecto de I+D+i aquella información que haya sido identificada como resultado de la actividad objeto de este acuerdo. Cada parte seguirá siendo propietaria de los conocimientos previos aportados. No se entienden cedidos a la otra parte, en virtud del presente acuerdo, ninguno de los conocimientos previos.

La memoria del proyecto en su punto 2.2.2 “Key results-proteccion and exploitation, en el apartado KER2. New MRT genetic resources in genebanks, dispone que “El material genético permanecerá bajo la propiedad del país donde se ha colectado”

En concreto, el material vegetal cedido por el Cabildo de Tenerife, seguirá siendo propiedad del mismo, no pudiendo registrarse ninguna de estas variedades como resultado de la ejecución del proyecto de I+D+i.

La información generada a lo largo de este Convenio es propiedad de ambas partes y será de plena disponibilidad de ambas organizaciones. Su utilización por ambas partes precisará de comunicación previa entre ellas, comprometiéndose a solicitar por escrito cualquier uso que se vaya a hacer de la misma ante la Comisión de Seguimiento, que deberá obtener respuesta en el plazo máximo de treinta días. Transcurrido dicho plazo sin obtener respuesta, se entenderá que el silencio es la tácita autorización para su difusión.

Cuando las partes deseen utilizar los resultados parciales o finales, en parte o en su totalidad, deberán reconocer y hacer constar la participación del personal que ha intervenido en este Convenio, así como entidad u Organismo de pertenencia. Se deberá informar del resultado de esta divulgación.

En caso de que alguna de las partes modifique los resultados, finales o parciales, o pretenda utilizarlos para un fin diferente al previsto se comunicará a la otra parte interviniente, quien se guarda la potestad de solicitar que se excluya su nombre de esa publicación, registro, etc.

Dado el carácter de entidad pública del Instituto Canario de investigación Agraria (ICIA) y de acuerdo al artículo 37 de la Ley 14/2011, de 1 de junio, de la Ciencia, la Tecnología y la Innovación, al menos una parte general de los ensayos ha de divulgarse, procurando que estas publicaciones científicas o técnicas resultantes de los trabajos, total o parcialmente estén disponible en acceso abierto. Siempre se mencionará el nombre de el/la o los investigadores o técnicos participantes y las entidades a las que pertenecen. El contenido de estas será aprobado por la Comisión de Seguimiento.

Décima.- Incumplimientos: consecuencias y criterios para determinar indemnización.

10.1. Régimen de Incumplimiento.

El incumplimiento de las obligaciones y compromisos asumidos por cada una de las partes en virtud de este Convenio, recogidos principalmente en sus cláusulas tercera (Obligaciones), octava (Confidencialidad) y novena (Propiedad Intelectual e Industrial), podrá dar lugar a la resolución del mismo, de conformidad con lo establecido en el artículo 51.2.c) de la Ley 40/2015, de 1 de octubre, de Régimen Jurídico del Sector Público (LRJSP), y dará lugar, en su caso, a la indemnización por los daños y perjuicios causados.

10.2. Incumplimientos leves.

1. Se considerará incumplimiento leve el cumplimiento defectuoso o extemporáneo de aquellas obligaciones que no impidan la consecución del objeto del Convenio y puedan ser subsanadas, tales como el retraso no reiterado e inferior a quince (15) días hábiles en el cumplimiento de las obligaciones de información, o la omisión de formalidades administrativas fácilmente subsanables.

2. En caso de incumplimiento leve, la Comisión de Seguimiento (cláusula séptima), a instancia de la parte cumplidora, requerirá formalmente a la parte incumplidora para que subsane su obligación en un plazo máximo de quince (15) días hábiles.

3. La reiteración de tres (3) incumplimientos leves, habiendo mediado requerimiento de subsanación, será considerada un incumplimiento grave a todos los efectos.

10.3. Incumplimientos graves.

1. Se considerará incumplimiento grave, que facultará a la parte cumplidora para instar la resolución del Convenio, cualquiera de las siguientes causas:

a) El incumplimiento de las obligaciones que, por su carácter esencial, impidan o dificulten gravemente la consecución del objeto del Convenio, definido en la cláusula primera.

b) Específicamente, se considerarán incumplimientos graves:

I. Por parte del Cabildo Insular de Tenerife: la negativa injustificada, el retraso reiterado o el incumplimiento manifiesto de su obligación principal de entregar el material vegetal en las condiciones pactadas (cláusula tercera), así como la no cumplimentación del Acuerdo de Transferencia de Material (ATM) si esto impide la ejecución del proyecto.

II. Por parte del ICIA: el incumplimiento de las obligaciones de gestión fitosanitaria y envío que, por negligencia acreditada, resulte en la pérdida de viabilidad, destrucción o imposibilidad legal de utilizar el material (cláusula tercera); el uso del material vegetal para fines distintos a los previstos en este Convenio (cláusula primera); o la vulneración de los derechos de propiedad del Cabildo sobre dicho material, en los términos de la cláusula novena.

c) La vulneración sustancial de las obligaciones de confidencialidad (cláusula octava) o de propiedad intelectual e industrial (cláusula novena) que irroque un perjuicio grave y demostrable a la otra parte.

d) La reiteración de incumplimientos leves, conforme al apartado 10.2.3

e) La negativa injustificada a cumplir con las instrucciones y acuerdos válidamente adoptados por la Comisión de Seguimiento en el ejercicio de sus funciones.

f) La concurrencia de causas de fuerza mayor, definidas en el apartado 10.5, que se prolonguen por un tiempo superior a seis (6) meses e impidan de forma irreversible la ejecución del proyecto.

10.4. Consecuencias del incumplimiento grave y criterios de indemnización.

1. El incumplimiento grave de las obligaciones y compromisos por una de las partes facultará a la parte cumplidora para instar la resolución del Convenio.

2. La resolución del Convenio por incumplimiento culpable de una de las partes conllevará la obligación de esta de indemnizar a la otra parte por los daños y perjuicios fehacientemente acreditados que dicho incumplimiento haya ocasionado.

3. Los criterios para determinar la indemnización, en cumplimiento del artículo 49.e) de la LRJSP, se basarán en los principios de responsabilidad patrimonial del sector público, debiendo cubrir tanto el daño emergente como el lucro cesante. A estos efectos, se considerarán:

a) El daño emergente: los costes directos, reales y acreditados en que la parte cumplidora haya incurrido como consecuencia directa del incumplimiento (ej. costes de reposición del material vegetal, gastos de gestión administrativa o de transporte que devengan inútiles).

b) El lucro cesante: específicamente, los fondos provenientes del proyecto HORIZON (referidos en la cláusula quinta) que la parte cumplidora se vea obligada a reintegrar a la Comisión Europea, o que deje de percibir, como consecuencia directa y demostrable del incumplimiento esencial de la otra parte.

4. La determinación de la existencia del incumplimiento y, en su caso, de la cuantía de la indemnización correspondiente se acordará, en primera instancia, en el seno de la Comisión de Seguimiento, como parte del proceso de liquidación del convenio previsto en el artículo 52 de la LRJSP. En caso de desacuerdo, las partes se someterán a la Jurisdicción Contencioso-administrativa, conforme a la cláusula décimo quinta.

5. Adicionalmente, se podrá determinar por la parte perjudicada la prohibición a la parte incumplidora de suscribir nuevos convenios con aquella durante un plazo de hasta dos (2) años.

10.5. Fuerza mayor.

1. Las causas de fuerza mayor eximirán de responsabilidad por el incumplimiento, pero no de la obligación de comunicar la causa a la otra parte interesada, a través de la Comisión de Seguimiento, a la mayor brevedad posible.

2. Se entiende como causa de fuerza mayor incendios, robos o destrozo del ensayo, situaciones fitosanitarias ajenas a los participantes, causas físicas y meteorológicas adversas o situaciones epidemiológicas declaradas por la Administración según sus competencias, así como otros de similar gravedad.

3. Si la causa de fuerza mayor impidiese el cumplimiento del objeto del convenio de forma definitiva o por un plazo superior a seis (6) meses, cualquiera de las partes podrá proponer a la Comisión de Seguimiento la extinción del convenio por imposibilidad sobrevenida de cumplimiento (artículo 51.2.b) de la LRJSP), procediéndose a la liquidación de las actuaciones realizadas hasta la fecha sin que proceda indemnización por esta causa.

Decimoprimera.- Causas de extinción.

El Convenio se extinguirá por el cumplimiento del plazo de vigencia y sus eventuales prórrogas. Igualmente podrá resolverse por cualquiera de las causas establecidas en el artículo 51 de la Ley 40/2015 de 1 de octubre, de Régimen Jurídico del Sector Público.

Las partes se comprometen a resolver de manera amistosa en el seno de la Comisión de Seguimiento cualquier desacuerdo que pudiera surgir en el desarrollo del presente Convenio, bien entendiendo que con anterioridad se aplicarán los criterios de buena fe y voluntad de llegar a acuerdos.

Los efectos de la extinción serán los previstos en el artículo 52 de la Ley 40/2015, de 1 de octubre, de Régimen Jurídico del Sector Público.

Decimosegunda.- Protección de datos de carácter personal.

Será de aplicación lo establecido en el Reglamento (UE) 2016/679 del Parlamento Europeo y del Consejo, de 27 de abril de 2016, relativo a la protección de las personas físicas en lo que respecta al tratamiento de datos personales y a la libre circulación de estos datos (Reglamento General de Protección de Datos) y la Ley Orgánica 3/2018, de 5 de diciembre, de Protección de Datos Personales y garantía de los derechos digitales. Las partes firmantes del presente Convenio se comprometen a tratar de manera confidencial los datos personales de los participantes, que le sean facilitados en el marco de la relación mantenida, a tratarlos de acuerdo con la normativa de protección de datos personales y a utilizarlos, exclusivamente, con las anteriores finalidades.

Asimismo, el deber de secreto y las limitaciones en su uso marcadas por la normativa de aplicación serán aplicables a cualquier información a la que se tenga acceso en la realización de actividades objeto de este Convenio, salvo aquella información que deba ser pública según lo establecido en la Ley 19/2013, de 9 de diciembre, de transparencia, acceso a la información pública y buen gobierno.

En el caso de cesión de datos personales, deberá firmarse un contrato entre el responsable y el encargado del tratamiento.

Decimotercera.- Publicación e inscripción registral del Convenio.

De conformidad con lo dispuesto en el artículo 29.2 de la Ley 12/2014, de 26 de diciembre, de transparencia y de acceso a la información pública, y en el artículo 20 del Decreto 11/2019, los convenios que se celebren por los órganos de la Administración Pública de la Comunidad Autónoma de Canarias y de los organismos y entidades vinculadas o dependientes de la misma, con otras administraciones públicas y otros sujetos, públicos o privados, así como las modificaciones, prórrogas y anexos o adendas a los mismos, deberán publicarse, en el Boletín Oficial de Canarias, dentro de los veinte días siguientes a su firma. Asimismo, se deberá proceder a su inscripción registral en los términos establecidos en el apartado 2 del citado artículo 20.

Decimocuarta.- Cláusula de exclusividad.

Los aspectos referentes a la exclusividad se atenderán a lo que mencione la memoria del Proyecto ROTATES.

Decimoquinta.- Régimen jurídico y jurisdicción competente.

El presente Convenio tiene naturaleza administrativa, rigiéndose en su desarrollo, efectos y para su interpretación por el ordenamiento jurídico administrativo, en concreto por lo establecido en la Ley 40/2015, de 1 de octubre, de Régimen Jurídico del Sector Público. Además, se estará a lo dispuesto en el Decreto 11/2019, de 11 de febrero, por el que se regula la actividad convencional y se crean y regulan el Registro Electrónico de Convenios del Sector Público de la Comunidad Autónoma y el Registro Electrónico de Órganos de Cooperación de la Administración Pública de la Comunidad Autónoma de Canarias y, en general, a lo dispuesto en las normas administrativas vigentes de desarrollo.

Por la especialidad de la materia, será de aplicación igualmente la Ley 14/2011, de 1 de junio, de la Ciencia, la Tecnología y la Innovación.

Está excluido del ámbito de aplicación de la Ley 9/2017, de 8 de noviembre, de Contratos del Sector Público, si bien se aplicarán los principios de esta ley para resolver dudas y lagunas que pudieran plantearse.

Las cuestiones litigiosas que se pudieran suscitar durante la vigencia del mismo y que no pudieran ser resueltas por la Comisión de Vigilancia y Control, serán sometidas a la Jurisdicción Contencioso-administrativa, siendo el órgano competente la Sala de lo Contencioso-administrativo del Tribunal Superior de Justicia de Canarias, de conformidad con su Ley reguladora 29/1998, de 13 de julio.

Y, en prueba de conformidad y aceptación, firman el presente Convenio, a 26 de noviembre de 2025.- Por el Instituto Canario de Investigaciones Agrarias, Janira Gutiérrez Peraza.- Por el Cabildo Insular de Tenerife, la Presidenta, Rosa Elena Dávila Mamely.

ANEXO



[HORIZON-CL6-2024-BIODIV-02-3-two-stage]-[Promoting minor crops in farming systems]

Proposal template Part B: technical description**ROTATES: MINOR ROOTS AND TUBERS FOSTERING AGROBIODIVERSITY AND ECOSYSTEM SERVICES**

#@APP-FORM-HERIAIA@#

List of participants

Participant No.	Participant organisation name	Short name	Country
1 (Coordinator)	Centre de Coopération Internationale en Recherche Agronomique pour le Développement - C.I.R.A.D. Epic	CIRAD	France (including Guadeloupe)
2	Institut National de Recherche pour l'Agriculture, l'Alimentation et l'Environnement	INRAE	France
3	Technologiko Panepistimio Kyprou	CUT	Cyprus
4	Université de Liège	UL	Belgium
5	Universidade da Madeira	UMa	Portugal (Madeira)
6	Universidade dos Acores	UAç	Portugal (Azores Islands)
7	Instituto Nacional de Investigação Agrária e Veterinária	INIAV	Portugal
8	Consiglio per la Ricerca in Agricoltura e l'analisi dell'economia agraria	CREA	Italy
9	Centre de Recherche de l'institut Lyfè	ILRIC	France
10	Instituto Canario de Investigaciones Agrarias	ICIA	Spain (Canary Islands)
11	Scotland's Rural College	SRUC	United Kingdom
12	Associação da Costa Oeste	ACOESTE	Portugal
13	European Network Of Living Labs Ivzw	ENoLL	Belgium
14	Inovacijsko Tehnološki Grozd Murska Sobota	ITC	Slovenia
15	Thomas Plants	TP	France
16	Geoponiko Panepistimion Athinon	AUA	Greece
17	Association Panakuh	PNKH	France (French Guiana)
18	Université Marien Ngouabi	UMNG	Republic of the Congo

1. EXCELLENCE #@REL-EVA-RE@#
1.1 Objectives and ambition #@PRJ-OBJ-PO@#

1.1.1 Project context and rationale

ROTATES aims to **sustainably enhance agrobiodiversity and ecosystem services** by introducing and promoting **starchy root and tuber crops**, currently considered as **minor crops and underutilised in Europe** despite increasing consumer demand. Minor starchy roots and tubers (MRT), namely sweet potato (*Ipomoea batatas*), yam (*Dioscorea* spp.), cassava (*Manihot esculenta*) and taro (*Colocasia esculenta*) are traditionally grown in tropical regions¹, yet many varieties have strong potential for temperate climates². They are already cultivated on a small scale in countries like Spain, Portugal, France, Italy, and Hungary. Locally, they are known as "**ancestral crops**" or "**grandparents' crops**". Previously more prominent, they have been overshadowed by major crops like potato and wheat^{3,4}. Classified as underutilised or orphan crops, they are the subject of limited research and receive little funding in Europe, while traditional processing methods constrain their further development^{5,6}. Our overall objective is to enhance the **resilience of European agrosystems** to climate change, **reduce the impacts of agriculture** on the environment, and **meet the rising demand for healthy foods**, by investigating the potential of the targeted crops, **sweet potato, yam, cassava, and taro**.

MRT crops are **traditionally grown in European outermost territories**, where they form part of local diets, because of the multi-cultural origins of the populations and favourable climates. Some EU regions, like French Guiana, are **diversity hotspots**, particularly for American yam (*Dioscorea trifida*) and cassava^{7,8}. Despite their cultural importance, local production and consumption are declining due to rising costs, technical lock-ins, and underdeveloped value chains⁹. Recent R&D has focused on cassava and sweet potato, but mainly in tropical zones located outside the EU^{10,11,12,13}. With advances in genomics, reference genomes and molecular marker resources are now available for many MRT crops, and breeding methods are evolving. Traditional phenotypic selection is increasingly being supplemented by molecular and genomics-assisted techniques¹⁴. The use of modern phenotyping technologies, such as near-infrared spectroscopy (NIRS), drones, and robotics, allows for large-scale and precise characterization of MRT germplasm¹⁵. Worth noting, a yam breeding program, and a taro phenotyping platform, are emerging in Europe.

MRT crops **have recently gained importance in Europe**, as there is a growing demand for these crops, including as food, non-food products (such as starch), and for animal feed¹⁶. The **per capita total demand has tripled** in the last decade to reach 1.1 Kg/capita/year and is mostly met by imports (FAOSTAT). In the last decade, on average, **domestic production**¹⁷ **has covered about 24% of the European demand**, although the percentage fluctuates with the crop, with imports increasing from 245,000 tonnes in 2010 to 820,000 tonnes in 2022. European production relies on intensive monoculture systems using so-called elite varieties, which has led to serious outbreaks of viruses and diseases, for example in sweet potato¹⁸, while at the same time eroding the sustainability of the agrosystems. This growing demand represents a significant opportunity to diversify EU agriculture, and the ideal momentum to introduce and disseminate agroecological practices as alternatives to current monoculture systems. This trend also opens up possibilities to expand the demand for MRT crops and to further diversify EU diets with healthy products. Moreover, as part of European **cultural heritage**, MRT crops have a significant advantage over exotic or unfamiliar crops; due to the ancestral knowledge that can be harnessed through innovative research and modern technology.

MRT crops have proven to be **efficient in both crop associations and in crop rotations** in many parts of the world^{19,20}. They require few or no pesticides and herbicides and are more resilient to climate change^{21,22}, meaning they are well placed to restore biodiversity to European agricultural systems, increase farmers' incomes, and offer alternatives to cereals for special diets. MRT crops will **likely have positive impacts on ecosystem service** provision: **i-**, their long growth cycle, large foliage, and reduced reliance on pesticides provide habitats for many species (including natural enemies of pests), **ii-** while protein/nitrogen-rich grain is the harvested product in the case of most crops (e.g., cereals), in MRT crops, carbon-rich roots are the harvested product, while the protein/nitrogen rich foliage is a by-product that has potential as animal feed or for soil amendment, **iii-** the extrafloral nectaries and exudates of MRT crops could be an alternative source of food for beneficial arthropods, and **iv-** the water demand of

¹ Lebot. *CABI* 2019

² Epping et al. *Planta* 2020

³ Kiem et al. *Plant Pathology* 2022

⁴ Grimaldi et al. *PLoS One* 2018

⁵ Chapman et al. *New Phytologist* 2022

⁶ Kamenya et al. *Theor Appl Genet* 2021

⁷ Olsen & Schaal *PNAS* 1999

⁸ Bousalem et al. *Genome* 2010

⁹ Colombet et al. *Eur J Nutr* 2021

¹⁰ Xue et al. *New Phytol* 2022

¹¹ Perez-Fons et al. *Plant Physiology* 2023

¹² Xue et al. *New Phytol* 2022

¹³ Perez-Fons et al. *Plant Physiology* 2023

¹⁴ Friedman et al. *Agriculture* 2018

¹⁵ Gouveia et al. *Appl. Sci.* 2020

¹⁶ <https://www.fao.org/faostat/en/>

¹⁷

<https://ec.europa.eu/eurostat/comext/newxtw eb/>

¹⁸ Fiallo-Olive *Microorganisms* 2022

¹⁹ Kando & Bitane, *IJSA* 2023

²⁰ Wang et al. *J Env. Manag.* 2024

²¹ Pushpalatha *Natural Resources Forum* 2023

²² Van Zonneveld et al. *PNAS* 2023

MRT crops is high during spring (leaf production), when more water is available, compared to cereals whose water demand peaks in summer (grain filling) when water availability is reduced. However, the net impact of these crops on soil structure remains to be determined, with the potentially positive impact of massive roots on soil decompaction potentially offset by soil disturbance during harvest.

MRT crops are **rich in starch and are gluten free**, plus many varieties have a **low glycaemic index (GI)**, and are rich in **health-promoting** compounds (anthocyanins and carotenoids)²³. They can have **dual purpose**, producing edible roots and fodder leaves in the case of cassava or sweet potato, and can serve multiple market segments: markets for fresh produce as well as industrial applications or multiple processing or consumer market segments, for example pasta and bread²⁴. They have a high potential for food processing²⁵. In the other hand, starchy MRT crops are primarily a source of energy due to their high carbohydrate content (70-85% starch), which accounts for 80-90% of their dry weight. These MRT crops can largely replace cereals in feed destined for three classes of livestock (small ruminants, cattle, and poultry) provided they are supplemented with a nitrogen source. Moreover, MRT by-product foliage has variable but generally high protein content, usually in the 15-30% DM range. Therefore, MRT by-products offer an excellent opportunity to enhance livestock feed resources by providing high-quality protein and energy, thus helping to **bridge the gap between feed supply and demand**. Producing MRT feed locally can significantly reduce feeding costs, and consequently benefit livestock farmers who often rely on expensive imported products that in addition, have a substantial environmental footprint.

The inclusion of MRT crops in this call will adequately **complement the cohort of minor crops**, minor cereals, legumes, traditional fruit trees, introduced into European agrosystems by the **four sister projects** (CROPDIVA, RADIANT, DIVINFOOD, BIOVALUE) funded by H2020. Additionally, this inclusion will extend the promotion of neglected and underutilized crops to Europe's outermost regions, thereby **broadening the geographical scope** beyond the areas covered by the sister projects.

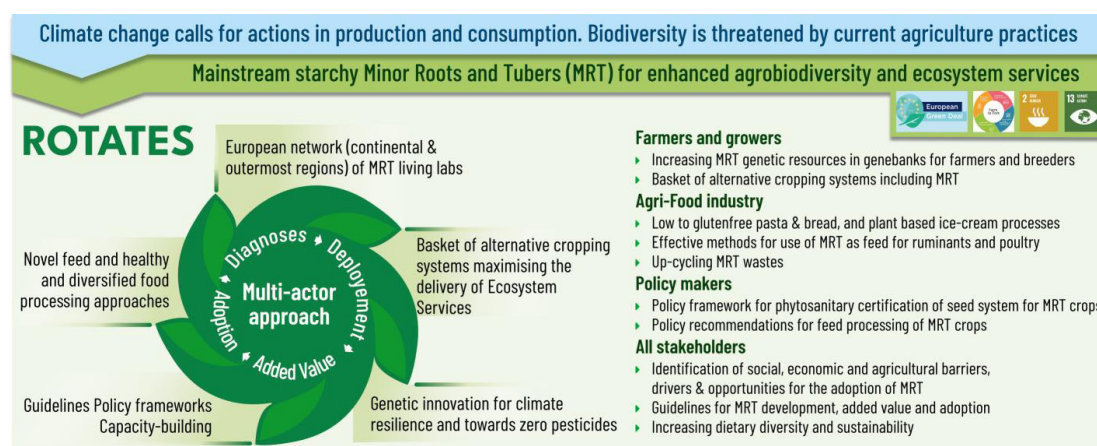


Figure 1: ROTATES overall concept

1.1.2. Project objectives and ambition

ROTATES will lay the foundations for the adoption and sustainable production of MRT crops by the different actors of the value chains beyond the timeline of the project itself. The overall objective is to integrate MRT crops as a lever for crop diversification in **five European continental countries (France, Portugal, Cyprus, Italy, Greece), including five outermost regions (Madeira, Canary Islands, Azores Islands, Guadeloupe and French Guiana)**, and one non-EU country (**Republic of the Congo**), where slash and burn practices dominate the production of MRT crops like in French Guiana), and, in so doing, to have a positive impact on resilience, biodiversity and adaptation to climate change. We will use an interdisciplinary and multi-actor approach (MAA), to develop agroecological practices based on MRT crops in both **conventional and organic agrosystems in 6 pilot Living Labs (LLs)** and establish **four MRT Breeding Hubs, one for each crop** for genetic innovation. ROTATES aims to generate **innovative food processing** and introduce MRT-based feeds in **livestock diets** to enhance food system diversity that

²³ Ferraro et al. *Crit. Rev. Food Sc. Nut.* 2016

²⁴ [https://www.cgiar.org/news-](https://www.cgiar.org/news-events/news/are-dual-purpose-crops-a-breeding-priority-the-case-of-cassava/)

[events/news/are-dual-purpose-crops-a-breeding-priority-the-case-of-cassava/](https://www.cgiar.org/news-events/news/are-dual-purpose-crops-a-breeding-priority-the-case-of-cassava/)

²⁵ Lebot et al. *J. Food Comp. Anal.* 2023

will benefit farmers, consumers and private companies. ROTATES will contribute to reducing Europe's **dependence on imports** by facilitating local production, giving added value to products, and will guide the proposal of a **policy framework for phytosanitary certification** of an MRT seed system. We will produce guidelines to foster the development of **MRT crops at value chain level** aimed at entrepreneurs, policy makers and other stakeholders (Figure 1).

SO.1. Identify levers along the value chain to increase MRT crop adoption by assessing opportunities and constraints, thus enhancing their use in farming systems and the feed and food industry. **(WP1 & 2)**

Currently, the challenges of expanding MRT crop value chains are highlighted through different perspectives: the food system lock-in concept (where established crops are prioritised, leading to less research and less economic viability for minor crops, making them less attractive to farmers²⁶); transition theory (which emphasises resistance to systemic change which hinders the adoption of alternative crops²⁷), and transaction cost theory²⁸ (where higher costs associated with minor crops due to their scale make them less competitive than mainstream crops).

ROTATES will develop tools to be used in the project (e.g., focus groups, surveys, semi-structure interviews) which will help to identify the factors which affect the adoption of MRTs at different stages of the chain, as well as identify opportunities for their expansion.

Measurable through six business case studies for MRT value chains for food and feed. Guidelines on value-chain opportunities and strategies. *The constraints and opportunities as well as the levers will be measurable and verifiable because they will be identified together with the help of the different actors that comprise the value chains concerned. Achievable thanks to the tools applied and interaction with participating value chain actors to highlight the possibilities for the uptake and further development of MRTs. Therefore, the identification of levers for the uptake of MRTs is achievable within the timeline of the project.*

SO.2. Co-design and co-develop novel agroecological management practices that incorporate the most suitable MRT varieties, and evaluate their impacts on agrobiodiversity and on farmers income diversification. **(WP3,4,6)**

Currently, agriculture is one of the main drivers of environmental degradation and biodiversity loss. Industrial agriculture with dominant monocultures has led to unsustainable agricultural systems characterised by high external energy inputs and high vulnerability to environmental and political shocks²⁹. European production of MRT crops relies on intensive monoculture systems using a few so-called elite varieties, which is threatening the sustainability of the agrosystems. Some production systems use crop rotation (with e.g., cereals, legumes or vegetables), but these are uncommon and their performance (including their provision of ecosystem services) poorly documented. In tropical forests, MRT crops are often produced in rich crop associations, but these could be optimized to improve their productivity and their sustainability (in particular reduce their impact on deforestation).

ROTATES will develop baskets of options integrating MRT crops (in both crop rotations and crop associations), in order to cater for the diversity of agroecologies considered as well as for the diversity of farms, with the aim of increasing total productivity and maximising the delivery of ecosystem services. The resulting cropping systems will combine MRT crops with cereals, legumes, cucurbits, vegetables and/or fruit trees in Europe, while the integration of cover crops and agroforestry shrubs will be privileged in the tropical forest sites. These options will be co-developed with farmers and other actors of the value chains represented in the **6 pilot Living Labs (LL)** within the consortium, and will be tested in a participatory manner. ROTATES will establish a **breeding hub for each crop** and develop breeding strategies to provide varieties adapted to these new agronomic practices.

Measurable through the methods proposed (on-station trials, on-farm adaptation and evaluation, modelling, etc) which will allow a robust comparison of the performance of alternative cropping systems based on MRT crops and usual farmer practices and their impact on biodiversity and soil health. *On-farm adaptation will be conducted in at least 60 conventional system and 30 organic farms in the 6 pilot LLs. Indicators of ecosystem services will be monitored and evaluated. The breeding hubs with their related genebanks and phytosanitary approaches which will provide genetic resources adapted to the agronomic practices developed in response to the different constraints.*

Achievable thanks to the methods proposed, which are realistic even within the lifespan of the project, given that two or more seasons of agronomy data will be collected in the pilot LLs. *The breeding hubs will be fully operational, leveraging the extensive European and international genetic resources available. Furthermore, the project will benefit from the partners' significant capacity and long experience in breeding MRT crops, and deep phenotyping for a wide range of biological traits. Additionally, the availability of sophisticated equipment and advanced technologies will further enhance our ability to achieve the project's objectives efficiently.*

²⁶ Meynard et al., *Agronomy for Sustainable Development* 2018

²⁷ Harich, *System Dynamics Review* 2010

²⁸ Williamson, *Handbook of Industrial Organization* 1989

²⁹ Agnoletti & Santoro, *Biodiversity and*

Conservation 2022.

SO.3. Unlock the potential of MRT crops, by co-identifying with value chain actors of uses of fresh and processed MRTs for feed and food markets adapted to European consumers (WP5 and 2)

Currently, in the context of a market that is increasingly attentive to the nutritional potential of foods, industries are increasingly looking for new raw materials to produce functional products. Furthermore, the gluten-free market is also increasingly expanding, so industries and consumers are looking for new formulations of baked products coming from flours and starches of origins other than cereals, perhaps mixed with gluten-free cereals and/or legumes. On the other hand, the feed industry is also very interested in new sources of high-energy raw materials to end their dependence on cereals and in inexpensive sources of protein to replace soya, currently an indispensable ingredient in balanced animal diets.

ROTATES will develop knowledge of the physico-chemical and technological properties and the production of different MRT food prototypes and animal feed. It will produce indications for the best industrial uses with high nutritional and sensorial quality products. Recipes will be developed using fresh and processed MRT, and the consumers acceptance will be assessed through surveys.

Measurable through three food products (pasta, bread and ice cream), use in feed (sheep, goats, cattle and poultry) and livestock products (meat, milk, cheese, eggs). A booklet and a video of recipes will be produced.

Achievable thanks to the capacity and long experience of the partners to process different kind of crops and the availability of a broad diversity of MRT crops and varieties within the consortium which will facilitate the identification of the most appropriate genotypes for transformation.

SO.4. Capacity building and dissemination to promote environmental, health and economic benefits of MRT crops to the diversity of value-chain actors involved in the project and to a wider audience through a network of living labs on minor crops. (WP4, WP6 & WP7)

Currently, the integration of minor crops in diverse agricultural systems is hindered by a lack of comprehensive training and dissemination mechanisms. While there is a growing interest in the environmental, health, and economic benefits of MRT crops, the knowledge and practices required for effective implementation are often fragmented and underutilised across the different regions.

ROTATES will develop a robust capacity-building framework that leverages pilot LLs and the multi actor approach (MAA) to bridge these gaps. Using a train-the-trainer approach, the project will deliver a structured training programme to partners for the deployment of tailored local training sessions to ensure that pilot LLs are equipped to replicate and adapt these solutions effectively.

Measurable through number of training sessions provided to trainers and the increase in partners' familiarity with MAA, LL methodologies, and stakeholder engagement practices. Three face-to-face training workshops on the innovative agroecological practices in each pilot LL, involving at least 25 participants including farmers, agricultural advisors and representatives of farmers associations. Co-learning through reciprocal visits of the pilot LLs between PNKH in French Guiana and UMNG in Republic of the Congo. Co-learning through meetings with Sister projects.

Achievable thanks to the structured approach of involving a wide range of stakeholders across different sectors. This approach ensures a comprehensive understanding of local needs and fosters a collaborative environment that supports the effective dissemination and adoption of MRT crop solutions.

1.1.3. Project R&I maturity

Currently available/in use on the market	ROTATES achievements
Scarce knowledge of the MRT genetic resources available in Europe. [TRL 2]	EU genebanks will be enriched with MRT genetic resources based on a gap analysis and, where appropriate, the introduction of new MRT resources from international genebanks. [TRL6]
No standard sanitation protocol is available for stakeholders in the areas where these crops are currently grown (although protocols have been published). [TRL 3]	Sanitation services for accessions of two target crops (sweet potato and yam) will be established in the framework of seed system implementation with suitable phytosanitary recommendations. [TRL6]
There are practically no MRT breeding programmes in Europe, except for one on yam in Guadeloupe and a taro phenotyping platform in Madeira.	Four breeding hubs, one for each target crop, will be implemented in European countries and outermost regions. They will be equipped for phenotyping, and researchers will be trained in appropriate breeding methods and strategies for these crops.
Current production of MRT crops is far below European market demand. Only 24% of the demand	Development of MRT based agroecological management practices maximising yield, tolerance to biotic and abiotic

is covered by MRT crops grown in Europe. [TRL 2]	stresses and the ecosystem services provided by these crops. [TRL 6-7]
MRT food products are available as raw materials or exotic foods. Processing methods exist in different tropical countries but are generally not suitable for the European market. [TLR 3]	Processing and enhancement of popular European foods—bread, pasta, and ice cream—by adding biofunctional compounds, lowering GI, and making them lactose-free, aligned with consumer tastes. [TLR5-6]
Practically no MRT crops are used feed for live-stock feed, except for peels used for animals in Guadeloupe and French Guiana, but with no studies of anti-nutritional and bio-functional components.	Enhanced knowledge and recommendations for the use of raw roots and tubers, leaves, peel and ensilage for ruminants and poultry and for the quality of by-products (meat, milk, cheese, eggs).
A small number of Living Labs already focus on minor crops in Europe. In ROTATES, pilot LLs expressed moderate familiarity with MAA and LL concepts (55.6% and 40%, respectively). Nevertheless, there is significant growth potential, as many are in early stages of co-creation and expanding engagement. [SRL 2-3]	Pilot LLs will receive training on stakeholder engagement, MAA principles, and LL setup. Through ENoLL ³⁰ , certification process, they will be evaluated against criteria to support their growth and certification as stable LLs. [SRL 7]
There is currently no network that incorporates the different value chain actors, farmers, seed companies, processors and researchers, concerned with MRT crops in Europe that could share knowledge, experience and resources.	Through ROTATES, which involves different actors of the value chain located in both continental Europe and in the outermost regions, as well as outside Europe, a network will be created to share knowledge and strengthen capacity concerning these crops at European level. Dissemination and alignment with the different missions to extend the MRT network.

#SPRJ-OBJ-PO\$#

1.2 Methodology #@CON-MET-CM@##@COM-PLC-CP@#

1.2.1 Overall methodology

ROTATES will conduct comprehensive, i- multi-disciplinary **DIAGNOSES** of current MRT uses, agronomic practices, varieties and value chain. The diagnoses will guide the activities for ii- **DEPLOYMENT** of novel MRT- based agroecological practices and the genetic improvement of locally adapted varieties to enhance crop productivity and quality of both (food and feed). Then, the knowledge, tools and products delivered will enable iii-**ADOPTION** of MRT through the creation of marketing avenues and capacity-building initiatives (Figure 2).

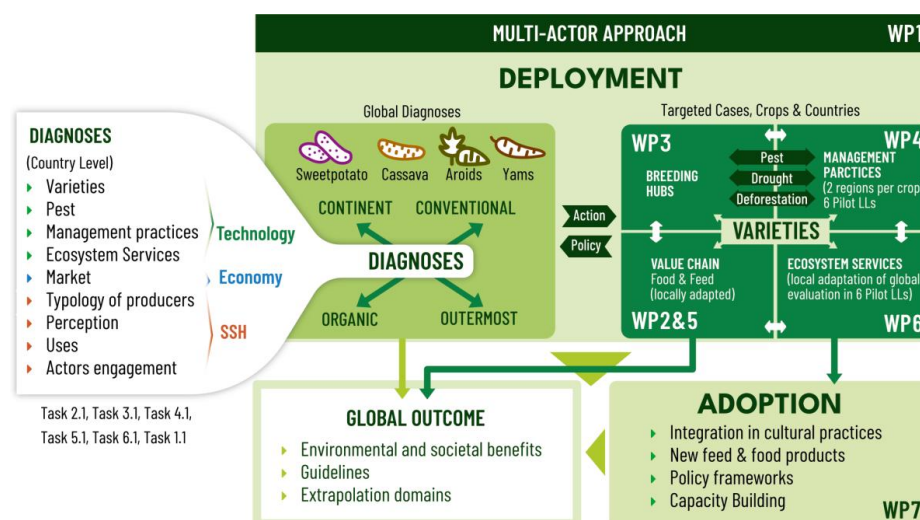


Figure 2: ROTATES methodology

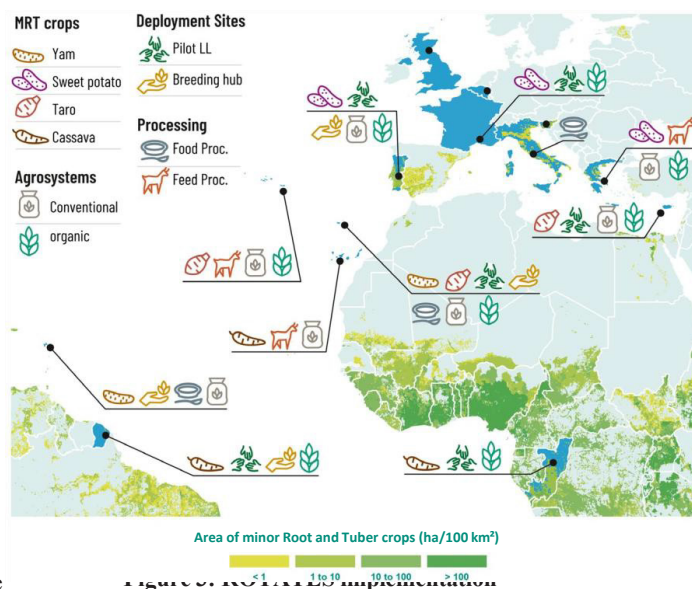
³⁰ Vervoort Proceedings of the XXXV IISPM

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➤ **Pilot LLs, Breeding Hubs, Case studies, crops and countries targeted for diagnoses, deployment and added value activities**

The **diagnoses will target the 10 countries** and outermost regions involved in the ROTATES consortium: Azores Isl., Canary Isl., Cyprus, France, French Guiana, Greece, Guadeloupe, Madeira, Portugal and Republic of the Congo (Figure 3).

Pilot Living Labs The development of **management practices**, and evaluation of their agronomic performance and their impact on ecosystem services will be implemented in **six pilot LLs**, three in continental Europe (**pLL1-Cyprus, pLL2-France, pLL3-Portugal**), two in overseas Europe (**pLL4-French Guiana, pLL5-Madeira**), and one in the Republic of the Congo a non-European country in the Global South endowed with forests and biodiversity of global significance (**pLL6-Congo**). Pilot LLs are **user-centred, open innovation ecosystems** based on systematic user **co-creation approach, integrating research and innovation processes** in real life communities and settings (using the Living Lab methodology detailed below). These pilot LLs were selected based on (i) the



diversity of contexts (agroecological conditions, and constraints and opportunities in the value chains, from production to consumption), (ii) the **availability of genetic material** to speed up the introduction and evaluation of the material, (iii) local **capacities and logistical conditions** for research and innovation, and (iv) the presence of both **conventional and organic systems** (SO2 & SO4). For each MRT crop, we selected at least two geographical regions with different constraints. Our choice was also influenced by the capacity of the partners and the opportunity to see the pilot LLs evolving into stable and long-term co-creation ecosystems and, possibly, into certified Living Labs.

Breeding Hubs ROTATES will set up **four breeding hubs**, i.e. sites that work in close collaboration with genebanks where genetic resources, a phenotyping platform and the skills required for breeding are available. **YAM HUB**: This breeding hub will be established in Guadeloupe where a **CIRAD** yam breeding programme is already underway and a worldwide yam collection at the **CRB-PT** genebank has already been genotyped^{31, 32} meaning budget savings will be available for the other MRT crops. The **TARO HUB** will be set up at **UMa** in Madeira, which hosts the **ISOPlexis** genebank, thanks to the taro diversity built by a former EU project (INEA²⁸), which is worth complementing. The **SWEET POTATO HUB** will be set up at **INIAV** in Portugal. Portugal and the Macaronesian islands have regulation facilities to exchange genetic material, thus providing the foundation for taro and sweet potato collections destined for European countries. A **CASSAVA HUB** will be set up at **PNKH** in French Guiana, chosen because it already hosts among the most important cassava genetic diversity³³.

Case Studies Since the direct use and processing of these MRT is not well known in Europe, the selection of the countries and outermost regions for food and feed activities was mainly based on **the consortium partners' experience** and their preliminary discovery work on the processing of MRT crops and on their post research collaboration with the food and feed industries to promote these products. Thus, processing MRT for **pasta** will be conducted by **CREA** in Italy, processing for **bread** by **CIRAD** in Guadeloupe, and processing for **ice cream** by **UMa** in Madeira. Their use as feed for **goats and poultry** will be researched by **ICIA** in the **Canary Isles**, for **cattle** by **UAc** in the **Azores**, and for **sheep** by **AUA** in **Greece**. Hence, investigation of the value chain will be conducted in **six of the case studies (CS)**, thereby ensuring the agrosystems where MRT will be grown and processed into food or feed are representative.

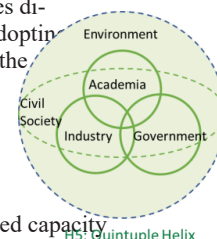
³¹ Arnau et al. *PlosOne* 2027

³² Sharif et al. *Annals of Botany* 2020

³³ Elias et al. *Economic Botany* 2004

Engagement with stakeholders and end-users: multi-actor approach (MAA) and Living Lab methodology

To ensure structured and long-term engagement of key stakeholders and end users at the local level, ROTATES will implement the Multi Actor Approach (MAA) and Living Lab (LL) methodologies as foundational methods (WP1, WP7). MAA, which is essential for agricultural innovation and crop diversification, integrates diverse perspectives to create sustainable solutions³⁴. ROTATES goes beyond MAA by adopting the LL approach, establishing real life environments for co-creation among key actors in the Quintuple Helix (5H) Model: Citizens, Government, Industry, Academia, and the Environment³⁵. This will help regions identify and leverage their economic niches and competitive advantages³⁶. A baseline survey conducted by ENoLL and ITC revealed that pilot LLs are still in the early stages of adopting MAA and LL methodologies, with limited co-creation activities and a focus on collaboration with industry (32%) and academia (30%) rather than with citizens and civil society (15%). To address these gaps, we will provide tailored capacity building actions using a train-the-trainer approach (WP7) to enhance the Pilot LLs' ability to engage key stakeholders, including researchers, farmers, farmer's groups and associations, advisors, food and feed processors, citizens and consumers, local rural communities, NGOs and government entities. ROTATES will use an integrated methodological approach to develop, implement, and monitor the MAA and LL framework throughout the lifespan of the project, ensuring active stakeholder engagement, structured co-creation, and a sustainable open innovation ecosystem, with continuous evaluation and refinement based on real-time feedback and performance metrics. The MAA will be implemented in 3 stages: 1: Framework development and Methodological Structuring (T1.1); 2: Action Plans and Implementation (T1.2). 3: Continuous Monitoring, Evaluation, and Iteration (T1.3) Active engagement throughout the project will support key policies and actions (CAP, EIP-AGRI, Green Deal/F2F, AgData, etc.), boosting the project's overall success (SO1 to SO4).



Holistic diagnoses of value chains

Because MRT are minor crops, **data on MRT crops are scarce in Europe**, and need to be complemented. To this end, we will leverage **consistent knowledge** on these crops available in **the main producing areas in Africa, Asia and South America**, two of these areas (French Guiana and the Republic of the Congo) are included in the project. For efficient adoption and sustainable production of MRT crops in Europe, we will conduct comprehensive diagnoses, combining desk studies and multi-stakeholder surveys and interviews, of the MRT genetic resources, of the performance of current cropping systems, and of the economic, social and cultural constraints and opportunities for the uptake of MRT crops along the value chain. Based on the results of our diagnosis, deployment and marketing strategies will be implemented aimed at the adoption and sustainable production of MRT crops (SO1). The aims of the desk study of available secondary information (scientific literature and extension documents, databases, regulations in force, etc.) will be to (i) understand the context in which the **food and feed value chains** operate or are recommended to operate³⁷, (ii) document existing **agronomic practices**, from planting to harvest, including a preliminary farm typology analysis, (iii) review the **provisioning, regulating, and supporting** ecosystem services associated with MRT-based agrosystems, and (iv) evaluate the potential of MRT for **food and feed production**. A **recommendation domain analysis**³⁸ will be performed to better understand the diversity of sites and the representativeness of selected sites to inform future scaling strategies (Figure 4A).

Primary data will be produced by surveys and interviews to complement the desk study. One objective of the surveys will be to **inform a farm typology analysis**³⁹ to understand the diversity of farms, and a **yield gap analysis**⁴⁰ to identify the constraints and opportunities specific to each farm type in pilot LLs where MRT crops are already produced (Figure 4A). The second objective of the surveys will be to assess the **genetic MRT material available** in farmers' fields in each partner country and to compare it to the global MRT diversity available in genebanks that hold these crops in trust. The third objective of the surveys will be to assess **disease pressure**. The fourth objective of the surveys will be to understand the current use of MRT crops in **food processing** and the use of MRT crop **by-products as animal feed**.

³⁴ Leclère et al. *Sustain. Dev* 2023

³⁵ <https://enoll.org/about-us/what-are-living-labs/>

³⁶ Ståhlbröst & Holst *The Living Lab Methodology Handbook*. Sweden, Centre for

Distance-spanning Technology. 2012

³⁷ Rattalino et al. *Agricultural and Forest Meteorology* 2017

³⁸ Muthoni et al. *Land Use Policy* 2017

³⁹ Hassall, et al. *Computers and Electronics in Agriculture* 2023

⁴⁰ Silva, et al. *European Journal of Agronomy*. 2017.

Field deployment of starchy MRT crops

➤ *Phenotyping of MRT diversity panels for traits targeted by ROTATES (WP3)*

Within the breeding hubs, each MRT crop diversity panel will be **phenotypically** evaluated in different environments, to capture the Genotype by Environment effect (GXE), for the traits which will be considered in management practices related to adaptation to agroecological conditions (i) **tolerance to drought and/or scarcity of water** for irrigation, (ii) **pest resistance**, (iii) **nutrient use efficiency** linked to zero deforestation and (iv) **weed resistance** through plant architecture and growth dynamics. **Field phenotyping** will consist of: (i) infrared gas analysers to measure photosynthesis and stomatal conductance to evaluate plant health; (ii) remote sensing by drones⁴¹ (NDVI, NDRE, GLI) to evaluate the health index and vegetation density, phenology, leaf temperature, leaf moisture content, and to predict productivity, (iii) morpho-agronomic traits (biomass, shoot length; size, shape and colour of the roots, etc); (iv) drought avoidance and phenotypic flexibility⁴²; (v) leaf relative water content and leaf area to apply Grime's CSR⁴³ model; (vi) water use efficiency measured as water productivity, i.e. the relationship between crop productivity and water use, or calculated using evapotranspiration data. **Laboratory phenotyping** will take place after harvesting and will include the (i) root-to-shoot signalling⁴⁴ (ABA, proline, the antioxidant enzyme ascorbate peroxidase) using spectrophotometry, and (ii) the aptitude for nutrient absorption and use to increase yields (nitrogen efficiency ratio, nitrogen use efficiency, nitrogen harvest index⁴⁵). The nutritional quality of shoots and roots/tubers (total minerals, protein, starch), using near infrared spectrometry (NIRS)⁴⁶, will be assessed both in the field and in the laboratory after harvesting. The detailed phenotyping data will make it possible to investigate the architecture of valuable traits **(SO2 to SO4)**.

➤ *Deciphering crucial genes and alleles for the traits targeted by ROTATES (WP3)*

MRT genetic resources will be enriched by introducing new varieties from neighbouring countries or international genebanks, thanks to the ongoing collaboration with international and CGIAR genebanks (CePaCT, CIP, Alliance Bioversity International-CIAT, IITA) after conducting a gap analysis. To capture the entire genetic diversity, which can later be used in breeding programmes, the newly collected and introduced accessions will be sequenced, at 50x coverage depth, using the Illumina NovaSeq 6000 System. The reads obtained from the sequencing will then be integrated in the pangenome of their respective species: *Colocasia esculenta*, *Manihot esculenta*, *Dioscorea sp.* and *Ipomea batatas*) using the *giraffe* tool in the *vg* software⁴⁷. The pangenomes will make it possible to capture both Single Nucleotide Polymorphisms and Structural Variants, including Copy Number and Presence-Absence of Variations. This strategy will enable us to **fully grasp the molecular basis of phenotypic traits of interests**. Ultimately, this strategy will facilitate the selection of accessions which harbour genomic/genetic variations underlying a trait of interest for breeding programs. Genotyping data will be coupled with large-scale phenotyping data for **genome-wide association** studies and **genomic prediction**. Trait heritability will be assessed, and genetic markers linked to those traits will be deciphered. For traits with a strong quantitative component, we plan to develop advanced phenomic and genomic prediction models, using both standard statistical methods and cutting-edge machine learning techniques, as appropriate **(SO2 to SO4)**.

➤ *Safe seeds: preventing disease outbreaks through an appropriate sanitary framework (WP3)*

Because starchy minor RT crops are propagated vegetatively, they are prone to accumulate diseases over generations, leading to significant losses in both quality and yield. An integrated seed health strategy is thus indispensable to build up sustainable seed systems for stakeholders and to ensure high yields and the development of value chain. In a vegetatively propagated crop, **the term 'seed' actually refers to the tuber or the root**. The potato seed system is the most widely studied seed system, which, in EU countries, is implemented using clear policies that are reviewed by the stakeholders involved. A scientific and context-based framework extending from the genebank to the farmer's field, will be designed for the **development of a clean seed system as a support for the sustainable production of MRT crops by farmers**. The presence of tuber-borne pathogens on collected MRT accessions will be evaluated using cutting edge (high throughput sequencing technologies) diagnoses. Combined with a cross-check of the availability of preferred germplasms in genebanks, the results will trigger a sanitation process, based on the combination of **meristem culture and thermotherapy**, targeting rare and elite germplasms to ensure the availability

⁴¹ Macedo et al. *Agriculture* 2023

⁴² Gouveia et al. *Emir. J. Food Agric.* 2020

⁴³ www.ecologycenter.us

⁴⁴ Gouveia et al. *Physiologia Plantarum* 2020

⁴⁵ Gouveia et al. *Emir. J. Food Agric.* 2020

⁴⁶ Gouveia et al. *Applied Sciences* 2020

⁴⁷ Jouni Sirén et al. *Science* 2021

of a diverse panel of clean seeds for further dissemination to farmers and for the exchange of material between countries. In addition, a thorough review of the literature related to the risk of pests, plus analyses, and the seed systems proposed in the Global South for the targeted crops and for clonal crops in the EU will be combined with stakeholder consultations to guide the proposal of a **policy framework for phytosanitary certification** of seed systems adapted to EU context and the requirements of IPPC international standards (such as ISPM⁴⁸) for at least one MRT crop (**SO2 & SO4**).

➤ **Co-construction of novel agroecological management practices based on MRT crops (WP4)**

Empirical agronomic research will be conducted in all 6 pilot LLs, which are characterised by different challenges: climate change and water scarcity in pLL1-Cyprus, pLL2-France, and pLL5-Madeira, pests and weeds in pLL2- France and pLL3-Portugal, and deforestation in pLL4-French Guiana and pLL6-Congo. The systems to be investigated will be crop rotations and crop associations (i.e., intercropping) that include MRT crops, but the performance of the system as a whole will be evaluated, not only the performance of MRT crops. We will use a system agronomy approach by adapting the **Describe Explain Explore Design (DEED)** approach in which we will **diagnose** current farming systems, **design and test** alternative MRT-based cropping systems on-station, **adapt** them on farm, **evaluate** them in a holistic and participatory way, and **scale** most promising ones (Figure 4B). Pilot LLs will be at the centre of this approach based on co-construction (strong link with WP1).

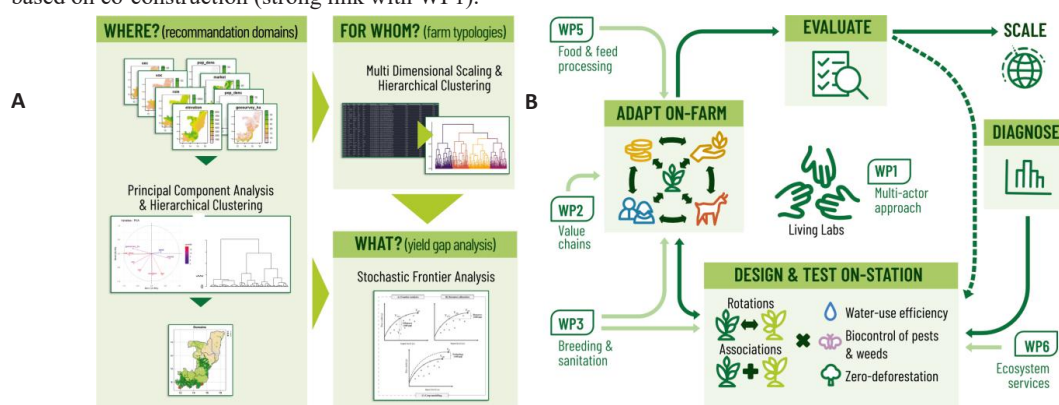


Figure 4: diagrammatic representation of the three steps of the agronomic diagnosis (A), and diagrammatic representation of the iterative and participatory approach used for the co-construction of novel agroecological management practices based on MRT crops (B). Connections between WP4 and other WPs (WP1, WP2, WP3, WP5)

The results of the agronomic diagnoses (desk study, recommendation domains, farm typologies and yield gap analysis, see above) will inform the **co-construction process within the pilot LLs**, based on the multi-actor approach (WP1), leading to the design of novel agroecological management practices based on MRT crops. These practices will then be tested on station and adapted on farm. The purpose of on station research will be to make detailed measurements in a controlled environment in order to better understand the **underlying mechanisms governing the performance of novel cropping systems** based on MRT crops. The purpose of on-farm research will be to assess how innovative cropping systems fit in diverse farm contexts, organic and conventional, and in particular the trade- offs their adoption may impose (for instance, in terms of labour). The purpose will also be to understand what adaptations and actions are necessary for their wide adoption. The performance of the most promising management practices will then be evaluated in the various pilot Living Labs^{49,50}. This evaluation will be informed by a **trade-off analysis of multiple criteria** (productivity, labour, provision of ecosystem services, etc., for the different types of farms and geographic domains identified⁵¹). Simulations by mechanistic models calibrated with the detailed measurement taken on-station will also be used, most likely using STICS, one of the rare models that can simulate crop associations^{52,53} (**SO2 & SO4**).

⁴⁸

<https://www.ippc.int/fr/coreactivities/standard-setting/ispms/>

⁴⁹ González-Esquivel et al. *Journal of Agricultural Sustainability* 2020

⁵⁰ Mottet et al. *Frontiers in Sustainable Food Systems* 2020

⁵¹ Klapwijk et al. *Current opinion in Environmental sustainability*. 2014

⁵² Vezy et al. *Agronomy for Sustainable*

Development 2023

➤ ***Monitoring and evaluation of ecosystem services supported by the integration of MRT crops in farming systems (WP4,6)***

The identification of the provisioning and regulating ecosystem services supported by the incorporation of MRT crops, will be addressed by key measurements across the different agrosystem components in the six pilot LLs compared with current local practices (SO2 & SO4). **Provisioning services** will be assessed based on **yield**. A key **regulating service, biological pest control**, will be assessed by monitoring pests, including nematodes, weeds and pathogens, compared with in controls, in collaboration with **WP3. Beneficial organisms** such as natural enemies of arthropods, soil micro-fauna, bees and wild bees as pollination indicators **will be sampled, sorted and taxonomically identified** through field surveys and DNA metabarcoding analysis. Deep metagenomic analysis will be used to assess the potential functional behaviour of the soil microbiota in taro in pLL1-Cyprus and in sweet potato in pLL2-France. The diversity and activity patterns of vertebrates will be investigated using camera trapping in the 6 pilot LLs. **Soil structure** will be analysed to assess water regulation (**WP4**). **Pesticide use** will be evaluated based on farm calendars. Soil health will be monitored using the BioFuncTool®, which rapidly and efficiently assesses the impact of agricultural practices on soil health in the field using simple, low-cost, yet robust methodologies. By exploiting these datasets, advanced modelling and GIS will enable the development of context-specific recommendations for irrigation, fertilisation, and pest management that maximise yield and minimise environmental impacts (SO2).

➤ ***Fostering the development of MRT value chains (WP1, 2)***

The potential role of MRTs in terms of food security, securing livelihoods and adapting to local environmental conditions has long been identified⁵⁴; however, only by fostering their value chains will it be possible to ensure that MRTs do have an impact⁵⁵. The information collected will be qualitative and quantitative and will facilitate the creation of value chains - or the enhancement of existing ones - for specific MRT food and feed products. In-depth case studies will be conducted based on the initial diagnoses, on the **economic, social, and cultural drivers and barriers** to the uptake of MRTs (SO1). It will also allow the preliminary mapping of value chains where possible (i.e., where a supply chain already exists) in order to identify **potential constraints** (e.g., financial constraints) and **opportunities** (e.g., economies of scale, satisfying consumers' interest in products with particular attributes). **Deliberative choice experiments** will be conducted **with farmers** to understand their preferences with respect to the characteristics of MRT crops identified in the other ROTATES WPs (1 focus group for each of 6 case studies including a deliberative choice experiment with farmers organised on site, i.e., 40 to 50 farmers each focus group, for food and feed). **Semi-structured interviews** will be conducted with **processors** to assess their **willingness to be involved in MRT value chains** (aspects to consider will be, for example, their views on the market, the existence of transaction costs or any technological barriers that could hamper uptake of the MRT crops) ((3-5 semi-structured interviews with processors for each of the 6 case studies, i.e., food and feed). Work with **consumers** will combine focus groups (1 focus group for each of the 4 case studies, i.e., Portugal, Cyprus, Guadeloupe, Italy with 20 consumers each to prepare consumer choice experiments) and **choice experiments** in four food case studies (4 surveys with consumers i.e., 350 consumers in Portugal, 100 in Cyprus, 50 in Guadeloupe and 1800 in Italy). The information collected will be compiled, analysed and reported in six business case studies. This work will **highlight the potential for success of different value chains** (e.g., combining the collected qualitative and quantitative information to simulate the profitability of different parts of the value chains as well as highlighting barriers and opportunities) and discuss the **most appropriate strategies** (e.g., assess the cost effectiveness of introducing these types of feed in livestock diets) (SO2 & SO4).

➤ ***Releasing the potential of MRT crops as food (WP5)***

The added-value of MRT crops will be explored through three traditional processed food products: **pasta, bread** (also in blends with gluten-free or with low gluten index cereals) and **plant-based ice cream**. To meet European standards including taste, nutrition, and method of production, the characteristics of MRT crops and of the varieties selected in terms of raw material processability, sensorial and nutritional quality of products, and proposed new recipes will be investigated. Flours will be produced, with the aim of use as a partial substitute for cereal flour or as is. Novel pasta formulations will be produced at laboratory scale and characterised in terms of their nutritional, technological/rheological and sensorial quality through biochemical analyses and non-destructive measurements (NIRS). Different types of bread will be produced using suitable composite flours and their sensory quality characterised. MRT-based milk alternatives will be produced to make ice cream^{56,57} and evaluate it. Sensory

⁵⁴ McMullin et al. Global Food Security 2020.

⁵⁵ Revoredo-Giha et al. *Agricultural and Food Economics* 2021

⁵⁶ Ferdaus et al. *Nutrients* 2023

⁵⁷ Kilinç et al. *J. Food Process. Preserv.* 2022

evaluation will be conducted with a panel of trained expert assessors (5 to 10) and untrained consumers (50 to 100 depending on the product). Collaboration with industry will be ensured by the partners. New recipes will be created using raw and processed MRT crops and will be evaluated by professional cooks and consumers in an experimental restaurant. The most promising recipes will be published in culinary booklets/guides (SO3 & SO4).

➤ ***Releasing the potential of MRT crops as feed (WP5)***

The use of MRT crop by-products as feed (**vegetative parts, roots and tubers**) will be evaluated in **one continental country (Greece) and in two outermost regions (the Canary Islands and the Azores)**. The assessment will involve different species, including **cattle, goats, sheep, and poultry**. The food products derived from livestock that consumed MRT feeds (**milk, cheese, eggs, and meat**) will be characterised to determine their nutritional and sensory quality (WP2). We will also develop methods to **conserve MRT by-products (fresh, dried, ensiled)** and determine their suitability for inclusion in livestock feed. In collaboration with WP1, to ensure the integration of MAA, untrained consumers (80 - 100 participants in each country) will perform a sensory evaluation of the different livestock products (meat, egg, cheese) based on appearance, aroma, flavour, overall palatability on a 9-point hedonic scale (ISO 11136, 2020). A trained 10-12-member sensory panel will perform a descriptive analysis, texture, odour, flavour and taste attributes depending of the product, following the ISO 8589 (2007) instructions. We will study the yield, production, and quality of livestock products. Finally, we will identify livestock **farmers and feed industries interested** in making use of the experimental results obtained throughout the lifespan of the project, using the **MAA methodology** (WP1 & 2) (SO3 & 4). Collaboration with industry will be ensured by the partners' connections.

Adoption of starchy MRT crops

➤ ***Marketing avenues (WP1 to 8)***

To support marketing avenues, co-construction processes in pilot LLs (WP1) will involve local stakeholders across the value chain, ensuring their standpoints are included in assessments of the value chain (WP2), of food and feed formulation (WP5), and in product acceptance and evaluation (WP2, WP5). This will ensure that products developed by the project meet stakeholder demand, and hence facilitate market entry. Given the growing demand for innovative processing methods and new recipes, product development and differentiation will be essential. Processing MRT crops into value-added products (e.g., low-gluten, low GI, lactose-free) and emphasizing their unique advantages, such as nutritional benefits and sustainability, will help position these crops in the market (WP5). The partners involved in WP5 already work closely with food companies, such as Barilla, and feed companies such as Graneros de Tenerife (see §3.2), thus ensuring close links with the market. Additionally, targeted marketing campaigns, social media, and farm-to-table events will educate consumers about the benefits and uses of MRT crops (WP5, WP7). We will create a framework and produce guidelines for a clean seed system to ensure a consistent supply of high-quality products (WP3), thereby strengthening the supply chain. Finally, ROTATES will provide guidelines, policy frameworks, and recommendations to policymakers on how to create a supportive environment for the integration of minor crops (WP2, 3,4,5,6). These activities will identify strategies and create **opportunities for the creation or expansion of existing MRT-based value chains** and increase possible **marketing avenues for farmers across Europe** (SO2 & SO3).

➤ ***Capacity building (WP1,3,4,5,6,7)***

The project prioritizes mutual learning and capacity building, especially for farmers and breeders, to promote the inclusion of minor crops in agrosystems. A tailored, agile capacity-building program (T7.1) will be co-designed with project partners, and implemented in alignment to the established methodological framework (T1.1), based on detailed needs assessments, to equip them and local stakeholders with the necessary skills to engage in value chains. The programme will adapt to evolving needs and project developments. This capacity-building initiative will use a three-fold approach: i) **Harmonized Approach**: Training will ensure consistent methods across all pilot sites and partners. This includes foundational lessons on how to create Living Labs (LLs), stakeholder engagement, and co-creation, aligned with the Methodological Framework. Participants will adapt these core concepts to their local contexts through practical applications. ii) **Train-the-Trainers (WP7)**: This will prepare project partners who run the pilot LLs, to promote MRT crop adoption within their ecosystems, focusing facilitation and pedagogical skills. iii) **Knowledge Sharing and Co-Learning (WP4)**: Field demonstrations in each pilot LL, workshops and collaborative learning sessions will cover agroecological practices, food safety, policy support, and business case studies. These sessions, including scenario-based exercises, will be tailored to local needs and supported by expert presentations and hands-on demonstrations (SO4). They will be preferably co-organised during existing farmer events to encourage the participation of stakeholders. ROTATES will use a mixed-method approach, including

workshops, mentoring, webinars, and field visits, to bridge theory and practice. The exact actions will be decided based on the results of the **needs assessment** (WP1). Using tools like serious games will help apply concepts in controlled environments. A continuous feedback mechanism will ensure the programme's adaptability, aligning with the **Monitoring Evaluation and Learning approach**, to enable the effective implementation of minor crop cultivation and distribution practices across diverse regions.

1.2.2 Relations with ROTATES context / environment

ROTATES partners are currently involved or have been involved in EU-funded projects or other international, and national research projects and initiatives linked to the project objectives. The table below summarises some of these projects and explains how ROTATES intends to seek complementarities and synergies with them.

Name (year)	Description	Link with ROTATES
IntercropValueES (2022-2026) HORIZON	IntercropValueES is developing a scientific research action for better understanding and modelling of intercrop functioning and a detailed analysis of lock-ins and levers at the value chain level.	ROTATES will collaborate closely with IntercropValueES, which is coordinated by CIRAD, to benefit from the approaches and methods developed.
RADIANT (2021-2025) Horizon 2020	RADIANT aims to promote crop diversification, environmental and agrobiodiversity preservation, and fair economic development through the valorisation of Underutilised Crops.	Experience in the AURORA Farms will be used to better tailor approaches to be used in Pilot LLs and other environments.
CROPDIVA (2021-2025) Horizon 2020	CROPDIVA aims to conduct in-depth study and promote six underutilised arable crops, create new value chains for the development of food and non-food products and assess the socio-economic impacts of the ecosystem services delivered.	The final decision support system for stakeholders developed by CROPDIVA could be tested by ROTATES.
DIVINFOOD (2022-2027) Horizon 2020	DIVINFOOD project aims to co-develop new marketing strategies and plant-based products from underutilised crops, benchmark agroecological farming techniques, to breed locally adapted cultivars, and to demonstrate business models for farmers.	ROTATES will ensure the exchange of knowledge and experience between LLs in the two projects.
BIOVALUE (2021-2025) Horizon 2020	The project aims to develop a modular simulation tool to analyse the link between biodiversity, the agri-food value chain, and consumer health.	The simulation tool developed in BIOVALUE could be useful for ROTATES (WP5, 6).
AGROSUS (2023-2027) Horizon 2020	The AGROSUS project aims to establish agroecological practices, with a focus on technology transfer and training for weed mitigation and management control.	Through UMa, ROTATES will collaborate with AGROSUS on tools and agroecological strategies for weed prevention and control (WP4).
RTB Breeding (2023-2024) extended for four years. BMG Foundation	The RTB Breeding project, with CIRAD as partner, aims to implement modernised approaches for the breeding of Roots, Tubers and Bananas in Africa.	ROTATES will use the most appropriate breeding methods developed and adapt them to the European context including challenges (WP3).
KOPEKOBAs project (2025-2029) CAFI-AFD	The KOPEKOBAs project implemented in the Republic of the Congo, with CIRAD as partner, aims at zero-deforestation farming and the sustainable use of biodiverse and carbon-rich ecosystems.	ROTATES will allow research on cassava from KOPEKOBAs to be deepened. KOPEKOBAs will also scale results from ROTATES in the Republic of the Congo.
SOILL (Support Structure for Soil Living Labs) 2024-2030 H2020	The SOILL framework partnership, supported by ENoLL, sets up and runs a support structure and services for the 100 Living Labs and light-houses of the Mission "A Soil Deal for Europe".	ROTATES will align with SOILL to enable knowledge sharing and benchmarking of results. Collaboration is also envisaged with Mission LLs within the agricultural sector.

1.2.3 ROTATES interdisciplinary approach

The implementation of ROTATES requires knowledges and methodologies from different disciplines gathered by the **18 partners representing complementary fields and disciplines** (detailed in §3.2) that are interwoven within the project activities and will be brought together and integrated in pursuit of its objectives, co-developing innovations in different fields. The diagnosis and development of agroecological cropping systems that include MRT crops and maximise their benefits will bring together agronomists, modelers, ecologists, entomologists, breeders and ecophysiologicals, as well as data scientists, social scientists and economists. The implementation of clean seed systems will involve phytopathologists, geneticists and biologists. The development of new feed and food products and of the value chains will engage food and feed scientists, socio-economists, breeders and agronomists.

Thanks to the use of the Living Lab methodology not only at pilot scale but also at project level, ROTATES will go beyond interdisciplinarity, to achieve trans-disciplinarity and a holistic approach.

1.2.4 Social science and humanities considerations

The core of the ROTATES approach is to address challenges through co-creation, through the active involvement of the stakeholders in a process of situated experimentation and collective learning for which it is essential to foster a culture of collective sense-making along the process. In line with the transdisciplinary approach, insights from key social and behavioural sciences (SSH), education sciences, humanities, as well as economics and business experts will be incorporated in the overall methodology and approach used by ROTATES to enable the understanding and addressing of the diverse needs and behaviours of community members.

In the framework of the MAA and LL methodology and as part of its trans-disciplinary approach, ROTATES will include SSH considerations both horizontally at project level and vertically (in Pilot LLs and value chain). Such an approach, led by social scientists and SSH partners, will play a critical role in data collection, analysis, engagement and validation throughout the lifespan of the project, from **co-design to scaling up**.

1.2.5 Gender dimension of ROTATES

Gender dynamics in agriculture involve a variety of roles, responsibilities, and access to resources that differ widely among individuals, extending beyond the binary. For instance, non-binary and gender-diverse individuals may face unique challenges that are often overlooked. Experiences can vary significantly based on intersecting factors such as socioeconomic status, ethnicity, and rural versus urban settings. Recognising and addressing these diverse gender experiences is essential to developing inclusive and effective agricultural innovations.

Even though the integration of gender dimension is not mandatory for this call, ROTATES will address gender from multiple angles—needs, acceptability, and replicability—to ensure that outcomes are inclusive, relevant, and effective for all genders. The ROTATES methodological framework, developed in WP1, will establish – alongside the multi-actor approach and living lab methodology – a foundation for a gender-responsive approach throughout the project's lifecycle. This includes incorporating gender-sensitive guidelines, practices, and tools to promote equitable participation and outcomes. ROTATES will implement key actions such as designing and applying gender-sensitive methods of data collection and analysis. When planning a strong participatory approach across different work packages, particular attention will be paid to fair and balanced participation in planned activities at project or pilot level. Data collection and analysis will be designed to account for gender differences and provide gender-disaggregated diagnostic outputs, thereby facilitating the production of recommendations and analyses that are sensitive to gender differences (if any). This will ensure that the co-created innovations are responsive to the needs of all genders.

1.2.6 Open science approach

Following the guidelines of Horizon projects, ROTATES underlines the importance of open access to scientific publications and research data. ROTATES will integrate open science practices to improve the quality, efficiency and responsiveness of its research results, as follows. **(i) Early and open sharing of research:** Scientific publications resulting from ROTATES, will be published in open access journals with an embargo period of 3-6 months. In addition, pre-prints may be available on platform (as pre-prints, HAL), Experimental social and economics protocols will be registered on the AsPredicted repository as recommended by HE guidelines. Datasets will be shared as early and openly as possible using trustworthy thematic repositories (e.g., the European Nucleotide Archive (ENA) for sequencing data) or institutional warehouses (e.g., Dataverse). **(ii) Research output management:** A first version of the Data Management Plan (DMP) addressing each ROTATES dataset will be released within six months after the beginning of the project and will be regularly updated in order to provide a clear and detailed description of the management of all research outputs. The data produced and considered open for re-use will be deposited in a trusted repository (e.g., ENA, Dataverse) in line with the FAIR principles Standardised metadata frameworks will be added to datasets (e.g. the MIAPPE standard - Minimum Information About a Plant Phenotyping Experiment-, the MIXS

standard -Minimum Information about any Sequence-, AACC standard, ICC standard or ISO, UNE norms). **(iii) Measures to ensure reproducibility and reuse of research outputs:** ROTATES will provide access to data or other results needed to validate our findings. Datasets will be archived (with persistent identifiers) on trusted thematic repositories or warehouse recognised by the community. The methodologies used during the project will be open and available to the public for the purpose of reproducibility. Workflows and software produced during the project will be made available on open access forges and saved in the Software Heritage archive. Appropriate licensing options will be attached to each shared output to support reuse. **(iv) Providing open access to research outputs:** All public deliverables and open research outputs, including scientific papers and datasets, will be made accessible on trustworthy thematic repositories (e.g., Zenodo, ENA, Dataverse) and accessible through API on the project website. Particular attention will be paid to avoid duplication of identifiers for appropriate identification, citation, and re-use.

(v) Participation in open peer review: In the case of scientific articles, the peer review process will be as open as possible, including practices such as open identities, open reports, and open participation. **(vi) collaboration with the stakeholders and the multi-actor approach** is a key aspect of ROTATES's overall methodology. All relevant stakeholders will be engaged in co-creation including citizens, customers and other end-users. This will be achieved, for example, through the work of pilot LLs, and product development food and feed companies.

1.2.7 Research data management:

The following table summarises project compliance with FAIR principles. Accessibility will be ensured by open access upon publication for all types of data. Re-usability is ensured with open licences – CC-BY-4.0 in all cases except for analytical workflows where GPLv3 will be applied.

Data	Findability	Interoperability	Curation
Genomic (Metabarcoding, Functional Genomics); type: text	European Nucleotide Archive (ENA)	<i>formats:</i> fasta, fastq, BED/GEF/GTF	CIRAD, CUT, UL WP3, & 6
Phenotypic; type: observational, images, text	Institutional Dataverse, BreedBase	<i>formats:</i> txt, csv, jpeg, tiff, png	UMa WP3
Spatial prediction; type: numerical (raster)	Institutional Dataverse	<i>formats:</i> tiff	CIRAD WP4
Analytical workflows; type: text	Forges (GitLab), saved on Software Heritage	<i>formats:</i> R, Python scripts	CIRAD WP3, 4
Biochemical Data and Rheological parameters; type: numerical	Institutional Dataverse	<i>formats:</i> txt, csv, MSD	CREA WP5
Livestock products type: images, text, numerical	Institutional Dataverse, Zenodo	<i>formats:</i> txt, csv, excel, tiff, png, jpeg	ICIA WP5
Community and population data; type: numerical, images	Institutional Dataverse	<i>formats:</i> txt, csv	CUT, CIRAD WP6
Soil analyses (BioFuncTool® data) type: numerical	Institutional Dataverse	<i>formats:</i> csv	INRAE WP6
Social, economic studies and surveys Pilot LLs needs and evaluation data; type: txt	Institutional Dataverse, Zenodo, AsPredicted	<i>formats:</i> txt, csv	ITC, INRAE, CIRAD, ENoLL WP1, 2, 4 & 7
Processing and use of MRT residues; type: images, text, numerical	Institutional Dataverse, Zenodo	<i>formats:</i> txt, doc, csv, excel, tiff, png, jpeg	ICIA WP5

1.2.8 Compliance with ethical principles and relevant legislation:

Members of the ROTATES consortium will ensure that the work undertaken within the framework of the project complies with EU ethical standards for research and innovation. MRT crops are food resources listed in Annex 1 of the ITPGRFA. The regulations and legislation that apply in each country will be carefully examined along with the management of personal data from social and economic studies and surveys to comply with ITPGRFA and Nagoya protocol. Research involving animals will be carried out in accordance with the EU applicable rules and legislation, including Directive 2010/63/EU on the protection of animals used for scientific purposes. Ethical self-assessment is provided in ROTATES PART A, detailing ethical issues, the ethical dimension of the objectives, methodology and impact and measures which will be taken to ensure compliance with ethical principles and relevant legislations.

1.2.9 Do No Significant Harm:

ROTATES will contribute to the environmental objective “protection and restoration of biodiversity and ecosystems” of Regulation (EU) No 2020/852 (EU Taxonomy Regulation). The results of an initial assessment indicate that the project does not include activities which could cause significant harm to any of the other five environmental objectives of the Regulation as described in article 17, and the assessment will continue throughout the lifespan of the project. More details are given in Ethical self-assessment in ROTATES PART A.

#\$CON-MET-CM\$# #COM-PLC-CP\$# #REL-EVA-RE\$#

2. IMPACT #IMP-ACT-IA@#**2.1 Project's pathways towards impact****2.1.1 Uniqueness of ROTATES towards expected outcomes**

ROTATES will produce evidence for all relevant stakeholders on the benefits – and site-specific actions to maximise these benefits – of MRT crops, which include positive environmental and socio-economic impacts as well as commercial opportunities. The figure 1 showing the ROTATES overall concept in section 1.1 introduces the key results and outcomes and identifies the direct beneficiaries. The following tables show how ROTATES will directly contribute to the six expected outcomes, namely: Increased evidence of the environmental benefits of minor crops (EO1); Farmers make use of a wider range of crops, and combination of crops (EO2); Minor crops are integrated in farming systems promoting their environmental benefits (EO3); Increased resilience and climate adaptation of farming systems vis-a-vis biotic and abiotic stresses (EO4); Feed and food industry make use of minor crops (EO5); Creation of new avenues for farmers and value chains through a wider range of products (EO6).

Basket of alternative and diversified cropping systems, integrating MRT	EO1; EO2; EO3; EO4
A set of recommended agroecological cropping systems that integrate MRT crops with specific guidelines for farm management practices and optimisation of their benefits will support the effective incorporation of MRT crops in farming systems. This will produce evidence for increased resilience and adaptation to climate change as well as to biotic and abiotic stresses (EO1). The inclusion of MRT crops in farming systems through intercropping or crop rotations will enable the use of a wider range of crops and combination of crops (EO2). The ecosystem services provided will be monitored and evaluated , enabling tailored management practices that improve yield, resilience, and sustainability (EO3). ROTATES will build farmers' and agricultural professionals' knowledge of the role played by minor crops in the provision of ecosystem services. Specific interventions will be suggested to farmers to overcome the constraints and challenges associated with the adoption of MRT crops (EO4). To ensure successful development of alternative and diversified cropping systems, ROTATES will identify new plant material and valuable traits and produce structured data . This will enhance breeding outcomes and result in genetic improvements in response to environmental and climatic challenges and ensure access to markets.	
KPIs: 20+ agroecological cropping systems tested, 10+ agroecological cropping systems recommended, 4 breeding hubs, scripts and workflows, guidelines identifying strategies for MRT value chains.	
Direct beneficiaries: The scientific community; Crop and livestock farmers; Farming associations; MRT breeders; Seed producers; Other projects and initiatives; Local decision-makers; Extension services (AKIS).	
New food/feed products incorporating MRT crops; Cost-effective animal feed in-EO5; EO6	corporate MRT crops; Up-cycling of MRT wastes
ROTATES will showcase food produced at the laboratory scale (gluten free pasta, pasta and bread with low gluten content and plant-based ice-cream - TRL3 to TRL5-6 - and feed products - ensiled, fresh, and dried - ready to be scaled up for the market in the near future, to food and feed industry stakeholders, paving the way for their use in other processed foods like biscuits and juices (EO5). It will produce knowledge on the nutritional and nutraceutical quality of MRT crops for use as livestock feed and on the nutritional, bio-functional and organoleptic quality of derived livestock products (milk, cheese, meat, eggs) (EO5). ROTATES will introduce new methods for efficient up-cycling of MRT waste that support the circular economy and test and present effective methods for the use of MRT crops as feed for cattle, small ruminants and poultry (EO6). To widen the range of products, ROTATES will create new recipes using raw and processed MRT. The most promising recipes will be published in culinary booklets / guidelines and short videos in at least three languages. In all, ROTATES will create new avenues for farmers and for value chains through a wider range of products (EO6).	
KPIs: 3 new food products and sources of feed for 4 livestock categories and by-products that can be incorporated in the food and feed industry. Recipe booklets and videos will be produced.	
Direct beneficiaries: Farmers, and breeders; Food and feed processors; Agrifood industry; Farming associations; Agricultural advisors, European citizens (consumers); NGOs; CSOs.	

New business case studies on raw and processed MRT crops produced in conventional and organic farming systems	EO2; EO3; EO5; EO6
ROTATES will enhance knowledge on research and innovation that will be delivered in the form of six business case studies and guidelines and that will: (i) allow understand barriers to the adoption and development of MRT crops at farm level and identification of key levers for their wider use by farmers (EO2 and EO3); (ii) identify and characterise the barriers that hamper crop diversification at supply chain level in Europe and that have to be overcome (EO3 and EO5) and (iii) identify effective strategies for the development of MRT value chains in Europe to remove barriers to their incorporation at farm and supply chain level (EO2 and EO3), increase demand for food and feed products (EO5) and identify business opportunities at all levels of the supply chain (EO6).	
KPIs: 6 business case studies with complete diagnoses of the challenges and opportunities for MRT development; and guidelines for the implementation of strategic policy interventions	
Direct beneficiaries: Crop and livestock farmers; Farming associations; Consumers; Food and feed producers; Food and feed processors; Seed producers; Genebanks; Policymakers; Local decision-makers; Extension services (AKIS); Other projects and initiatives.	
Structured open access data, scientific findings and new tools to unlock the full potential of MRT crops + increasing MRT genetic resources in genebanks for farmers and breeders	EO1; EO3; EO4; EO5
ROTATES will deliver open access databases on the performance of agroecological practices based on MRT crops in conventional and organic agriculture and leverage dashboards which support data collection and/or data packaging and monitoring in the 10 countries and the five outermost regions involved in the project. These data will provide evidence of the effect of incorporating MRT crops on increased resilience and adaptation of farming systems to biotic and abiotic stresses (EO4). Models including data on biodiversity, soil health, and water regulation for the effective management of MRTs which allow an increase in the environmental benefits of MRT crops will be delivered (EO1). Models and tools will be developed to predict (i) the suitability for processing of raw materials and flours derived from the MRT crops tested in the project, and (ii) the nutritional, technological and sensorial quality of the resulting products (EO5). Scientific findings, data and workflows will be published as peer reviewed papers in open-access journals and warehouses to share the knowledge and reach a wider readership (EO1, 4, 5). MRT genetic resources will be enhanced and conserved in seven genebanks that are accessible to farmers and breeders - assessed, enriched and sanitised -, as evidence for the contribution of MRT crops to a wider range of crop genetic resources and for the promotion of agrobiodiversity in production systems (EO3). Passport data including the performance of different varieties will be included in the genebank databases thereby facilitating access to well-documented genetic resources (EO1).	
KPIs: Datasets deposited in public repositories accessible under FAIR principles (cf. §1.2.7); 6 open access databases on the performance of agroecological practices based on MRT in conventional and organic systems. MRT in 7 EU genebanks; 20+ peer-reviewed scientific articles; 10+ papers at international scientific conferences.	
Direct beneficiaries: The scientific community, Farmers and breeders, Agricultural advisors, Genebanks, The agrifood industry, Seed producers, Living labs, other EU projects and initiatives	
Policy framework for phytosanitary certification of seed systems for MRT crops; Policy Recommendations for MRT along the value chain; Guidelines for MRT development, added value and adoption	EO2; EO3; EO4; EO5; EO6
ROTATES will provide phytosanitary recommendations for seed systems. The recommendations will increase farmers' knowledge about the propagation and distribution of safe genetic materials. In addition, the availability of pathogen-tested material for farmers, seed suppliers and breeders will be improved. Safe material will lead to the use of a wider range of crops by farmers (EO2). ROTATES will create a policy framework for a system of phytosanitary certification for MRT crop seeds showing how the material can be safely shared between countries, as well as policy recommendations on supporting measures for MRT crop adoption both locally and in Europe, which will provide policymakers with evidence for the environmental benefits of incorporating MRT crops in agrosystems (EO3). ROTATES will provide information and recommendations to farmers and livestock producers on the use and on the processing of feed made from MRT crops (EO5). This is intended to increase knowledge on crop development, management and conservation of MRT crop by-products in the form of technical and information guidelines and brochures (EO6). In this way, they will also become familiar with the nutritional and sensory characteristics of the livestock products obtained.	
KPIs: One policy framework, 4 Policy briefs, Guidelines for MRT use as food and MRT processing; a Phytosanitary certification system for the seeds of at least two MRT crops.	
Direct beneficiaries: Policymakers; Local decision-makers; Farming associations; Chambers of commerce; (Agrifood) Industrial decision-makers.	

Methodological guidelines for MAA and MRT-focused LLs; Recommendations for MAA and MRT-focused LLs	EO2; EO3; EO6
ROTATES will establish a European multidisciplinary network of LLs on MRT crops involving all the actors of the value chain to enable the sharing of knowledge, data and material at international scale. For the first time, it will have brought together actors from continental countries and outermost regions, involving them in promoting MRT crops (EO2, 3, 6). Knowledge exchange will be enabled through workshops and other events gathering all the members of the network and sister projects, to create a sustainable structure and improved understanding among MRT value chain stakeholders for better feedback to policy makers and better interconnection between EU continental and outermost regions. Stakeholder involvement will result in higher societal acceptance of the solutions identified during the project. For an impact within the lifetime of the project and beyond, ROTATES will implement new co-creation approaches under the umbrella of MAA methodology, horizontally and vertically at local level, for the pilot LLs. Guidelines and a MAA strategic framework for the creation and running of LLs will be disseminated across Europe, supported by specific events (EO2, 6). KPIs: 100+ living labs on new management practices; one network on living labs on MRT crops; A total of 250 stakeholders engaged, ensuring balanced representation with at least 50 from each helix from quintuple helix model. At least half the pilot LLs will evolve into permanent, stable, and potentially certified LLs. Direct beneficiaries: Farmers and breeders; the Agri-food industry, Agricultural advisors, the Scientific community, Policymakers, Extension services (AKIS), Civil society, LLs, Other EU projects and initiatives.	

2.1.2 Uniqueness of ROTATES towards wider impacts

<i>Practices in agriculture and forestry will be developed and improved to support and make sustainable the use of biodiversity and a wide range of ecosystems services.</i> ROTATES will develop and promote agroecological practices in both organic and conventional systems, by providing a basket of alternative and diversified cropping systems, integrating MRT crops in European mainland and outermost regions. Our activities will lead to crop diversification that encourage the use of diverse MRT varieties, intercropping and crop rotation to improve climate resilience, reduce pest outbreaks, and improve soil health. We will promote agroforestry systems in pLL4-France that incorporate trees and MRT crops in agricultural landscapes to enhance agrobiodiversity. Through the activities which will be implemented in the two emblematic rainforests, The Amazon (French Guiana) and the Congo (Republic of the Congo), we will contribute to Sustainable Forest Management (SFM). We will produce evidence of actions that increase agrosystem biodiversity and operationalize European ambition to only import deforestation-free products from the two countries (with lessons that can be applied to other forested countries that trade with Europe). We will contribute to increasing agro-food biodiversity and to the objectives of the Green Deal and the Farm to Fork strategies. <i>Biodiversity, ecosystem services and natural capital will be mainstreamed in the society and economy: e.g. they will be integrated into public and business decision-making; approaches for enabling transformative changes to tackle societal challenges will be built including by deploying nature-based solutions (NBS).</i> ROTATES will invest in Capacity Building, by providing training programmes for farmers, food and feed actors and the different stakeholders of the value chain, on sustainable practices, agrobiodiversity management, food and feed processing and on monitoring ecosystem services. Planned dissemination actions will play a pivotal role in sharing knowledge, data, and innovations. It will ensure Multi-Stakeholder Collaboration so that actors have a voice in decision-making processes and benefit from the sustainable management practices, provisioning and regulating services. Consumers will be included in the different stages of the project as actors of the value chain. Policy framework for phytosanitary certification of seed systems for MRT crops; Policy Recommendations for MRT along the value chain; Guidelines for MRT development, added value and adoption will be delivered by ROTATES. During training sessions, messages will be delivered on the importance of including agrobiodiversity in decision-making processes. We will deliver six business case studies on raw and processed MRT for food and feed produced in conventional and organic systems and in this way, improve the understanding of the economic value of MRTs. Mainstreaming of agrobiodiversity through the integration of MRT crops and the ecosystem services provided will also be conducted through international cooperation and partnerships. We will create a network on MRT crops connecting different European countries. This partnership will include continental Europe and outermost EU regions where activities will be conducted in a balanced manner.

2.1.3 Scale and significance

Agriculture occupies more than 38% of the world's terrestrial surface⁵⁸ and is the main driver of an estimated 30% of biodiversity loss worldwide⁵⁹. MRT crops have the potential to reduce biodiversity loss by 2035, to preserve agrosystems, and to enhance conservation and restoration actions.

ROTATES will reach out to 40,000+ farmers (80% conventional and 20% organic farmers). More than 200 farmers will be involved in testing and capacity building actions. We expect that 50%+ of the farmers who are directly involved in the project will grow MRT crops and adopt new farm management practices after the end of the project. ROTATES will increase the adoption of MRT crops in European agricultural systems and support organic farming. Specifically, we will:

- Quintuple the use of MRT crops produced in Europe in agrosystems and in food and feed value chains by 2035.
- Triple per capita consumption of MRT-based food in Europe compared with current levels by 2035, with benefits for public health.
- Double the proportion of production in organic farming for MRT crops which will reduce the use of chemical pesticides and fertilisers, reduce nutrient losses, and increase water use efficiency by 15% by 2035.

2.1.4 Identified requirements and potential barriers

Environmental barriers: Extreme weather conditions or unforeseen agricultural effects of climate change. Mitigation can be achieved through collaboration with other sites and farms as well as extension of the duration of the project. **Political barriers:** In some countries there is a general low level of agricultural innovation and government subsidies that do not support agroecology. Discussions with policymakers should lead to different subsidy measures.

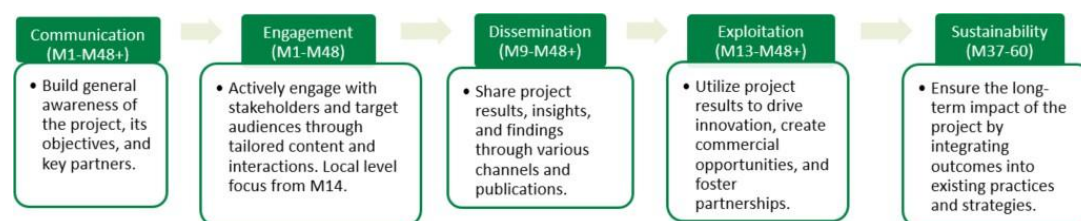
Legal obstacles: New regulations or restrictions on the distribution of genetic material may be introduced during the lifespan of the project. If adaptations are needed, they will be implemented.

Economic obstacles: (i) The current higher cost of cultivation of MRT crops than of other crops and the integration of these new species into culinary and food habits could be problematic. However, the aim of agronomic research will be both to increase crop yields and reduce production costs. The LLs will highlight the healthiness of the MRT products and how they can be incorporated in dietary habits. (ii) competition for the land between agriculture and urbanisation, (iii) push for cash crops and limited public and private support for MRT crops, (iv) MRT crop shelf- life, which is very important for market access and raw material availability, could be a problem. The consortium will network with ongoing European and international projects on this topic to implement the latest innovations.

Cultural barriers: The cultivation of certain MRT crops may be completely unknown to farmers. The implementation of the MAA approach, showcasing of the benefits of MRT crops and their promotion should help avoid this issue. Additionally, the fact that some of the pilot LLs will be located in regions where the minor crops were grown in the past will also facilitate the re-emergence of these crops.

2.2 Measures to maximise impact - Dissemination, exploitation and communication #@COM-DIS-VIS-CDV@#

To support the pathway to impact, ROTATES will develop a comprehensive and innovative strategy for the dissemination, exploitation, and communication (DEC) of the project outcomes. This strategy will be carefully crafted to engage a wide range of stakeholders, including farmers, industry partners, policymakers, and the consumers. To this end, five levels of DEC actions have been identified and are summarised in the Graph below.



2.2.1 Target group and stakeholders

A preliminary stakeholder analysis has already been conducted to assess the primary target groups to get involved to achieve effective communication, use and dissemination of the project results, and these stakeholders were grouped in the five categories of the 5H model: academia, industry, government, civil society, and environment. This analysis is summarised the following table, specifying the key actors, the proposed value, and the relevant Key Exploitable

⁵⁸Ramankutty et al. *Global Biogeochemical Cycles* 2008

⁵⁹Sala et al. *Science* 2000

Results (KER) detailed in §2.2.2, for each Helix.

5H	Key Actors	KERs
Public sector	National/regional agricultural ministries; Environmental protection agencies; Local governments; Research funding bodies; Regulatory bodies; Policy makers, Public Agricultural advisors.	KER 6, 7, 8, 9, 10
Industry	Agricultural businesses; Private agricultural advisors, Food/Feed (processing) companies; Seed producers; Genebanks; Innovation hubs; Exporters and importers.	KER 3, 4, 5, 8, 9, 10
Academia	Research institutions: Agricultural and Environment universities; Socio-economic departments, Food science, Feed science, Agricultural colleges; Agri-food vocational educational institutes.	KER 8, 9, 10
Civil society and users	Farmers and breeders; Citizens and consumers; Agricultural cooperatives and farmer associations; Consumer advocacy groups; NGOs; Community Groups; local rural communities.	KER 1, 2, 8, 9, 10
Environment	Environmental Organisations; Sustainability experts; Climate Research Institutions.	KER 2, 6, 7, 8, 10, 11

2.2.2 Key results – protection and exploitation

ROTATES outcomes and impacts will be achieved via the delivery of KERs that represent the starting point of the various pathways to impact through their DEC. The table below presents a preliminary analysis of the expected results with high exploitation potential and that contribute directly to the impact of ROTATES. For each KER, the table identifies: (i) the most relevant target groups, (ii) the partner(s) with ownership, and interested in further KER exploitation (Expl); (iii) the applicable protection measure and an exploitation route; (iv) the relevant WP.

KER1. Description of cropping systems (10+), Scientific publications with open data, analytical workflows	CIRAD, INIAV, UMa, INRAE, CUT, PNKH, UMNG, ENoLL
C: Farmers and breeders; A: Academia	WP3, 4, 6
Protec/Expl: Copyright, CC-BY- 4.0, workflows: GPLv3 Further research and MRT promotion will be done by researchers and farmers' associations, agricultural advisers. The farmers will be using the description of the cropping systems to be implemented. Scientific data will be transformed and translated into improved training, practical demonstrations and guidelines for farmers to support adoption. Protocols, datasets and scientific papers on social, economic, and cultural information for MRT adoption will be used throughout the value chain. The experimental economic surveys and protocols will be available via open access to the scientific community/academia for replication and improvement of the scientific methodology. Information on social, economic and cultural barriers and enablers of the development of MRTs will be freely usable by academia.	
KER2. New MRT genetic resources in genebanks (20+ per crop)	CIRAD, PNKH, UMa, INIAV
I: Farmers and breeders; A: Academia	WP3
Protec/Expl: The genetic material will remain under the ownership of the country where it is collected. The genetic material and the passport data will be provided to the genebanks. The data will also be deposited in the genebanks' data repository. Genebanks will supply the genetic material to research organisations, farmers and breeders free of charge. If replication is needed, costs will be charged with the cost of shipment. Further research and testing will be done by researchers, and further testing, and incorporation of the MRT will be done by farmers and breeders.	
KER3. New food products incorporating MRT crops	CREA, UMa, CIRAD
I: Agrifood industry; Pasta-bread-ice-cream companies, A: Academia, I: Farmers and breeders	WP5
Protec/Expl: New food products will reach the lab experimental stage and the validation stage in pilot plants belonging to the project partners. Patent or other IP protection such as utility model will be considered (probably after the project ends). Research publications will be produced for open access with an open CC license. Knowledge will be used for further research by the Research & Innovation departments of companies. After scientific publications, public manifestation of interest for the commercial exploitation of the results will be opened for the industry. Third parties can also express their interest in exploiting the food formulations (pasta, bread and ice-cream companies will be contacted – such collaboration is already underway, with for example, Barilla, Fornovecchino, Verdi Terre, Eocene, Pirro, FRIIC – Gelados de Fruta da Madeira.	

KER4. Feed processes of MRT by-products	ICIA, UAc, AUA
I: Agrifood industry, A: Academia, I: Farmers and breeders	WP5
Protec/Expl: Patents or other IP protection such as utility models will be considered (probably after project ends). Research publications will be produced for open access. Knowledge will be used for further research by researchers on MRT processing. After scientific publications, public manifestation of interest for the commercial exploitation of the results will be opened for the industry. Feed companies can then do their own testing and put the products on the market. Farmers and breeders will be integrating MRT in their livestock feeding practices.	
KER5. Training material, web-based content	All and ENoLL, ITC
A: Academia, I: Agrifood industry, E: Environment, I: Farmers, G: Government, C: Civil society & Users	WP1,2,4,5,6,7
Protec/Expl: Copyright (diagrams etc), CC-BY-NC-SA 4.0 Knowledge will be used in further training courses and educational programmes (vocational, adult education and master classes). Farmers will be using the description of cropping systems to be implemented for their own purposes. The training material will remain available on the project's website and Horizon Results Platform.	
KER6. Policy framework and recommendations for phytosanitary certification of seed systems for MRT crops	UL, INRAE, INIAV, UAc, CUT
G: Policymakers; relevant authorities	WP3
Protec/Expl: Copyright, CC-BY- 4.0 Evidence-based policymaking - where local government and relevant national authorities will be applying the framework. KER2 will serve as evidence.	
KER7. Policy recommendations /briefs for MRT along the value chain	All partners
G: Policymakers; local decision-makers; farming associations, I: Industry, C: Civil society – customer groups	WP2
Protec/Expl: Copyright, CC-BY- 4.0 Evidence-based policymaking – where value chain members, e.g. industry and farmers' associations will be applying the findings. The policy recommendations for the implementation of strategic interventions and concrete solutions (e.g. labels, incentives, types of investment, market targets, types of partnerships) to foster the development of MRTs will be used for implementation.	
KER 8. Business case studies and guidelines for MRT development, added value and adoption	CIRAD, ICIA, UAc, AUA, ENoLL and ITC
All targets, G, I, A, C, E	Relevant WP
Protec/Expl: Copyright, CC-BY- 4.0 Business case studies will be made available and presented to the farmers and intermediaries in LLs and other innovation hubs and clusters. The case studies will include a complete diagnosis of the challenges to and opportunities for the development of MRTs in their area. Guidelines for the implementation of strategic policy actions and concrete solutions (e.g. labels, incentives, types of investment, market targets, types of partnerships) to foster the development of MRTs will be identified for implementation by governments (G), industry (I) and civil society (C) and environmental organisations (E).	
KER 9. Methodological guidelines for MAA and MRT-focused LLs and Recommendations for MAA and MRT-focused LLs	ENoLL, ITC
All targets, G, I, A, C, E	WP1, WP7
Protec/Expl: Copyright, CC-BY-NC-SA 4.0 Knowledge will be used in other living labs and other agri-food projects implementing MAA; Results will be incorporated in internal service portfolios of ENoLL and used in future training programmes by ENoLL and ITC. The guidelines will be available on the project website and Horizon Results Platform. Evidence-based decision-making and use of recommendations in other Living Labs and other agri-food projects that implement MAA; Support the positioning of LLs in the EU agenda to promote co-creation, evidence-based open innovation and growth of initiatives at European level adopting the MAA and LL approach.	

2.2.3 Dissemination & communication (D&C)

The D&C strategy in the DEC plan will be designed in line with the 5W+1H approach to effectively reach out to target groups and stakeholders, thereby prolonging the impact beyond the end of the project, making project results known to those who could benefit from them or who could support further development and deployment, thus increasing sectoral coverage and use of the proposed solutions (“**Why**”). The plan will further define the key results and outputs of the project (“**What**”), map the target groups in more detail (“**Who**”), identify the most appropriate channels (“**Where**”), and define the specific activities to undertake (“**How**”) and their time plan (“**When**”).

Dissemination will actively support and promote the future successful exploitation of the project results, whose uptake and use will be measured during and after the end of the project. The plan is based on a multichannel strategy, combining offline and online communication, with specific and ambitious targets and metrics/KPIs, and a detailed plan included in the DEC plan (D7.2) to be submitted at the first semester of the project.

Dissemination	Goals, target indicators
ROTATES events	Final ROTATES conferences [1] and hands-on workshops [1+] on project results will be organised alongside a major EU-level event (see below), to promote awareness and create opportunities for collaboration at European level
MAA events	The co-creation of events will engage members from all helixes in the pilot regions
Capacity-building events	Training materials will be openly available potentially in more languages depending on the final capacity-building programme in WP4, 7
EU/International-level events	Participation in EU-level events with poster, stands, and/or presentations for promotion of project and results and for networking [10+]. <i>Forum for the Future of Agriculture (ForumforAg), European Biodiversity conference, European Mission Soil Week, Open Living Lab Days, ISPIM Conference, EU Agri Research conference, etc.</i>
Scientific publications	The results of the project will be disseminated to the scientific community by means of publications in open access and high impact journals [20+], presentations at conference and in conference proceedings [10+], as well as in open and FAIR datasets [5+]. <i>Indicative conferences: International Triennial Symposium of the International Society for Tropical Root Crops (ISTRC); Plant & Animal Genome conference (PAG); The European Federation of Food Science and Technology (EFFoST); European Federation of Animal Science (EAAP), European Association of Agricultural Economists (EAAE), AI4SoilHealth.</i> <i>Indicative journals: Nature Food (IF 23), Molecular Plant (IF 17.1), Nature Plants (IF 15.793), Science Advances (IF 11.7), LWT (IF 6), Field Crops Research (IF 6.1), Food policy (IF 6.8), Agricultural Economics (IF 4)</i>
Industry meetings and presentations	The results of ROTATES will also be disseminated to the industrial community through presentations, articles, and information to be disseminated by all ROTATES partners involved in industrial networks as well as at key relevant events. <i>Indicative industry meetings: Paris International Agricultural show, The International Pastaria festival, Food Technology and Processing, International Production & Processing Expo (IPPE)</i>
Policy events and meetings	At EU level, the public sector will be addressed through policy recommendations and replicable guidelines to stimulate adoption of the results of the project and effective uptake across Europe. Joint statements with other Soil Mission and Agrifood projects will help disseminate the recommendations and the guidelines.
Public sector at local level	Specifications for dissemination activities for the private sector at local level will be defined by the individual pilot LL/value chains in alignment with the DEC plan. Other European LLs and farming associations will support the spread of MAA methods, best practices and training materials.
Local community	Each partner will undertake local dissemination actions, the specifications about these actions will be defined in the individual action plans of the pilot LL/value chains <i>e.g. AISTEC _Associazione Italiana Scienza e Tecnologia Dei Cereali_ Meeting (held every two years – all countries will identify).</i>
EC Tools	Horizon Results Platform, Zenodo and EU Academy platforms will be used for wider dissemination of the key reports, recommendations and training materials. The Horizon Results Booster services may further support the dissemination and exploitation of the 6 business case studies.
Joint dissemination actions with clustering projects	All projects funded under the topic will join forces and implement clustering actions (joint campaigns, attendance or organisation of events, joint statements etc) <i>SOILL and PREPSOIL (Mission Soil), FARCLIMATE (Mission Climate), CLIMAGEN (NEB), BlueMissionMed (Mission Ocean), Agroecology partnership</i> Dedicated resources and tasks (T7.4) are envisaged for cross project cooperation.

The DEC actions will include the four sister projects funded under H2020, and the sister project which will be funded by the same call (HORIZON-CL6-2024-BIODIV-02-3-two-stage), several European initiatives funded under Horizon Europe, PRIMA, etc., as well as with other projects implemented in the outermost EU regions and in Africa.

Communication. Aside from dissemination actions to boost exploitation and impacts of ROTATES, the project and its results will be presented to a wider European and international audience. See a short overview below:

Comm. tool	Target communities	Performance indicators
Public website	General public, investment groups, end-users	+2000 visits by the end of the project
Social media	LinkedIn, X, Instagram and YouTube relevant stakeholders	2000+ followers on social media investment groups, end-users, public sector, related initiative, and other EU funded projects
Information material	All external stakeholders	Printed: at least 15 events where the roll-up is displayed, and 500 flyers distributed.
Press releases and newsletter	General public, investment groups, end-users	Biannual press-releases and quarterly newsletters
Partners' websites and newsletters	Existing networks	At least 30 links/articles by the end of the project
Newspapers, magazines, e-media	All external stakeholders	5+ articles concerning the project published in non-specialised publications
Videos and interviews	All external stakeholders	5+ videos and 3 specific videos on the new recipes based on MRT crops, and interviews on the website and in the media

2.2.4 IP management

The knowledge generated in the project will be managed according to the rules of the DESCA Consortium Agreement (CA), which will be signed at the beginning of the project (WP8). The CA will address background and foreground knowledge, ownership, proprietary third-party components, protection, use and dissemination of results and access rights. In line with the CA, ROTATES background and foreground IP will be managed under Task 7.3 "Exploitation, replicability, recommendations and policy contribution". The most appropriate protection strategy and measures will be selected during the project, but ROTATES partners want to pursue open science solutions. Preliminary plans for result protection are included in **Key results – protection and exploitation**. The choice will depend on the nature and specific characteristics of the results under consideration and the objectives of the IP owner(s) at the time. The Data Management Plan will explain how research data will be managed in accordance with the FAIR principles. ROTATES supports open access to scientific publications, models, tools and data. In this way, the results will be made available for further research. The background information and knowledge contributed to the project by each participant will be listed in the CA. Background information will be made freely available to other participants for the implementation of the project tasks. Foreground knowledge is the property of the partner who created it. Each partner is responsible for ensuring that its obligations under the CA in relation to results are met by planning with any third parties who may claim rights to them. Results produced jointly by two or more participants will be shared between the participants who carried out the work. The terms of joint ownership, protection and sharing of ownership, and the costs of any protection, will be agreed in writing in a joint ownership agreement. Each participant is responsible for investigating the possibilities of protecting results that can be commercially or industrially exploited. In deciding on protection, the participant must take into account its own legitimate interests and the interests of the other participants.

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2.3 Summary

KEY ELEMENT OF THE IMPACT SECTION

SPECIFIC NEEDS	EXPECTED RESULTS	D & E & C MEASURES
• Need for MRT cropping systems, supported by data, knowledge, tools, varieties, to shift production towards agroecological and climate smart systems, thereby reducing pesticide use and nutrient losses. • Need to diversify farmers' income by increasing the value of MRT by increasing the demand for them throughout the value chain, and by mitigating the factors that hamper their use. • Need for tailored recommendations for different stakeholders (including policymakers) on the safe exchange of MRT material, on mainstreaming agroecological practices, on developing value chains and on novel processing of MRT. • Need to strengthen European MRT capacity building in breeding, farming and in food and feed development. • Need for a coordinated network for MRT crops in Europe, as the current disconnect among seed companies, processors, farmers, and researchers is hindering progress and innovation.	• Basket of alternative and diversified cropping systems that maximise ecosystem services, including MRT with 10+ recommendations for agroecological cropping systems • Structured open access data, scientific findings, prediction models and new tools to deliver the full potential of MRT crops • New food (3 categories of healthy food) and feed products (MRT processing for 4 livestock categories) incorporating MRT crops; animal feed incorporating MRT crops; Up-cycling of MRT wastes • 6 Business case studies showing how to develop MRT-based value chains _raw and processed food and feed produced in organic and in conventional agrosystems • Policy framework for phytosanitary certification of seed systems; Policy Recommendations for MRT along the value chain; Policy briefs, Guidelines for development of the MRT value-chain. • Methodological guidelines for MAA and MRT-focused LLs; Recommendations for MAA and MRT-focused LLs	Exploitation: Commercial exploitation of feed and food products; Incorporation of MRT crops in agrosystems on other farms in Europe; Advancing knowledge in other agro-food innovation hubs and in the scientific community; Advancing policies and evidence-based policymaking on phytosanitary certification of seed systems for MRT crops and for the development, added value and adoption of MRT. D&C to farmers, agri-food industry and different value-chain actors: workshops, agricultural and industrial fairs; living lab events, ROTATES news and events, capacity- building programme. Meetings with bigger food/feed companies. D&C to scientific community: scientific articles; presentations and posters at scientific events; datasets, GIS, dashboards, workflows, working groups and advisory group work in projects, final conference. D&C towards policymakers: replicability, promotion of guidelines at events and meetings, working groups and expert groups, roundtables Communication towards society: website, social media, press, magazines, promotion of text/video/audio material, partner newsletters.

TARGET GROUPS	OUTCOMES	IMPACTS
• Farmers, breeders, seed producers, Genebanks, Agricultural advisors • Agricultural research organisations, colleges and universities • Food and feed (processing) industry. • Consumer advocacy groups; Agrifood NGOs; Community Groups; local rural communities • Regulatory and policymaking bodies; policymakers, research funding bodies; environmental and agri-food agencies • End-users, citizens	Deepened understanding of key drivers and solutions for overcoming barriers to adopting MRT crops and crop diversification. Expanded crop diversity, rotation strategies, and advanced agroecological management practices in agrosystems. Increased genetic diversity in genebanks with pathogen-free materials readily accessible to farmers, seed suppliers, and breeders. Enhanced awareness and expertise among farmers and breeders in using MRT crops, effective crop rotation and intercropping, and distribution of safe genetic material. Strengthened local value chains through the inclusion and processing of MRT crops. Greater recognition of living labs, with pilot LLs progressing towards full compliance with ENoLL criteria.	Scientific impacts: Production of new data and insights to develop innovative MRT farming practices. Drive advances in breeding for climate resilience, pest and weed resistance, and market access. Inform future strategies for scaling MRT integration in European agrosystems. KER1; KER2; KER3; KER4; KER5 Economic and Technological impacts: Strengthen and diversify value chains for food and feed products, drive economic diversification and unlock niche market opportunities. Enhance agricultural resilience and productivity by empowering farmers with a broader crop portfolio supported by a robust MRT seed system. Accelerate economic growth and sustainability through increased technological investments in MRT crop research and innovation. KER3; KER4; KER8, Societal impacts: Enhance dietary diversity and sustainability, to improve human health and well-being. More effective informed policies to streamline the inclusion of MRT in existing farming systems and the exchange of MRT materials. KER1; KER6; KER7, KER9 Environmental impacts: Crop combinations supported by a clean MRT seed system leading to sustainable use of biodiversity and effective delivery of a wide range of ecosystem services; strategies for deforestation-free management practices. KER1; KER2; KER6

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3. QUALITY AND EFFICIENCY OF THE IMPLEMENTATION #@QUA-LIT-QL@##@WRK-PLA-WP@#

3.1 Work plan and resources

3.1.1 Overall structure of the ROTATES work plan

ROTATES's work plan is structured in WPs that give the actors (**WP1**) and the value chain (**WP2**) a central role to ensure the involvement of the stakeholders and users in co-creation, co-innovation and co-validation along the value chains, in WPs dedicated to the deployment of starchy MRT in the European agrosystems to ensure their benefits (**WP3, 4 and 5**) and a **WP7** dedicated to capacity building, dissemination, exploitation and communication to ensure the adoption of MRT crops beyond the lifespan of the project. To ensure efficient management of the project and that EU requirements are met, **WP8** will be dedicated to project management and coordination.

3.1.2 ROTATES planning

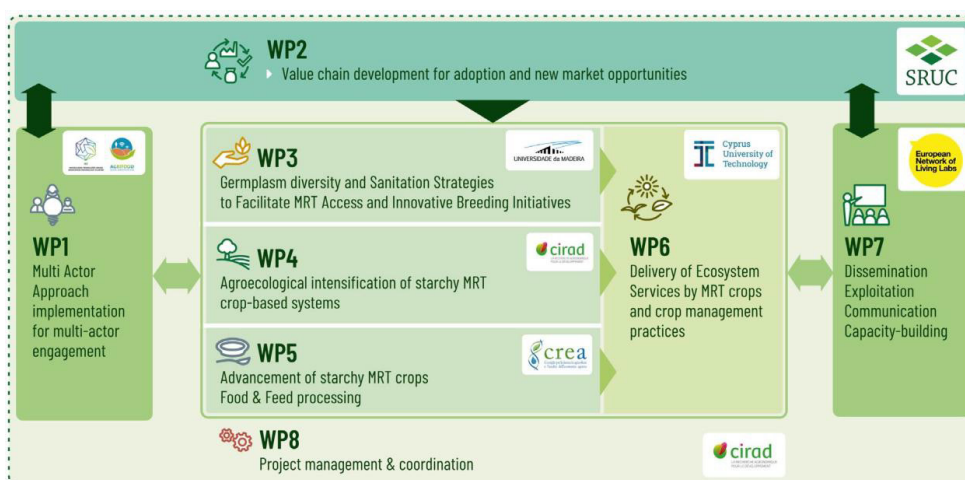


Table 3.1a: List of work packages

No	Work Package Title	No. Lead Particip.	Name Lead Particip.	Person-Months	Start Month	End month
1	Multi Actor Approach implementation for multi-actor engagement	15	ITC	69.5	1	48
2	Value chain development for adoption and new market opportunities	12	SRUC	55.6	1	48
3	Germplasm diversity and Sanitation Strategies to Facilitate MRT Access and Innovative Breeding initiatives	6	UMa	170.5	1	48
4	Agroecological intensification of starchy minor root and tuber crop-based systems	1	CIRAD	93.5	1	48
5	Advancement of starchy Minor Root and Tuber crops for Food and Feed Processing	9	CREA	112.5	1	48
6	Delivery of Ecosystem Services by MRT crops and crop management practices	4	CUT	78.5	1	48
7	DEC and Capacity building	14	ENoLL	52	1	48
8	Coordination and Management	1	CIRAD	51.5	1	48
			TOTAL	686		

Workpackages/Tasks	Planning		Year 1				Year 2				Year 3				Year 4												
	Leader	M Start	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	
Multi Actor Approach Implementation for multi-actor engagement	1	IC	M	48																							
	2	ENALL	1	12																							
	3	FC	10	48																							
	4	FC	13	48																							
1) MAA & L.L. methodological approach 2) Action plans and implementation 3) Monitoring & Evaluation	1	ENALL	1	12																							
	2	FC	10	48																							
	3	FC	13	48																							
	4	FC	13	48																							
Value chain development for adoption and new market opportunities	1	SRUC	1	48																							
	2	SRUC	1	9																							
	3	NRAE	10	25																							
	4	NRAE	24	42																							
1) Diagnostic of MRT value chains operational contexts 2) In-depth research on 6 representative MRT value chains 3) Development of strategic recommendations for the 6 MRTs value chains 4) Production of guidelines for MRTs' value chain development	1	SRUC	42	48																							
	2	SRUC	42	48																							
	3	SRUC	42	48																							
	4	SRUC	42	48																							
Germpasm diversity, Sanitation Strategies for MRT Access and Innovative Breeding	1	UMA	1	48																							
	2	UMA	1	36																							
	3	UMA	6	48																							
	4	UMA	3	48																							
1) Germplasm collection and biodiversity preservation of MRT 2) Deployment of advanced phenotyping methods and genetic studies 3) Clean seed system for sustainable production	1	UMA	1	48																							
	2	UMA	1	36																							
	3	UMA	6	48																							
	4	UMA	3	48																							
Agroecological intensification of starchy minor root and tuber crop-based systems	1	Grand	1	48																							
	2	Grand	1	12																							
	3	UT	5	48																							
	4	UT	5	48																							
1) Agronomic diagnostic of current MRT production systems 2) Design and on-station testing of alternative MRT-based cropping systems increasing water 3) Design and on-station testing of alternative MRT-based cropping systems promoting the 4) Design and on-station testing of alternative MRT-based cropping systems promoting zero- 5) On-farm adaptation and scaling of MRT-based cropping systems through Living Labs 6) Training farmers on the innovative cropping systems co-created and validated	1	IN																									

Table 3.1b: Work package description

Work package number	1
Work package title	Multi Actor Approach implementation for multi-actor engagement
Objectives ▪ Develop a framework to implement the MAA ▪ Implement the MAA, especially in pilot LLs and monitor their progress ▪ Establish a dynamic workflow for continuous communication between this WP and others to ensure the MAA at all levels.	
Description of work Task 1.1: ROTATES MAA & LL methodological approach (M1-M12; Leader: ENoLL (6PM), contributors: all Partners) Create a comprehensive and structured ROTATES methodological framework (D1.1) to effectively guide the MAA and LL approach throughout project implementation aimed not only at supporting the involvement of multiple actors in the co-creation process (MAA) but also structuring this approach in the context of a long-term and stable open innovation ecosystem (LL). This framework will detail the principles, methods, and tools required to implement the MAA within the project, with particular emphasis on pilot levels. In parallel, the framework will define the ROTATES monitoring framework, with detailed KPIs and structures to monitor the level of alignment to both MAA and LL methodologies, as well as the impact of these in the specific pilot/context. The creation of the framework will follow these steps: desk research and analysis of the literature, state of the art and tools; co-creation with WP and Pilot partners, including mapping stakeholders; validation of the approach and tools through workshops to increase acceptance; and preparation of iterative guidelines. Task 1.2: ROTATES MAA & LL Action Plans and Implementation (M10-M48; Leader: ITC (12PM), contributors: all Partners) Development of a dynamic and flexible workflow that fosters ongoing communication and collaboration within the project, ensuring the MAA principles are consistently applied at all levels through regular updates, feedback loops and adaptive management practices to maintain alignment and responsiveness to stakeholder needs and project goals: Deploy the MAA, especially in pilot LLs, with the definition of Action Plans by M14 for each Pilot and with the definition of key performance indicators (KPIs) against which progress and success will be monitored and measured, ensuring clear objectives and strategies; alignment of strategies during implementation to maintain coherence and focus; co-creation workshops in pilot LLs and ecosystems to engage stakeholders and obtain feedback; re-assessment of Action plans (M30) based on continuous monitoring to enable necessary adjustments and improvements. Task 1.3: Monitoring of the implementation of ROTATES MAA & LLs (M10-M48; Leader: ITC (14PM), contributors: all Partners) Continuous monitoring and evaluation of the pilot LLs' implementation of the MAA will be conducted to assess the effectiveness of the MAA, identify challenges, and collect feedback for iterative improvements. This process will include establishing the baseline to start the pilot LL (M12) with gathering initial data on current conditions, stakeholders' perspectives, and existing processes in the pilot areas; conduct an interim evaluation (M28) to assess progress compared with the established KPIs, identify challenges and barriers; and undertake a final evaluation report and make recommendations (M44, D1.2 in M46) for scaling up the MAA, addressing identified challenges, and enhancing future implementations.	
Work package number	2
Name of the work package	Value chain development for adoption and new market opportunities
Objectives ▪ Provide a diagnosis of the current situation of the use of MRT crops for food and feed in 10 countries ▪ Identify levers of the adoption of MRT crops by analysing opportunities and barriers along selected food and feed value chains ▪ Extract lessons that can be applied to the development of MRT value chains.	

Description of work

Task 2.1 – Diagnosis of MRT value chain contexts (M1-M9, Leader: SRUC (3PM), contributors: all Partners)

This will be a desk-based study based on secondary information concerning the economic, social and cultural barriers to and enablers of the uptake of MRTs at value chain level. It will be complemented with interviews with 5 to 10 key informants suggested by partners. The WP will provide background information for the 10 countries involved in the proposal (the crop and its use are indicated in brackets): Portugal (organic sweet potato - food), the Azores (conventional taro – feed), Cyprus (conventional taro – food), Guadeloupe (conventional yam – food), Canary Islands (conventional cassava – feed), Italy (One of the MRT grown in conventional system food), France (organic sweet potato – food), Greece (organic sweet potato – feed), French Guiana (conventional cassava – food) and the Republic of the Congo (conventional cassava – food). A report will present the diagnosis concerning all the cases (D2.1). These results will be shared with the scientific team and the Pilot Living Labs (LL).

Task 2.2 – In-depth research on 6 representative MRT value chains (M10-M25, Co-leaders: INRAE & SRUC (8PM), contributors INIAV, CUT, UAc, CIRAD, ICIA, CREA)

Detailed information will be collected for the six business case studies: **CS1)** Organic raw sweet potato for food products in Portugal; **CS2)** Conventional taro for feed in the Azores; **CS3)** Organic raw taro for food in Cyprus. **CS4)** Bread made from conventional yams in Guadeloupe; **CS5)** Conventional cassava for feed in Canary Islands; **CS6)** Pasta made in Italy from one of the four MRT crops in a conventional system. The specific information collected will depend on the characteristics of the case study concerned. The following work will be carried out: **Farmers' activities** - Focus groups including a deliberative choice experiment with farmers organised on site with 40 to 50 farmers per case study (CS1, CS2, CS3, CS4, CS5). This activity will coincide with the project activities of the pilot LLs. **Activities undertaken by food processors** - Semi-structured interviews with processors and a qualitative analysis for the case studies (CS4, CS5, CS6). **Consumer focus groups** - Five focus groups with 20 consumers to prepare the economic choice experiments and to perform a qualitative analysis (CS1, CS3, CS4, CS6). **Consumers' economic experiments** - Economic choice experiments with consumers (online questionnaire, choice experiment, laboratory experiment) and quantitative analysis to establish consumers' willingness to pay for MRT food products (CS1, CS3, CS4, CS6). **Activities undertaken by feed producers** - Semi-structured interviews with feed producers' customers for two case studies and qualitative analyses (CS2, CS5).

Task 2.3 - Development of strategic recommendations for the 6 MRT value chains (M26-M42, Co-leader INRAE & SRUC (8PM) in coordination with WP1 and WP7)

Based on the information and on analysis of the information produced in Task 2, a collection of 6 business case studies will be prepared (Deliverable 2.2). These business case studies, in coordination with partners with WP1 and WP7, will be presented to relevant stakeholders in the pilot Living Labs (LL) to highlight the potential and appropriate strategies to ensure the success of value chains and to collect feedback on results.

Task 2.4 - Production of guidelines for the development of MRT value chains (M43-M48, Co-leader SRUC & INRAE (8PM)) The guidelines will identify strategies and opportunities to promote the development of MRTs at the value chain level. These guidelines will contain key lessons drawn from the case studies and will target entrepreneurs, policy makers and other stakeholders (D2.3).

Work package number	3
Work package title	Germplasm diversity and Sanitation Strategies to Facilitate MRT Access and Innovative Breeding Initiatives
Objectives - Identify new MRT materials and valuable traits for breeding programmes and increase adaptation of cultivation to agroecological conditions, climatic constraints, and production quality. - Streamline the selection of MRT hybrids by identifying and combining favourable alleles, thereby enhancing the efficiency and precision of the breeding process. - Design integrated system approaches to eliminate and control seed-borne pathogens	
Description of work Task 3.1: Germplasm collection and characterisation for the conservation of rare and elite MRT crops (M1-M36; Co-leaders: CIRAD & UL (25,5PM), contributors: all Partners) A preliminary diagnosis identified 378 accessions held in genebanks in countries involved in the project. For an accurate diagnosis, a standardised questionnaire will be sent to the genebank in each partner country (CRB-PT⁶⁰,	

⁶⁰ [Centre de Ressources Biologiques Plantes Tropicales de Guadeloupe](#)

CRB-PP⁶¹, ISOPlexis⁶², CBA⁶³, BPGV⁶⁴, ARI⁶⁵, CCBAT⁶⁶, GGB⁶⁷) to collect information on MRT accessions and their characteristics. Farmers will also be surveyed to gather information on the traits and origin of the accessions, potential diseases and the conditions required for maintaining MRT accessions in the field. Germplasm will be collected and maintained in the field or *in vitro* by partners in the 10 regions targeted by the diagnoses. The collected MRT crop materials will be the subject of (i) phytosanitary analyses to detect the presence of seed-borne viruses by UL using high-throughput sequencing (on at least 100 accessions as individual or pooled samples and on uncharacterised samples from WP4) and standard targeted tests to evaluate the pressure and spread of diseases. In addition, the literature on sanitation protocols and pathogen diagnostic tests will be reviewed and included in the proposed system(s); (ii) lean phenotyping by **each partner**, focusing on basic morpho-agronomic traits and (iii) whole genome sequencing by **CIRAD** to assess the genetic diversity available in each country and to identify useful gene pools for crop improvement and for the new management practices. The data produced will be merged with the data produced in WP2, 4 and 5 to precisely identify the germplasm that is needed to fill the gaps that could be obtained from neighbouring countries or international genebanks, while accounting for spatial distribution, environmental conditions, agronomic performances and uses.

Task 3.2: Deployment of Advanced Phenotyping and Genetic Studies in Plant Breeding (M6-M48; Co-leaders: UMA & CIRAD (24,5PM), contributor: TP)

The germplasm resources gathered within Task 3.1, will be used to establish diversity panels in **YAM HUB** (Guadeloupe), in **TARO HUB** (Madeira), in **CASSAVA HUB** (French Guiana), and in **SWEET POTATO HUB** (Portugal). Specific traits for abiotic challenges, i.e. the best MRT plant health, quality of production, and water use efficiency under extreme climatic conditions will be investigated in Guadeloupe and Madeira. In French Guiana, the cassava diversity panel will be phenotyped for nutrient absorption and yield. In addition, specific traits for biotic challenges concerning weed resistance will be assessed across all panels. The sweet potato diversity panel will be phenotyped for pest resistance in Portugal. Detailed phenotypic data from these 4 diversity panels will be collected and analysed using remote sensing technologies, nondispersive infrared technology, non-destructive chemical analysis technology (near-infrared spectroscopy, NIRS), and laboratory assessment of whole-plant nutritional quality and root-to-shoot signalling, among others. Genotyping data of the 4 diversity panels will be combined with large-scale phenotyping data for genome-wide association studies and genomic prediction. Similarly, NIRS data will be used to develop predictive models through phenomic selection. The data produced will be merged with the data produced in WP 4, 5 and 6 to identify varieties whose traits are adapted to the agricultural management practices developed, ecosystem resilience, and food and feed quality production.

Task 3.3: Developing an integrated system approach for the elimination and control of seed-borne pathogens (M3-M48; Leader: UL (15PM), contributors: UAc, INRAE, INIAV)

The development of an integrated system approach will rely on desk studies, data generation and stakeholder consultation, including experts. Task 3.3 will develop a proof-of-concept for sweet potato, the fastest growing ROTATES crop. First, the desk study will review the abundant literature on seed systems with a double lens: (i) existing seed systems for potato in the EU and (ii) previously proposed seed systems at local scale or in the Global South. This will define a “playground perimeter” for stakeholder and expert consultation (relying on national and regional networks of partners) as seed systems must be adapted and compatible with local conditions and stakeholders. In addition, the integrated approach will rely on the epidemiological data collected in Task 3.1 and in WP4 that will be analysed together with literature on seed-borne pathogens. Furthermore, at least two sanitation protocols, for sweet potato and yam, will be applied to 10 accessions of each crop to produce elite clean materials that will be deposited in genebanks and be available to farmers and breeders.

Work package number	4
Work package title	Agroecological intensification of MRT crop-based systems
Objectives	To develop a basket of novel management practices/alternative cropping systems that incorporate the best suited varieties, minimise the use of pesticides while simultaneously maximising ecosystem services.
Description of work	Task 4.1: Agronomic diagnostic of current MRT production systems (M1-M12; Leader: CIRAD (10PM), contributors all Partners) This WP will start with a diagnostic of current MRT crop-based systems in the six pilot LLs. This diagnostic, which will be conducted in parallel to diagnoses in other WPs, will inform the design of alternative cropping systems

⁶¹ [Centre de Ressources Biologiques Plantes Pérennes de Guyane](#)

⁶² [Banco de Germoplasma - Universidade da Madeira](#)

⁶³ [Biotechnology Centre of Azores](#)

⁶⁴ [Banco Português de Germoplasma Vegetal](#)

⁶⁵ [Agricultural Research Institute of Cyprus](#)

⁶⁶ [Centro de Conservación de la Biodiversidad Agrícola de Tenerife](#)

⁶⁷ [Greek Genebank](#)

(specifications/criteria for these systems), serve as a baseline against which the performance of alternative cropping systems will be evaluated, and will allow for the delineation of spatial recommendation domains and farm types for up-scaling of the most promising alternative cropping systems developed in this WP. The first step in the diagnostic will be documenting current practices using existing data complemented by a participatory diagnostic conducted in pilot LLs. To better understand how representative the six selected pilot LLs actually are and to inform future scaling strategies, we will delineate spatial recommendation domains using open-access spatial products and empirical data where available. Similarly, using survey data, we will create a farm typology that captures the diversity of farms, and the constraints and opportunities specific to each type of farm. We will also analyse yield gaps in MRT crops (in the pilot LLs where the crop concerned is already present) using survey data and crop cut data, to quantify yield gaps and to identify the determinant of the gaps.

Task 4.2: Design and on-station testing of alternative MRT-based cropping systems to increase water use efficiency in irrigated systems (M5-M48; Leader: CUT (6PM), contributors: INRAE, UMa)

Informed by the diagnostic stage, crop rotations and crop associations combining MRT crops with legumes, cucurbits, vegetables and/or fruit trees will be tested on-station to increase water use efficiency, in Cyprus, France and Madeira where this was identified as a major challenge. Both the improvement of systems that already include MRT crops (taro in Cyprus and Madeira, sweet potato in France), and the introduction of MRT crops in systems where they are currently absent (yam in Madeira) will be considered in this task. The initial selection of crops for rotations and intercropping will be based on the findings of the diagnostic (Task 4.1). Using an iterative approach, the performance of the entire system (all crops included in rotations and associations) will be evaluated each season and the systems will be refined for the following season. The main performance indicators will include yield, and apparent water use efficiency (kg of harvested product by mm of rain-water + irrigated water). Ecosystem services will also be evaluated in these trials (strong linkage with WP6).

Task 4.3: Design and on-station testing of alternative MRT-based cropping systems promoting the biocontrol of pests and weeds (M5-M48; Leader: INIAV (8PM), contributor: INRAE)

In France and Portugal where pests and weeds represent a major challenge to production, MRT crops in rotations and in associations, will be tested on-station. This task will focus on sweet potato in both countries. As in Task 4.2, crop rotations and crop association combining MRT crops with legumes, cucurbits, vegetables and/or fruit trees will be designed and tested on-station in at least 2-3 soil types during 3 years. As for Task 4.2, an iterative approach will be used. The main performance indicators will include yield, weed pressure (estimated through remote sensing or android apps), weed abundance and diversity (measured in quadrats), pest abundance and density (assessed through colour traps and net swapping), incidence of viruses transmitted by insect pests (assessed through PCR and/or ELISA), and incidence of soil borne diseases (assessed through morphological observations). As in Task 4.2, ecosystem services will also be evaluated in these trials.

Task 4.4: Design and on-station testing of alternative MRT-based cropping systems promoting zero- deforestation (M5-M48; Leader: UMNG (7,5PM), contributor: PNKH)

Cassava-based cropping systems specifically designed to reduce deforestation will be tested on-station in the Republic of the Congo and in French Guiana. Cropping systems adapted to both recent clearing (e.g., short improved fallow using *Cajanus* before planting cassava) and to older fields (e.g., association of perennial crops e.g. *Stylosanthes*, *Sesbania*, – and annual crops e.g. *Centrosema*, *Pueraria*) will be developed. The same iterative approach as in Task 4.2 and Task 4.3 will also be used. The main performance indicators will include yield, weed pressure (estimated through remote sensing or android apps), weed abundance and diversity (measured in quadrats), and soil health (using BioFuncTool®) (linked to WP6). As in Task 4.2 and Task 4.3., ecosystem services will be evaluated in these trials.

Task 4.5: On-farm adaptation and scaling of MRT-based cropping systems through pilot Living Labs (M13-M48; Leader: CIRAD (9PM), contributors: CUT, PNKH, INRAE, UMNG, UMa, INIAV)

On-farm research will be conducted in parallel to on-station research in the 6 pilot LLs: **pLL1-Cyprus, pLL2- France, pLL3-Portugal, pLL4-French Guiana, pLL5-Madeira, pLL6-Congo**. While the purpose of on-station research will be to collect detailed measurements in controlled environment in order to better understand the underlying mechanisms (e.g., water and nutrient dynamic, pests and weeds dynamics), the purpose of on-farm research will be to assess how innovative cropping systems fit in diverse farm contexts, and to understand what adaptations and interventions will be required for their wide adoption. In each of the 6 pilot LLs targeted by this WP, on-farm research will take place in a network of at least 10 farms, in organic and conventional system, selected from the typologies delineated in Task 4.1., to ensure their representativity. Based on the data collected, a trade-off analysis between multiple criteria (productivity, labour, provision of ecosystem services, etc) for the most promising alternative MRT-based cropping systems will be conducted for the various farm types and spatial domains identified. Finally, a participatory holistic impact assessment of the most promising alternative MRT-based cropping systems will be conducted in the 6 pilot LLs.

Task 4.6 Training of farmers in the use of novel agroecological practices co-created and validated (M13-M48; Leader: UMa (8PM), Contributors: CUT, PNKH, INRAE, UMNG, INIAV, CIRAD)

Field demonstrations will be organised regularly for farmers and growers in each pilot LL. Three face-to-face training workshops per pilot LL will be organised with the support of WP1 and WP7 involving at least 25 participants including farmers, agricultural advisors and representatives of farmer associations. The first demonstration will be scheduled following the agronomic diagnostic and at the beginning of the process of co- design. A second demonstration at mid-term will focus on demonstrating the practices under development on-station and on-farm and collect farmers' feedback to make any necessary adjustments. The final field demonstration will provide training on the most promising innovative agroecological practices developed in POTATES – including on the provision of

Work package number	5
Work package title	Advancement of starchy Minor Root and Tuber crops Food and Feed Processing
Objectives	▪ Assess, valorise and implement MRT crops as functional food and feed for different animal species ▪ Assess the aptitude of MRT crops for processing in pilot plants ▪ Evaluate livestock products obtained with diets based on MRT crops ▪ Assess consumer acceptance of MRT-based foods ▪ Develop novel food and feed products for industry, farmers and the general public

Description of work**Task 5.1 – Physical and phytochemical characterisation of MRT. (M1-M24; Leader: INRAE (4PM), Contributors: CIRAD, UAc, UMa, CREA, ICIA, AUA)**

The species and the genotypes most suitable for food and feed will be identified from both diagnoses and consultation. The diagnoses will involve multiple methodological approaches: (i) consultation and analysis of available and relevant literature and databases on the use of tubers for food and whole plants for feed; (ii) detailed analysis of the regulations in force; (iii) search for complementary information in government agricultural services and biodiversity plant conservation centres; (iv) implementation of field and on-farm visits and surveys of MRT target species (M1-M12). After which raw materials and flours from different selected genotypes will be characterised in terms of **physical and nutritional parameters** (M6-M24). **Rheological and technological characterisation** (pasting properties (CIRAD), **falling number and farinograph** (CREA), non-destructive measurements of native and blended flours (MRT/low gluten index cereals, i.e., einkorn/ gluten free-cereals, i.e., sorghum and corn), obtained using different drying pre-treatments (vacuum freeze-drying, hot air, CREA/CIRAD/UMa) and by different pilot plant milling procedures (ultra-fine grinding, stone millers, CREA) will be performed. **Nutritional** (proteins (CREA), fibres (CREA), starch (CREA), *in vivo* **glycaemic index** on mice (UA)), small sugars-amino acids-organic acids (AU), bio-functional (phenols and polyphenols (UA), antioxidant activity (CREA), anthocyanins and carotenoids (UA) and **antinutritional** (trypsin inhibitor activity (AU), calcium oxalate crystals (AU) , cyanogenetic glycosides (AU), acrylamide (CREA) compounds will be analysed both on tubers and other feed-usable parts of the crops such as leaves and stems and/or on processed products (pasta, bread and ice cream, eggs, milk and meat). To complete this general nutritional analysis, **species-specific animal feed** nutritive values of different species (small ruminants and poultry) will be determined (ICIA).

Task 5.2. Food processing (M25-M36; Leader: CREA (12PM), Contributors: CIRAD, UMa)

Fresh and dry pasta, different types of bread and flat breads and different ice cream formulations will be produced from the genotypes selected in task 5.1, and in **different blend formulations with cereals with a low or zero gluten index (einkorn, sorghum and corn)**. CREA will produce pasta in a pilot plant using different temperature (high 80°C and low 50°C and humidity) conditions. Sensory analyses will be performed by a panel of trained assessors along with instrumental assessment of cooking quality (optimal cooking time, firmness, stickiness, water absorption, cooking loss). The processing suitability of the various composites flours prepared for **bread making** will be evaluated (by CIRAD) based on the results of sensory analysis of breads, the pasting properties of blends and dough rheology. **Ice cream formulations** will be evaluated (UMa) in terms of different cooking times, cooking media (water, milk, vegetable drink), cooking modes (freeze-drying vs oven vs frying pan), different percentages of incorporation of the core ingredient (e.g., 10%, 15%, 20%), and use of flour (assess the need for a hot or cold process) (M24-M36).

Task 5.3. Feed processing and quality assessment of by-products (M1-M45; Leader: ICIA (10PM), Contributors: UAc, AUA)

Research will be conducted to assess the potential use of MRT by-products in animal feed (M1-M40). Taro and yam will be evaluated in Greece, sweet potato and cassava in the Canary Islands, sweet potato and taro in the Azores.

Different uses and methods of conservation (fresh, dried - hayed or pelleted - and ensiled) will be developed and evaluated. Different drying methods will be evaluated (forced-ventilation oven drying, open-air drying) with and without pelleting. Subsequently, these results, combined with those obtained for **silage suitability** and the **fermentative quality** index of MRT foliage, roots and tubers, will identify the characteristics of the best potential quality products to be used in the feed evaluation phase. Once the previous step has been completed, the inclusion of the experimental feeds with the best feeding potential in the diet will be tackled. Firstly, **palatability and ingestion** tests will be conducted to determine the optimum level of inclusion in the diet. Next, diets with **different degrees of inclusion** will be fed to target species, ruminants (goats, sheep, cattle) and monogastric species (poultry) for evaluation. In ruminants, yield indicators, milk production, the physicochemical and sensory quality of cheese by both expert (trained 10-12-member sensory panel) and consumer panels (80 - 100 participants) in each country, will be studied. In monogastric species (poultry) results will be obtained for **performance parameters**, meat and egg production and quality. In addition, their health and sanitary status will be monitored based on biochemical blood parameters. This specific assessment of the livestock products obtained is very important as it will add value to MRT crops as feed and provide a first insight into consumer acceptance.

Task 5.4. Consumer acceptance of MRT-based foods and livestock products of MRT-fed animals. (M24-M48; Leader: ILRIC (7PM), Contributors: CREA, UMa, ICIA, AUA & UAc)

ILRIC will **develop four recipes for each MRT crop**—two using raw tubers and two using processed forms, e.g. flour. Creative sessions with chefs and a dietician will be organised to design culinary prototypes that align with ROTATES requirements and appeal to consumers at the same time. These prototypes will be further refined during culinary sessions held at the COOKLAB, where detailed technical sheets will be produced, including lists of ingredients, step-by-step preparation instructions, descriptions of the final prototypes, and photographs of both the preparation process and the finished dishes. **Key food products—pasta, ice cream, and bread—will be evaluated** by sensory characterisation with gastronomy experts, including a pastry maker, a bread baker, and culinary chefs. Over six sessions, these experts will assess the processed products in order to identify their strengths and weaknesses, and provide critical feedback for optimisation. The **sensory quality of animal products, such as eggs and milk, derived from MRT-fed animals** will also be assessed by comparing standard products with those from tuber-fed animals, and by evaluating their acceptability by 100 consumers per product in laboratory-based tests using both raw ingredients and prepared recipes, like rice pudding. Additionally, ILRIC will conduct **consumer tests with 100 participants in a real dining setting** at its restaurant, to collect feedback on liking, acceptability, strengths, and weaknesses of the recipes used for all the products evaluated. Finally, a **professional photographer will document the selected tubers, their culinary transformation, and the final plated dish**. Alongside WP5 partners, ILRIC will produce a guide and three short videos to showcase the innovative use of MRT crops (D5.5)

Work package number	6
Work package title	Delivery of Ecosystem Services by MRT crops and crop management practices
Objectives ▪ Building strong know-how on ecosystem services assessment ▪ Establish an accurate baseline of current biodiversity status in pilot LLs ▪ Record/map and analyse above- and belowground biodiversity ▪ Assess soil health using cutting edge methodologies combined with simple, cheap, not disturbing methodologies	
Description of work Task 6.1: Diagnosis of baseline ecosystem services (M1-M4; Leader: CUT (11PM), Contributors: UMa, PNKH, INRAE, UMNG, INIAV, CIRAD) A comprehensive baseline assessment will be undertaken to understand the current state of biodiversity, soil health, water regulation, and carbon sequestration in the study areas (All partners). This task begins with the collection of existing literature and historical data relevant to the study area and MRT crops from various sources, including academic databases, relevant agencies, local agricultural records, and the testimonies of farmers and other stakeholders. It will be complemented by baseline field surveys to collect data on species diversity, soil health, and water quality. Representative sampling sites in the study areas will be selected randomly, and systematic surveys will be conducted to assess vegetation, soil health, nematode and microbial biodiversity, and soil health indicators, using BioFuncTool® to measure carbon dynamics, nutrient cycling, and soil structure maintenance (D6.1). To ensure consistency and accuracy across the project, training workshops will be held at the project's kick-off meeting, where work package (CUT) and task leaders (CUT, CIRAD, INRAE) will provide educational materials and hands-on training in methods such as nematode isolation, characterisation, and the use of the BioFuncTool®.	

Task 6.2: Impact of MRT crops on biodiversity (M5-M45; Leader: CIRAD (3,5PM), Contributors: UMa, PNKH, INRAE, UMNG, INIAV, CUT)

Biodiversity will be assessed to understand the impacts of different crop management approaches developed in a related work package (WP4). Biodiversity indicators, both above- and belowground, will be measured and analysed. **Above-ground biodiversity** will be assessed by measuring natural enemy biodiversity and the natural regulation of pests using non-selective interception traps and selective traps for bioindication. Experimental sites and farmers' fields will serve as study areas where systems of sentinel prey are established to empirically record pest regulation in the field. Simultaneously, camera traps will be used to identify vertebrate species that use the plots as habitats and to determine their status as pests, beneficial, or neutral agents. This task will also examine how the surrounding landscape, climatic variables, and agricultural practices influence the introduction of starchy MRT crops in the agroecosystem (all pLL partners). **Below-ground biodiversity** will be assessed by analysing microbial genetic diversity through metabarcoding using next-generation sequencing technologies (CUT, INRAE, INIAV). Soil samples will be collected at various depths and locations in the study areas, and nematodes will be identified and counted using microscopy, with further analysis using DNA barcoding for precise identification (D6.2).

Task 6.3: Impact of MRT crops on soil health (M1-M45; Leader: INRAE (9PM), Contributors: UMa, PNKH, UMNG, INIAV, CUT)

To identify the impacts of crop management on soil health services, soil samples will be collected and analysed to measure organic matter content, nutrient levels, and microbial activity, using BioFuncTool®. Pesticide use will be surveyed by farmers based on their calendar, compared to the baseline and monitored throughout the project (all pLL partners). The findings will be complemented by functional metagenomic analysis, which reveals the diversity and functions of microbial communities involved in nutrient cycling, organic matter decomposition, and the maintenance of soil structure (INRAE, CUT, INIAV). This knowledge (D6.3) will be harnessed to manage soil health more effectively, particularly by engineering microbial communities in the rhizosphere for improved plant performance.

Task 6.4: Data integration and modelling (M36-M47; Leader: CUT (10PM))

Data from biodiversity assessments, soil health monitoring, agronomic management (including water use; WP4), and meteorological stations will be combined. Integration will be achieved through a Geographic Information Systems (GIS) platform, which will integrate the data spatially. **Advanced modelling techniques**, including **geostatistical analysis**, **machine learning** (models like Random Forest, Gradient Boosting, or Neural Networks), and **crop simulation models**, will be used to predict crop responses to different environmental conditions. This will enable the development of tailored management practices that enhance yield, resilience, and sustainability. The resulting data will be **visualised on interactive dashboards** that will display real-time or historical spatially-explicit data on soil health, crop performance, phenological status, and weather patterns (D6.4)

Work package number	7
Work package title	Dissemination, Exploitation, Communication & Capacity Building
Objectives	
▪ Design and implement an agile capacity building programme tailored to the needs of pilots, partners, and stakeholders ▪ Raise awareness and maximise visibility among the key target groups; ▪ Build and engage a community of stakeholders and establish synergies with other relevant initiatives. ▪ Assess and maximise the exploitation potential, replicability, impact and added value of project results and outputs.	

Description of work**T7.1 – Agile Capacity Building (M1-M48, Leader: ENoLL (10,5PM), Contributors: ITC, CIRAD, UMa, CUT, PNKH, UMNG, INRAE)**

A dynamic capacity building plan will be designed and implemented to support Partners and Pilots in their activities, enabling the agile integration of new knowledge and tools in response to emerging needs. The plan will be designed and regularly revised following sensemaking methodology, MEL and system thinking approaches, based on accurate and regular needs analysis and first enabling correct understanding of key methods and approaches (MAA/LL, WP1), and then supporting harmonised pilot implementation to the related open innovation ecosystems. While ENoLL will mainly lead the definition of the plan and methodology, ITC will be the core supporting partner and will co-design and implement the training programme in alignment with WP1 activities and results. **Key steps:** (i) preliminary alignment training in MAA/LL principles and methods (M1); (ii) assessing the needs of the partners and pilots (M2-M4) and related ecosystems (M4-9); (iii) co-design of agile capacity building programme (D7.1, M9); (iv) implementation of training activities (action plan designs, monitoring framework, train-the-trainers, etc.);

(v) revision of capacity needs following interim monitoring (M24 and adaptation of future training. **Main results:** Agile capacity building programme and methodology, with appropriate tools and training material

T7.2 – Dissemination and communication (M1-M48, Leader: ENoLL (8PM), Contributors: all Partners)

A unified and accurate Dissemination & Communication strategy and targets will be established and implemented to maximise the visibility and awareness of project activities and results supporting their exploitation within and beyond the consortium. **Key steps:** (i) Design of Communication/Dissemination toolkit (M1) to ensure the consistent visual appearance of the project (ROTATES visual identity [logo, colour guide, templates], website/portal, social media accounts, flyers, banners, project presentations); (ii) launch of the ROTATES website (M2); (iii) Collective identification and selection of content/results/activities to be disseminated; (iv) Mapping of stakeholders and target audiences and definition of the key messages; (v) Definition of the DEC plan (D7.2, M6) with detailed planning of D&C activities (publications, events, etc.), guidelines and distribution of responsibilities, and preliminary collaboration plan with key initiatives (T7.4); (vi) Implementation of D&C activities, regular monitoring and fine-tuning of actions to match the timing of key projects and delivery of the results. Targeted communication and dissemination actions will inform and engage specific stakeholders and audiences through several channels e.g., conferences, fairs, workshops, webinars, scientific publications, stakeholder meetings, demonstration events, the contents of the ROTATES website, short videos, press releases, and a final ROTATES event. **Main results:** Project branding, D&C material, DEC plan.

T7.3 – Exploitation, replicability, recommendations, and contribution to policy (M4-M48, Leader: ENoLL (6PM), Contributors: all Partners)

This task will focus on the one hand, on mapping and planning the exploitation, sustainability, and maximisation of the impact of the results of the ROTATES project, and, on the other, on guiding the replicability of the approach and results, stimulating their adoption and effective uptake across Europe and beyond through impact assessment, recommendations drawn from the monitoring actions and active contributions to policy making. **Key steps:** (i) analysis of project KERs and identification of main steps in exploitation to be implemented during & after the project ends; (ii) definition of a preliminary plan for exploitation to be included in D7.2 (M6); (iii) analysis of interim monitoring of result and drafting preliminary recommendations for replication and policy making; (iv) regular assessment of results, ownership and exploitation pathways; (v) definition of post-exploitation steps; (vi) analysis of final monitoring results, (vii) delivery of replication/transferability guidebook including the results of replicability analysis, and the provision of a set of ROTATES recommendations (D7.3). **Main results:** Long-term sustainability and exploitation plan, Recommendations, Replicability guidebook.

T7.4 – Synergies and collaboration with related initiatives (M1-M48, Leader: CIRAD (2PM), Contributors: all Partners)

T7.4 will align with the dissemination effort to set up and sustain an active community of stakeholders, drawing from partners' existing networks and collaboration with both ongoing and previous projects, including the sister projects. In parallel, the consortium will seek systematic and sustainable cooperation with other related projects, initiatives, and clusters to mutually benefit from increased visibility and research alignment. Collaborative activities envisaged include regular updates on achieved results and methods, joint presentations of the projects and their impacts (e.g., workshops, webinars, symposia, public engagements), etc. The task will be coordinated by CIRAD in close collaboration with ENoLL, with all Partners taking responsibility for good cooperation and dissemination of the project results to linked initiatives. **Key steps:** (i) mapping stakeholders and relevant initiatives, (ii) connecting with such initiatives and reaching agreement on collaboration pathways (including definition of MoU or similar collaboration agreements) (D7.2), (iii) planning and implementing joint actions, (iv) regular alignment meetings with relevant sister initiatives for knowledge exchange and joint planning. **Main results:** Map of stakeholders to engage in participatory activities and in ensuring long-term sustainability.

Work package number	8
Work package title	Project Coordination and Management
Objectives The overall objective of this WP is ensuring professional coordination of scientific, legal, ethical, financial and administrative management of ROTATES project.	
Description of work Task 8.1 Monitoring progress and quality control (M1-M48; Leader; CIRAD (7PM), Contributors: all Partners) The Coordinator will establish an internal team to ensure smooth and timely operation of the ROTATES project. This task will ensure the ROTATES project functions efficiently and that information exchange between partners is adequate. A set of comprehensive work plans will be presented, dealing with each of the WPs. These Work Plans will be put together by the WP leaders at the start of ROTATES and include precise definitions of the partner's actions, risks and time scales for each WP. The coordinator will assemble the inputs in the project's overall Detailed Work Plan. Finally, the task will include quality assurance, documentation and information provision, which includes setting up SharePoint areas and GDPR compliance. Task 8.1.1 Reporting This task involves reporting on the technical, scientific and financial aspects of the project. This will see the cooperation of all the Partners under CIRAD coordination. The Partners shall provide the Coordinator with a detailed report on their technical and scientific activities, included the release of publications, together with the other results obtained by the consortium to enhance the visibility and impact of the project. Task 8.2 Administrative and financial coordination (M1-M48; Leader; CIRAD (7PM), Contributors: all Partners) The purpose of this task is to ensure efficient day-to-day running of the project through sound and effective administrative and financial management. In close collaboration with the financial unit, the Project Coordinator will handle communication with partners' financial units to monitor contracts. They will establish and maintain financial records and coordinate submission of financial statements by all the Project Partners. Task 8.3 Research Data Management (M1-M48; Leader; CIRAD (2,5PM), Contributors: all Partners) ROTATES' data management policy is to provide open access to most of the research datasets generated by the project. The data management strategy will be defined with all the Partners, in order to share the same objectives in terms of data sharing and exploitation. A first version of the Data Management Plan (DMP) covering each ROTATES dataset will be delivered in month 6. The DMP will specify all data curation, storage, and reporting procedures in accordance with FAIR principles. The task will also provide a server to host data from all WPs for the duration of the project. The DMP will be updated as needed during the project. Task 8.4 Coordination and organisation of Consortium Meetings (M1-M48; Leader: CIRAD (7PM), Contributors: all Partners) CIRAD will communicate regularly with its partners, respond to technical and other project management related queries in a timely fashion, and distribute the EU contribution to the partners. An effective communication mechanism will be set up including regular e-mail exchange between the partners, the decision-makers appointed in the General Assembly (GA) and members of the Advisory Board (AB). Timesaving, environmental-friendly and cost-effective solutions such as Teams will be used to enable meetings to be held online. In addition to day-to-day communication, CIRAD will organise regular consortium meetings with the WP Leaders, GA and AB to discuss the progress of the project and plan the next steps, as well as specific workshops to define the details of specific activities with the appropriate members of the Work Package concerned. Task 8.5: Ethics coordination (M1-48, Leader: CIRAD (2PM), Contributors: all Partners) All the partners who undertake field sampling or conduct social studies among local communities will obtain upfront approval from their institutional ethical review boards. CIRAD will engage its internal or external specialists in the different ethical domains to support ethics management. They will oversee all activities that may be of ethical concern, alone or in combination with other activities, and will also make sure there is no risk of ethical dumping. Finally, they will also ensure that ethical standards are respected for the publication of results.	

Table 3.1c: List of Deliverables

Number	Deliverable name	Short description	WP No.	Lead particip.	Type	Dissemination level	Delivery date
D1.1	MAA & LL methodological framework	Framework for implementation and monitoring of the MAA and LL approaches	1	ENoLL	R	PU	M12
D1.2	ROTATES MAA Performance	Detailed evaluation of the performance of ROTATES' MAA, based on MAA KPIs.	1	ITC	R	PU	M46
D2.1	Value-chain Context diagnoses	Value-chain diagnoses for the 10 target regions	2	SRUC	R	PU	M9
D2.2	Six business case studies	Business opportunities for the creation of a MRT supply chain	2	INRAE SRUC	R	PU	M42
D2.3	Guidelines for the development of the MRT value chain	Strategies and opportunities to promote the development of the MRT value chain.	2	SRUC INRAE	R	PU	M48
D3.1	Distribution of MRT diversity	Distribution of MRT diversity in the 10 countries and 5 outermost regions	3	CIRAD	R	PU	M24
D3.2	Assessment of seed-borne pathogens	Epidemiological survey of disease epidemics	3	UL	R	PU	M36
D3.3	Phenotypes of MRT diversity panels	MRT material with valuable traits for breeding programmes	3	UMa	R	PU	M36
D3.4	Genetic architecture of targeted traits	Genetic studies and the development of predictive models	3	CIRAD	R	PU	M48
D3.5	Review of the literature on sanitation protocols	Diagnostic tests and prevalence/severity of seed-borne pathogens of MRT crops	3	UL	R	PU	M17
D3.6	Recommendations for phytosanitary certification	Use of an integrated system approach to ensure a healthy seed system and sustainable production	3	UL	DEM	PU	M48
D4.1	Agronomic diagnostic	Agronomic diagnostic of current MRT production systems	4	CIRAD	R	PU	M12
D4.2	Evaluation of MRT for water use efficiency	Plot-level performance of MRT-based cropping systems to increase water use efficiency	4	CUT	R	PU	M24 M36 M48
D4.3	Evaluation of MRT for the biocontrol of pests and weeds	Plot-level performance of MRT-based cropping systems that advance biocontrol of pests and weeds	4	INIAV	R	PU	M24 M36 M48
D4.4	Evaluation of MRT for zero-deforestation	Plot-level performance of MRT-based cropping systems that promote zero-deforestation	4	UMNG	R	PU	M24 M36 M48
D4.5	On-farm adaptation of MRT	Farm-level performance of the most promising MRT-based cropping systems	4	CIRAD	R	PU	M24 M36 M48
D4.6	Training workshops	Final report on training farmers in the use of the new cropping systems	4	UMa	R	PU	M15 M28 M42

D5.1	Food & feed diagnoses	Diagnoses of physical and phytochemicals characteristics of MRT crops for food and feed	5	INRAE	R	PU	M15
D5.2	Nutritional & nutraceutical analysis	Biochemical characterisation of MRT	5	INRAE	DATA	PU	M30
D5.3	Food quality	Technological and sensorial quality of the 3 processed foods	5	CREA	R	PU	M40
D5.4	Feed quality	Characterisation of feed processing techniques and live-stock products	5	ICIA	R	PU	M45
D5.5	Booklet and videos of new recipes	Recipes using MRT as ingredients	5	ILRIC	DEC	PU	M46
D6.1	Diagnoses of current ecosystem services	Baseline data report and field survey	6	CUT	R, DATA	PU	M6
D6.2	Biodiversity assessment	Monitoring above- and below-ground diversity in the 6 pilot LLs	6	CIRAD	R	PU	M24 M46
D6.3	Soil health assessment	Monitoring soil health in the 6 pilot LLs	6	INRAE	R	PU	M24 M46
D6.4	Data integration & modelling	Actionable recommendations for sustainable crop management practices.	6	CUT	R	PU	M47
D7.1	ROTATES agile capacity building programme	ROTATES capacity-building methodology, aligned with a agile, iterative approach.	7	ENoLL	R	PU	M9
D7.2	DEC plan	Strategy, goals, activities, targets, guidelines, and materials for ROTATES WPs	7	ENoLL	R	SEN	M6
D7.3	Replicability guidelines and planning	A guide for replicating ROTATES methodologies	7	ENoLL	R	PU	M48
D8.1	Management and Quality Plan	Project structures, roles, quality, progress monitoring, risk analysis and activities	8	CIRAD	R	PU	M6
D8.2	Data Management Plan	Data management life cycle of the data generated and collected	8	CIRAD	DMP	PU	M6 M36 M48

Table 3.1d: List of milestones

Mill. Nb.	Milestone name	Work package(s)	Due date	Means of verification
M1	Collation of secondary data	WP3,4,5,6	M4	Report, database
M2	Mapping of key stakeholders and initiatives	WP1,7	M4	Stakeholder database
M3	Biochemical parameters that meet food and feed processing requirements	WP5	M6	List of varieties of the most appropriate quality for processing
M4	Identification of partners' and pilots' needs for training	WP1,2,4,6,7	M8	Report on training needs
M5	Action Plan and Baseline monitoring	WP1	M10	Action plan for each pilot LL
M6	Establishment of on-station trials	WP4,6	M18	Report, database
M7	Identification of a network of farms for adaptation	WP4,6	M18	Report, database
M8	Diversity panels for yam, taro, cassava, and sweet potato in their respective	WP3	M24	List of varieties of each crop introduced in each breeding

	breeding hubs			hub
M9	Mid-term evaluation analysis for revision of capacity building and recommendations	WP1,2,4,6,7	M24	Evaluation report
M10	Consultations with stakeholders and experts on adapting seed systems to local conditions	WP3	M30	Report on feedback provided by stakeholders and experts
M11	Validation of business case studies	WP1,2	M38	Feedback from the workshop

Table 3.1e: Critical risks for implementation #@RSK-MGT-RM@#

Description of risk	Work package(s)	Proposed risk-mitigation measures
Difficulty in managing a wide and international consortium [L: Low, S: Low -]	WP8	CIRAD staff have extensive experience in EU project management. If the risk occurs, robust risk monitoring will be set up with detailed guidelines, regular meetings, to ensure partner engagement.
Lack of response to surveys in pilot countries on the barriers and needs analysis consortium [L: Low, S: Medium]	WP1,2,7	Engagement trainings will be carried out to support partners and increased follow-up will be organised or a new strategy adopted in the case of lack of responses.
Difficulties to engage the different actors in co-creating and piloting activities in pilot LL, and Value Chains. [L: Low, S: Medium]	WP1,2, 4, 6,7	Dedicated value proposition messages will be defined for each pilot LL to support the building of trust. A dedicated D&C campaign will be designed to elicit interest among the actors.
Delays in receiving genetic resources from international genebanks, particularly for taro and sweet potato, could result in insufficient material to establish on time the Breeding Hubs	WP3	Initiate proactive communication with international genebanks to track genetic resource availability and shipping timelines. Secure alternative sources from national or regional genebanks and research entities as contingency plans for needed varieties.
Some technical risks could arise in formulating 100% MRT-based dry pasta and bread, due to technological issues caused by the processing of the dough.	WP5	These unexpected events can be mitigated by applying different processing techniques and/or mixing with cereal flours with gluten or low to no gluten content.
Drought or other climatic events that affect on-station and on-farm trials [L: Low, S: High]	WP3,4,5,6	Activities will be adapted to mitigate the impact of climate on the pilot LLs concerned. If needed, activities will be rescheduled.
Planting materials are lacking for on-station and on-farm trials [L: Medium, S: High]	WP4,6	Multiplication of material will be undertaken by seed producers in response to the need for planting material for field trials.
Delays in project timetable [L: Low, S: Medium]	All WPs	Project partners agree to apply contingency plans (tailored to match the exact circumstances) including re-allocation of resources, parallel execution of tasks and re-scheduling of activities.
Changes of key personnel in the consortium [L: Low, S: Medium]	WP8	Partners will be required to provide substitute staff with equivalent (or higher) qualifications and experience. The substitute staff will be informed in detail about the project, their role and duties.

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Table 3.1f: Summary of staff effort

	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WP8	Total PM
1- CIRAD	2	2	41.5	21.5	3	4	3	24.5	101.5
2 – INRAE	2	24.5	3	8	4	14	1.5	1.5	58
3 – CUT	2	1	1	9		35	3	1.5	52.5
4 – UL	2	1	30.5	1			1	1.5	37
5 – UMa	2	1	16	12	6	5	2	1.5	45.5
6 – UAc	2	9	12	1	9		0.5	1.5	35
7 – INIAV	2	6	19	11	0.5	12	3	1.5	55
8 – CREA	2	1	1		22.5		0.5	1.5	28.5
9 – ILRIC	2	1	1		9		1	1.5	15.5
10 – ICIA	2	1	3	2	39.5		1	1.5	50
11 – SRUC	2	6	1				0.5	1.5	11
12 – ACOESTE	2	1	10	3			0.5	1.5	18
13 – ENoLL	8.5	1	1				27	3	40.5
14 – ITC	29	1	1				4.5	1.5	35
15 – TP	2	1	13				0.5	1.5	18
16 – AUA	2	0.5	1.5	0.5	19		1	1.5	26
17 – PNKH	2	1	14	14		4.5	1.5	1.5	38
18 – UMNG	2	1	1	10.5		4	1.5	1.5	21
Total PM	69.5	55.6	170.5	93.5	112.5	78.5	52	51.5	686

Table 3.1g: ‘Subcontracting costs’ items

Participants	Subcontracting Cost (€)	Description of tasks and justification
1-CIRAD	75,000€	In T6.2: Taxonomic identification of trapped samples (15k). In T3.1: MRT Genome sequencing for pan-genome approach (50k). In T8.5: Ethical advisor to provide support concerning exchanges of biological material or any other critical issue (10k)
5-UMa	31,000€	In T4.5: Implementation of a pilot Living Labs through the development of educational and social activities (20k). In T5.2: Development of new formulations of MRT-based milk alternatives ice-cream by an ice-cream maker company (11k)
9-ILRIC	10,000€	In T5.4: Video capsule of new recipes to showcase the innovative use of MRT crops
10-ICIA	16,400€	In T5.3: Specific root and plant physicochemical analyses (5k), analyses of ensilability and fermentative quality of silages (5,4k), analyses of amino and fatty acids in livestock products (6k)
11-SRUC	40,000€	In T2.2: Marketing company for the implementation of economic experiment surveys (online questionnaire, choice of experiment, laboratory experiment) conducted with consumers in case studies.
15-TP	6,739€	In T3.1&2: Renting land for field trials and laboratory analyses
Total	179,139€	

Table 3.1h: 'Purchase costs' items

Participants	Cost (€)	Justification
1-CIRAD	Travel and subsistence (TS)	99,174€ WP3: Follow up meetings (Guadeloupe – France – 3 (9k), trips to Madeira & Greece for phenotyping (5k), Participation in congresses (3.5k), local field trips within French Guiana – 28 days (per diem & car rental, 6k). WP4: Field trips (10.1k). WP5: Partner follow-up meetings (13.5k). WP6: Trips to support Partners– 10 (22.5k). WP8: Travel by advisory board– 6 (9.6k), Trips for the purpose of governance– 10 (20k)
	Equipment	3,200€ WP4: Weather station (1.6k), Auger (1.6k).
	Other goods, works and services (OGS)	153,790€ WP3: Open access publications (5k), Laboratory & field consumables (30k), field workers (10k), transportation of samples (4k). WP4: Soil analysis (15k), Field workers - 45 days (15,6k), Probe tubes – 80 (4,8k). WP5: Chemical products, laboratory supplies and devices (13k), Consumables for processing devices (10,5k). WP6: Traps for entomofauna & shipment (6k), camera traps & shipment (8k). WP9: Financial audit (5,7k), Kick-off meeting (8k), graphic designer (10k), Project management software (8k).
	Total	256,164€ Remaining (<15%): 43,600€
2-INRAE	TS	37,000€ WP2: Travel required for case studies & to present results to stakeholders (8) (18,5k). WP8: Participations in project meetings (4) - 2 pax (18,5k)
	OGS	34,692€ WP1: Stakeholder engagement activities (1,5k). WP3: Introduction of plants in in-vitro culture (9k). WP4: Pilot LL workshops (3) (7,1k). WP6: Insect traps, phone Apps, barcoding of bacteria & fungi, 3 bacterial gene functions, etc.(7,1k). WP7: Conference & publication fees (10k).
	Total	71,692€ Remaining (<15%): 12,175€
3-CUT	TS	10,000€ WP8: Participation in annual project meetings (4) - 2 pax (10k)
	OGS	43,112€ WP4: Pilot LL workshops (3) (7,1k). WP6: Consumables for biodiversity and soil health assessments (chemicals, plasticware, molecular biology kits-NGS) (30k), BioFuncTool® kit (6k).
	Total	53,112€ Remaining (<15%): 7,350€
5-UMa	TS	10,000€ WP8: Participation in annual meetings (3) - 2 pax
	Equipment	55,000€ WP3: Perkin Elmer SPECTRUM TWO NIR (44k), Greenhouse for phenotyping taro's tolerance to extreme climatic conditions (11k)
	OGS	14,612€ WP4: Pilot LL workshops (3) (7,1k). WP8: Hosting of an annual project meeting (40 pax, catering & local transport)
	Total	79,612€ Remaining (<15%): 8,452€
6-UAc	TS	6,800€ WP8: Participation in annual meetings– 2 pax (4)
	OGS	15,000€ WP3: Consumables for collection & in vitro preservation of sweet potato & taro germplasm (3k), molecular biology consumables for genetic diversity analysis (4k), consumables for screening viruses on sweet potato and taro germplasm and for the sanitation of sweet potato (8k)
	Total	21,800€ Remaining (<15%): 3,800€
7-INIAV	TS	38,400€ WP7: Participation in conferences (4) - 2 pax (24k). WP9: Participation in project meetings (4) - 4 pax (14,4k)
	Equipment	15,451€ WP6: Stereoscopic microscope for identification of functional groups of arthropods (5,6k), Digestor (digiblock for sample preparation) (9,8k)
	OGS	89,462€ WP3: Renting of farmers' fields for SP varieties (4k), chemical reagents and consumables (HPLC analysis) (6k), orthophoto map analysis (8k), field assays to monitor seed-borne diseases (4k). WP4: Renting of farmers' fields for experimental trials (4k), Sweet potato plants (3k), Monitoring of pests, diseases and weeds (2k). WP4: Pilot LL workshops (3) (7,1k). WP6: Chemical reagents and small-scale lab material, DNA extraction kits, sequencing (around 160Gb), bioinformatic analysis, BioFuncTool® kit (28,4k), chemical reagents and consumables for analytical determination of nutrient contamination in water (19k). WP7: Open access publications - 2 (4k).
	Total	143,313€ Remaining (<15%): 21,362.44€

8-CREA	TS	7,800€	WP8: Participation in 3 project meetings– 3/4 pax
	Equipment	3,696€	WP5: Industrial potatoes peeler/slicer
	OGS	25,500€	WP5: Enzymatic kits, reagents, laboratory devices for the characterisation of physical & phytochemicals (8,5k), Chemical reagents, lab devices and consumables for food processing (5k). WP8: Hosting of an annual project meeting (40 pax, catering & local transport) (12k).
	Total	36,996	Remaining (<15%): 6,500€
9-ILRIC	TS	5,000€	WP8: Participation in project meetings (4) – 1 or 2 pax
	OGS	36,110€	WP5: Culinary ingredients (6,3k), Head baker & pastry chef services (11k), <i>Maître d'hôtel</i> service & incentives for testing consumers (10,6k), Graphist (7k), Photographer (4,5k) for culinary booklet.
	Total	41,110€	Remaining (<15%): 6,000€
10-ICIA	TS	8,600€	WP7: Participation in international conferences – 2 (4,8k). WP8: Participation in annual meetings (4) - 2 pax (3,8k)
	OGS	47,180€	WP3: Collection and shipment of samples (13,1k). WP5: Local irrigation system (4k), Chemical reagents for analysis of feed nutritive value (2,5k), Chemical reagents and miscellaneous material for experimental microsilos/dehydrated products (2k), Experimental farm silos (1k), Goats feeding (6,5), Biochemical blood reagents (2k), Materials required for dairy laboratory & sensory analysis of cheese (1,2k), Hens and broilers (1k), Poultry feeding (3,8k), Vaccinations, treatments (0,8k), Materials for egg and meat analyses (2,5k). WP7: Open access publication, conferences registration fees, brochures (5,8k)
	Total	55,780€	Remaining (<15%): 8,800€
11-SRUC	TS	20,000€	WP2: Trips for diagnosis – 1 pax, 5 days (5k), Travel to case study locations (2) - 2 pax, 3 days (9k). WP8: Participation in annual meetings (4) - 2 pax (6k).
	Total	20,000€	Remaining (<15%): 3,000€
13-ENoLL	TS	28,500€	WP7: 5 capacity-building trips – 2 pax (10k), 6 trips for dissemination & communication activities – 1 or 2 pax (10,5k). WP8: Participation in annual meetings (4) - 2 pax (8k)
	OGS	24,000€	WP7&8: Room rental, catering, printing, video production, translations for dissemination & communication activities and final project meeting (24k)
	Total	52,500€	
15-TP	Equipment	12,862€	WP3: Culture and conservation room, cold room, autoclave, reverse osmosis, greenhouse equipment
	OGS	32,711€	WP3: Energy consumption (electricity, gas) and consumables for maintenance and multiplication of in-vivo genetics (sowing, pot transfer, cuttings, etc.) (24,4k), laboratory consumables for multiplication and maintenance in vitro (disinfection, culture media, trays, test tubes, etc.) (8,3k)
	Total	45,573€	Remaining (<15%): 7,106€
16-AUA	TS	20,400€	WP3: Trips to collect material/phenotyping (5) (8,4k). WP8: Participations in annual project meetings (4) - 2 pax (12k)
	OGS	25,082€	WP5: SP cultivation, harvesting, evaluation for sheep feed (15,4k), Consumables for analyses of animal products (9,7k)
	Total	45,482€	Remaining (<15%): 6,800€
17-PNKH	TS	6,000€	WP4: Travel from French Guiana to the Republic of Congo for exchange learning – 2 pax
	OGS	43,112€	WP3: Land preparation for planting, collection, and maintenance (15k). WP4: Field consumables for experimental trials (15k), Pilot LL workshops (3) (7,1k). WP6: Biofunc-tool® kit (6k).
	Total	49,112€	Remaining (<15%): 4,000€
	TS	23,500€	WP4: Consolidating methodologies and sharing experience in French Guiana, 2 pax, 14 days, France 2 pax, 7days (15k). WP8: Participation in annual project meetings (4) - 1

18-UMNG			pax (8,5k).
	Equipment	7,800€	WP4: Spectrophotometer (MicroNIR)
	OGS	40,112€	WP4: Field workers (27k), Pilot LL workshops (3) (7,1k). WP6: Biofunctool® kit (6k).
	Total	71,412€	Remaining (<15%): 11,300€

Table 3.1i: ‘Other costs categories’ items

4-UL	Cost (€)	Justification
Internally invoiced goods and services	7,000€	High throughput sequencing (HTS) of nucleic acid extracted from the sampled plants (WP3)

Each partner will provide adequate resources to fulfil their roles in the project as shown in Table 3.1f. The total budget is 5 251 K€ and the requested grant is 4 999 763€. CIRAD, INRAE, ICIA are providing 251k€ as own resources contribution. The distribution of the budget and of the grant is well balanced among the different types of Partners with 49% for Research institutes, 32% for Universities, 7% for SME, 4% for Farmers organisations and 8% for the NGO. The types of costs are also well distributed with 54% of direct personnel costs, 24% for purchases, and 19% for indirect costs.

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3.2 Capacity of participants and consortium as a whole #@CON-SOR-CS@# #@PRJ-MGT-PM@#

Consortium description

The consortium encompasses 18 partners from nine European countries, five outermost regions and one African country. It has all the necessary skills and competences to successfully carry out the activities defined by the project and to meet the call requirement. ROTATES embraces a multidisciplinary and multi-actor approach. Thus, it brings together scientists representing different disciplines and stakeholders from all levels of the value chain (Farmers, Processors, Marketing personnel, Consumers). The consortium comprises five Research Institutes including one that is a digital innovation hub (ITC), six Universities, two Farmers’ associations, two SMEs (one seed company Thomas Plants, and one food and hospitality Institute (ILRIC institute, former Paul-Bocuse Institute) and one NGO (ENoLL –living lab network).

Consortium disciplinary and inter-disciplinary knowledge

The consortium covers different fields and disciplines and together represents the entire value chain.

	Research Institutes						Universities							Farmers' association		SME	NGO	
	CIRAD	INRAE	INIAV	CREA	ITC	ICIA	CUT	UL	Uma	Uag	AUA	UMNG	SRUC	PNKH	ACOESTE	TP	ILRIC	ENaLL
Multi-Actor Approach	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Living Labs	x	x	x		x		x		x		x			x	x			x
Socio-economy		x											x				x	
Genetics and Breeding	x			x					x					x				
Phytopathology		x	x					x		x						x		
Agronomic practices	x	x					x		x			x		x	x			
Entomology	x			x			x											
Soil health														x	x			
Feed science & processing						x	x				x							
Food science & processing	x				x												x	
Data processing and modeling	x	x	x	x		x	x	x	x	x	x	x	x			x	x	
Dissemination, Exploitation & Communication	x				x									x	x		x	

Consortium partners’ main contribution to the project

CIRAD, the ROTATES project coordinator (**WP8 leader**), is a public industrial and commercial enterprise focused on development from local to global scales. CIRAD is affiliated with the University of Montpellier, a Shanghai World University top-ranking institution in Ecology. With over 100 years of experience in Root and Tuber crops across Africa, the Pacific region, Latin America, and European outermost regions, CIRAD will leverage its expertise in

global agrobiodiversity, agroecology, and its deep understanding of EU biodiversity policies (it has been involved in more than 100 European projects). CIRAD will **lead WP4**, thereby ensuring the sustainability of the introduction of MRT in the European agrosystems. It will contribute its expertise on entomology and ecology (**WP6 task leader**), on RT crop biodiversity and breeding (**WP3 task leader, Yam Hub**) and on food quality (**contributes to WP5**).

INRAE, a national research institute for sustainable agriculture, food, and the environment, has over 50 years of expertise in Root and Tuber Crops in Guadeloupe. It manages 430 in-vitro yam accessions, including 80 in the field. INRAE **leads WP2** on experimental economics and social sciences, **contributes to WP3** on yam viruses and sanitation, and **leads Task 5.1** on physico-chemical properties of Tropical RT. It hosts the MetaboScope platform, has experience in agroforestry, and will provide facilities for on-site field assays (**pLL2-France**).

INIAV, the State Laboratory of the Ministry of Agriculture and Fisheries, supports sustainable agricultural development through scientific and policy research. It manages plant and animal germplasm databases and national reference collections. In the ROTATES project, INIAV will contribute its expertise on sweet potato viruses and sanitation, on setting up a **Sweet Potato Hub (Task 3.4)**, **leads Task 4.3** on experimental agroecology for biological control, and **contributes to WP6** on ecosystem services, particularly biodiversity (**pLL3-Portugal**).

CREA, Italy's leading agri-food research institute belongs to the Ministry of Agriculture, focuses on agro-industrial processes and sustainable agri-food supply chains. CREA specialises in enhancing the quality of agri-food products, particularly pasta and bakery goods made from wheat, minor cereals, and legumes. With over 40 years of experience, CREA will use its advanced labs for biochemical and technological analysis **WP5 leader**.

ICIA, the leading agricultural research centre in the Canary Islands, focuses on subtropical and arid-adapted crops. The Animal Production Unit, with over 40 years of experience improving Canarian livestock, particularly goats and poultry, will assess the potential of using MRT crop residues for high-nutritional animal feed. ICIA will also evaluate the quality of derived livestock products such as milk, cheese, meat, and eggs, **leads Task 5.3**.

UAc plays a key role in research and innovation in agriculture and agro-industry, particularly in areas prioritized by RIS3 Azores 2022-2027. Known for its strong R&D capacity, UAc focuses on agriculture, biotechnology, genetic characterisation, and animal nutrition. In the ROTATES project, UAc will contribute expertise in feed processing (**contributes to Task 5.3, case study in WP2**) and virus sanitation (**contributes to Task 3.4**), with a focus on traditional crops like sweet potato and taro.

AUA, Greece's largest agricultural university, specialises in agricultural and environmental sciences. The Laboratory of Nutritional Physiology and Feeding excels in animal nutrition and in evaluating alternative feed resources. In ROTATES, AUA will contribute to identifying and characterising local sweet potato species (**contributes to WP3**), and determining their nutritional and antinutritional value in sheep feed (**Case study WP2 & 5**).

CUT, Cyprus' only academic institution, was founded in 2004. The Laboratory of Agricultural Microbiology and Biotechnology at CUT has an experienced research team focussed on microbiology from farm to fork, supporting the One Health concept—healthy soils, plants, animals, and foods. In ROTATES, CUT leads WP6 on microbial ecology and biotechnology, will use remote sensing for agriculture (**Task 6.4**), and will conduct field trials for both research and demonstration (**contributes to WP4, pLL1-Cyprus**).

The UL plant pathology laboratory has more than 30 years of experience in the study of plant pathogens on clonally propagated crops, including potato, cassava, banana. As **WP3 task leader**, the laboratory will mobilize and share with members of the consortium its cutting-edge expertise gathered in previous EU and international projects for pathogen detection and for the development of healthy seed systems. The laboratory is equipped with all necessary materials and has access to the university sequencing platform.

UMa focuses on agro-diversity, genetic resources, and food technology, and participates in national and international research projects. It belongs to the FAO Regional Network of Germplasm Banks and contributes to the EMPHASIS European Infrastructure and Portuguese Plant Phenotyping platform, UMa-ISOPlexis supports genetic resources strategies and local varieties. In the ROTATES project, it **leads WP3**, it will implement a **Taro Hub**, **contribute to WP4 & WP6** through the development of management practices that promote water use efficiency, and provide expertise in food processing (**contributes to task 5.2**) and through the **pLL5-Madeira (WP1, 4, 6)**.

SRUC is a knowledge-based organisation that supports land-based industries and rural communities. Recognised for its impactful veterinary and agricultural research, SRUC collaborates with the University of Edinburgh, ranked top in the domain in the UK. SRUC focuses on policy-led research, particularly in social science applied to the rural economy, environment, and society. In ROTATES, SRUC will be responsible for the activities on the development of MRT value chains (**WP2 leader**).

PNKH, a non-profit association founded in 2014, works on the transmission and promotion of traditional Amerindian knowledge and supports family agriculture in French Guiana. In ROTATES, PNKH will implement a **Cassava-HUB** and will be involved in the development of agroecological management practices (**contributes to WP3, WP4, WP6**), **pLL4-French Guiana**.

ACOESTE, a non-profit agricultural association founded in 2015, supports and trains farmers across Madeira. It organises agricultural events and provides public counselling services. In ROTATES, ACOESTE will conduct surveys (**contributes to WP3, WP4, WP6**), maintain a collection of elite MRT cultivars and establish local pilot living labs with intercropping systems from associated farmers (**contributes to WP1, WP4 & 6**).

The TP Group is a leading French producer of vegetable and fruit planting material for professional market gardeners. Operating across two agroclimatic areas: Humid Atlantic (Brittany) and dry Mediterranean (Provence), it manages around 40 hectares of production facilities, including greenhouses, a grafting room, and a molecular biology lab. The TP Group has developed a breeding programme focused on sweet potato, aiming to offer farmers productive varieties suited to a variety of European climates. In ROTATES, TP Group **contributes to WP3 & 6**.

ILRIC conducts research on food and hospitality, focusing on pleasure and health. It offers a doctoral programme, collaborates with universities, and develops industry knowledge through researcher-expert cooperation. The centre emphasizes transferring scientific knowledge to the food and food service industries for public well-being. In the ROTATES project, ILRIC will **lead Task 5.5**, contributing expertise in culinary development, product evaluation by experts, and consumer testing.

ITC, a non-profit Business Support Organisation in North-Eastern Slovenia, supports cross-sectoral innovation and ICT in rural sectors. ITC focuses on transforming SMEs, food system actors, and farmers into "Smart" entities, fostering a resilient and sustainable rural innovation ecosystem. It manages the Green Point Living Lab and the European Digital Innovation Hub - DIH AGRIFOOD, and provides support for the Multi-Actor Approach (MAA). ITC has developed an MAA implementation methodology for Living Labs and innovation ecosystems, and will **lead WP1** in the ROTATES project.

ENoLL, the European Network of Living Labs, is an international association of certified Living Labs specialising in user-driven and open innovation ecosystems. With expertise in creating and certifying Living Labs, ENoLL facilitates knowledge exchange and partnerships among its 550+ members worldwide. ENoLL will **lead WP7** on capacity building and **contribute to WP1** on the methodological framework.

UMNG in Republic of the Congo, through its agronomy school has expertise in agronomic practices and management with a strong tradition of stakeholder involvement (particularly that of farmers). UMNG has researchers who will provide technical knowledge on growing cassava and on agricultural practices in the Congo. The country has a favourable environment for agronomic diagnosis and field experimentation (**pLL6-Congo, WP4 task leader**). As in French Guiana, cassava is the main root and tuber crop in the Republic of Congo. Both regions are facing problems of deforestation, which are threatening the forests of the Amazon and the Congo. The research activities to be carried out in the Republic of the Congo and French Guiana, the exchange of knowledge and experience, and the reciprocal pilot LLs visits will benefit both regions.

➤ *Industrial/commercial involvement in the project to ensure exploitation of the results*

ROTATES aims to have an impact beyond the timeline of the project. The activities concerning the value chain are designed to account for the different actors and the different products to be obtained from MRT, including raw materials processed into food (pasta, ice-cream and bread) and by-products from feed processing. CREA already works in close collaboration with large-scale pasta-sector companies (e.g. Barilla) as well as small companies (Fornovecchino, Verdi Terre, Eocene, Pirro). ICIA works in close collaboration with industrial feed companies such as Graneros de Tenerife, Compañía Canaria de Piensos S.A (CAPISA) and Cereales Archipiélago. ILRIC collaborates with company restaurants, with industry and with distributors. This also applies to the partners working on food (UMa and CIRAD) and those working on feed processing (UAc and CUT). The partners and their associated industrial partners will be involved in the economic, social, and cultural studies to

ensure the uptake of MRT. ROTATES will implement a gender sensitive, multi-actor approach ensuring, co-creation, co-design, co- implementation and co-exploitation of ROTATES outcomes. The involvement of farmers through the pilot LL will ensure the sustainable use of MRT beyond the project lifespan while taking the importance of income for farmers into account (provisioning services and avenues for value chain).

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